

HelixTerminator — API & Database Specification

Document 07: API and Database

Project: HelixTerminator

Backend: Go 1.25, Gin Gonic framework

Module: helixterminator.io/core

Database Stack: PostgreSQL 17.2 (primary), SQLite (dev/embedded), Redis 8 (cache/sessions)

Architecture: Database-per-service microservices (25 services)

API Style: REST (external clients) + gRPC (internal service-to-service)

Version: 1.0.0

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1. API Design Principles

1.1 Overview

HelixTerminator exposes a versioned HTTP/REST API for all external clients (web application, desktop client, CLI, third-party integrations). Internal microservice communication uses gRPC for low-latency, strongly-typed contracts. This document is the authoritative specification for both surfaces.

The REST API adheres to RFC 7231 (HTTP Semantics), RFC 7807 (Problem Details for HTTP APIs), RFC 8288 (Web Linking), and the OpenAPI 3.1 specification. All timestamps are RFC 3339 / ISO 8601 in UTC. All identifiers are UUID v4 (RFC 4122).

1.2 REST Conventions

Resource Naming

Resources are named as **plural nouns** in lowercase, with words separated by hyphens when needed. The URL structure follows a hierarchy that mirrors the domain model.

Pattern	Example	Meaning
/api/v1/{resource}	/api/v1/hosts	Collection
/api/v1/{resource}/{id}	/api/v1/hosts/abc-123	Single resource
/api/v1/{resource}/{id}/{sub}	/api/v1/hosts/abc-123/connections	Sub-collection
/api/v1/{resource}/{id}/{action}	/api/v1/sessions/abc-123/resize	Action on resource

Rules: - **Never** use verbs in resource URLs (wrong: /getHosts, /createHost). Use HTTP methods for actions. - **Always** use lowercase. Never camelCase or PascalCase in URLs. - Use hyphens (-) not underscores (_) in URL path segments. - Trailing slashes are **not**

allowed. `/api/v1/hosts/` returns 301 Moved Permanently to `/api/v1/hosts.` - UUIDs in path parameters use lowercase hex with hyphens: `550e8400-e29b-41d4-a716-446655440000`.

HTTP Methods

Method	Semantics	Idempotent	Safe	Body
GET	Retrieve resource or collection	Yes	Yes	No
POST	Create resource or invoke action	No	No	Yes
PUT	Replace resource entirely	Yes	No	Yes
PATCH	Partial update (JSON Merge Patch, RFC 7396)	No	No	Yes
DELETE	Remove resource	Yes	No	Optional
HEAD	Same as GET but no body (used for existence checks)	Yes	Yes	No
OPTIONS	CORS preflight; also lists allowed methods	Yes	Yes	No

PATCH uses JSON Merge Patch semantics: send only the fields you want to change; omitted fields remain unchanged; set a field to `null` to clear it. Full replacement operations use PUT.

HTTP Status Codes

2xx Success

Code	Name	Usage
200 OK	Standard success	GET, PUT, PATCH with body in response
201 Created	Resource created	POST that creates a resource; Location header present
202 Accepted	Async operation started	Long-running jobs queued
204 No Content	Success, no body	DELETE, POST actions with no response body
206 Partial Content	Range response	Binary downloads with Range header

3xx Redirection

Code	Name	Usage
301 Moved Permanently	URL changed	Trailing slash removal
302 Found	Temporary redirect	OAuth callback redirects
304 Not Modified	Conditional GET hit	ETag / If-None-Match

4xx Client Errors

Code	Name	Usage
400 Bad Request	Malformed request	Syntax error, missing required field, type mismatch
401 Unauthorized	Not authenticated	Missing or invalid token; include WWW-Authenticate header
403 Forbidden	Authenticated but not authorized	Insufficient permissions for the resource/action

Code	Name	Usage
404 Not Found	Resource does not exist	Also returned deliberately for privacy (e.g., vault of another user)
405 Method Not Allowed	Wrong HTTP method	Include Allow header listing valid methods
406 Not Acceptable	Cannot satisfy Accept header	Server cannot produce the requested content type
409 Conflict	State conflict	Duplicate unique constraint, optimistic lock failure
410 Gone	Resource permanently deleted	Soft-deleted resource accessed after retention period
413 Content Too Large	Body exceeds limit	File upload too large
415 Unsupported Media Type	Wrong Content-Type	Send JSON without Content-Type: application/json
422 Unprocessable Entity	Semantic validation failure	Well-formed JSON but business rule violation
429 Too Many Requests	Rate limit exceeded	Include Retry-After header

5xx Server Errors

Code	Name	Usage
500 Internal Server Error	Unhandled exception	Should never be exposed; always log with trace ID
502 Bad Gateway	Upstream service failure	Microservice unreachable
503 Service Unavailable	Maintenance / overload	Include Retry-After
504 Gateway Timeout	Upstream timeout	gRPC or DB timeout

1.3 Versioning Strategy

HelixTerminator uses **URI path versioning**. The version segment is the second path component after the API root:

`https://api.helixterminator.io/api/v1/hosts`

`https://api.helixterminator.io/api/v2/hosts`

Version lifecycle:

Phase	Duration	Behavior
Current	Indefinite	Fully supported, receives bug fixes and features
Deprecated	12 months	Supported, deprecation notice in response headers, no new features
Sunset	3 months	410 Gone with migration guide URL

Deprecated endpoints include:

Deprecation: Sat, 01 Jan 2028 00:00:00 GMT

Sunset: Sat, 01 Apr 2028 00:00:00 GMT

Link: <<https://docs.helixterminator.io/migration/v1-to-v2>>;
rel="deprecation"

Breaking vs. non-breaking changes:

Non-breaking (no version bump required): - Adding new optional fields to response bodies - Adding new optional query parameters - Adding new endpoints - Adding new enum values to non-exhaustive enumerations

Breaking (require new version): - Removing fields from responses - Changing field types - Changing authentication requirements - Removing endpoints - Changing pagination semantics

Version routing in Go/Gin:

```
v1 := router.Group("/api/v1")
{
    v1.Use(middleware.AuthRequired())
    hosts := v1.Group("/hosts")
```

```

    {
        hosts.GET("", hostHandler.List)
        hosts.POST("", hostHandler.Create)
        hosts.GET("/:hostId", hostHandler.Get)
        hosts.PUT("/:hostId", hostHandler.Update)
        hosts.DELETE("/:hostId", hostHandler.Delete)
    }
}

v2 := router.Group("/api/v2")
{
    v2.Use(middleware.AuthRequired())
    // v2-specific routing
}

```

1.4 Pagination

All collection endpoints that may return more than 20 items support cursor-based pagination. Cursor-based pagination is preferred over offset-based pagination because:

1. It is stable: inserting or deleting records between pages does not cause items to be skipped or duplicated.
2. It scales: no `COUNT(*)` or `OFFSET` scans required; just `WHERE id > $cursor LIMIT $pageSize`.
3. It is efficient for large datasets (e.g., audit logs with millions of rows).

Cursor Pagination Request Parameters

Parameter	Type	Default	Description
cursor	string	(none)	Opaque cursor returned by previous page
limit	integer	25	Number of items per page (1–100)
sort	string	created_at:desc	Sort field and direction, colon-separated
direction	string	next	next or prev — direction of

Parameter	Type	Default	Description
			cursor traversal

The `cursor` value is a **base64url-encoded JSON object** containing enough state to reconstruct the query position. It is opaque to clients.

Example cursor payload (before encoding):

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "created_at": "2026-01-15T10:30:00Z",
  "sort_field": "created_at",
  "direction": "next"
}
```

Encoded:

```
eyJpZCI6IjU1MGU4NDAwLWUyOWItNDZkNC1hNzE2LTQ0NjY1NTQ0MDAwMCIsImNyZW
F0ZWRFYXQiOiIyMDI2LTAxLTE1VDEwOjMwOjAwWiIsInNvcnRfZmllbGQiOiJjcmVh
dGVkX2F0IiwizGlyZWNoaW9uIjoibmV4dCJ9
```

Cursor Pagination Response Envelope

All collection responses use a standard envelope:

```
{
  "data": [...],
  "pagination": {
    "cursor_next": "eyJpZCI6...",
    "cursor_prev": "eyJpZCI6...",
    "has_next": true,
    "has_prev": false,
    "limit": 25,
    "total_count": 1847
  }
}
```

`total_count` is only included when the query cost is acceptable (small collections). For large tables it is omitted or approximated.

Offset Pagination (Legacy / Admin)

Certain admin endpoints that require random access (e.g., audit log export with page jumps) support offset pagination:

Parameter	Type	Default
page	integer	1
per_page	integer	25 (max 100)

Response:

```
{
  "data": [...],
  "pagination": {
    "page": 3,
    "per_page": 25,
    "total_pages": 74,
    "total_count": 1847
  }
}
```

1.5 Filtering, Sorting, and Searching

Filtering

Filters are passed as query parameters. Multi-value filters use repeated parameters or comma-separated values.

```
GET /api/v1/hosts?status=active&os=linux
GET /api/v1/hosts?tag=production,staging
GET /api/v1/audit/events?
event_type=login_success&event_type=login_failure
```

Range filters use `_gte`, `_lte`, `_gt`, `_lt` suffixes:

```
GET /api/v1/audit/events?
created_at_gte=2026-01-01T00:00:00Z&created_at_lte=2026-02-01T00:00:00Z
```

Boolean filters accept `true/false`:

```
GET /api/v1/hosts?jump_enabled=true
```

Sorting

The `sort` parameter accepts a comma-separated list of `field:direction` pairs:

```
GET /api/v1/hosts?sort=name:asc
GET /api/v1/hosts?sort=created_at:desc,name:asc
```

Allowed sort directions: asc, desc. Invalid field names return 400 Bad Request.

Full-Text Search

The q parameter activates full-text search (backed by PostgreSQL pg_trgm trigram indexes or tsvector full-text search):

```
GET /api/v1/hosts?q=prod-web
GET /api/v1/snippets/search?q=git+rebase
```

Search scoring is returned in _score when q is present.

Field Selection (Sparse Fieldsets)

To reduce response payload, clients may request specific fields:

```
GET /api/v1/hosts?fields=id,name,hostname,status
```

This is a performance optimization hint; the server may return additional fields if they are always required (e.g., id).

1.6 Error Response Format

All errors follow RFC 7807 “Problem Details for HTTP APIs”. The Content-Type is application/problem+json.

```
{
  "type": "https://errors.helixterminator.io/v1/validation-error",
  "title": "Validation Error",
  "status": 422,
  "detail": "The request body contains invalid field values.",
  "instance": "/api/v1/hosts",
  "trace_id": "7f3a9b2c-1d4e-5f6a-8b9c-0d1e2f3a4b5c",
  "errors": [
    {
      "field": "hostname",
      "code": "required",
      "message": "hostname is required"
    },
    {
      "field": "port",
      "code": "range",
      "message": "port must be between 1 and 65535",
      "value": 99999
    }
  ]
}
```

```
}  
]  
}
```

Standard error type URIs:

Type URI	HTTP Status	Description
errors.helixterminator.io/v1/validation-error	422	Field-level validation failures
errors.helixterminator.io/v1/authentication-required	401	No valid authentication credentials
errors.helixterminator.io/v1/forbidden	403	Authenticated but lacks permission
errors.helixterminator.io/v1/not-found	404	Resource does not exist
errors.helixterminator.io/v1/conflict	409	Conflicting state (duplicate, optimistic lock)
errors.helixterminator.io/v1/rate-limit-exceeded	429	Rate limit hit
errors.helixterminator.io/v1/internal-error	500	Server error (sanitized — no stack trace)
errors.helixterminator.io/v1/upstream-error	502	Dependent service unavailable
errors.helixterminator.io/v1/timeout	504	Request processing exceeded deadline

1.7 Authentication

All authenticated endpoints require a Bearer token in the Authorization header:

Authorization: Bearer eyJhbGciOiJIJZERTQSI6InR5cCI6IkpXVCJ9...

HelixTerminator uses **EdDSA (Ed25519) signed JWTs** (RFC 8037).
Token structure:

```
{  
  "iss": "https://auth.helixterminator.io",  
  "sub": "550e8400-e29b-41d4-a716-446655440000",  
}
```

```
"aud": ["helixterm:api"],
"exp": 1751120400,
"iat": 1751116800,
"jti": "unique-token-id",
"scope": "api:read api:write",
"org_id": "org-uuid",
"session_id": "session-uuid",
"mfa_verified": true
}
```

Token lifetimes:

Token Type	Lifetime	Storage
Access token	15 minutes	Memory (client)
Refresh token	30 days (sliding)	HttpOnly Secure cookie or secure storage
API key token	No expiry (until revoked)	Hashed in DB
Session token (WebSocket)	Duration of connection	Redis

API Keys use the format `htk_v1_<base62-secret>` and are authenticated via:

Authorization: Bearer

`htk_v1_abcdefghijklmnopqrstuvwxyz0123456789ABCDEFGHIJ`

or

X-API-Key: `htk_v1_abcdefghijklmnopqrstuvwxyz0123456789ABCDEFGHIJ`

1.8 Rate Limiting

Rate limits are enforced at multiple layers:

Layer	Limit	Window	Scope
Global (unauthenticated)	60 req	1 minute	Per IP
Global (authenticated)	1000 req	1 minute	Per user
Auth endpoints	10 req	1 minute	Per IP
Sensitive operations	5 req	15 minutes	Per user
AI endpoints	100 req	1 hour	Per user

Layer	Limit	Window	Scope
File upload	50 req	1 hour	Per user
WebSocket connections	20 concurrent	—	Per user

Rate limit response headers (always present on every response):

```
X-RateLimit-Limit: 1000
X-RateLimit-Remaining: 847
X-RateLimit-Reset: 1751117460
X-RateLimit-Policy: authenticated;q=1000;w=60
Retry-After: 47
```

On 429 Too Many Requests:

```
{
  "type": "https://errors.helixterminator.io/v1/rate-limit-
    exceeded",
  "title": "Rate Limit Exceeded",
  "status": 429,
  "detail":
    "You have exceeded the rate limit of 1000 requests per 60
    seconds.",
  "retry_after": 47,
  "limit": 1000,
  "window_seconds": 60
}
```

Rate limiting uses a **sliding window counter** stored in Redis with the key pattern `rl:{scope}:{identifier}:{window_start}`.

1.9 HATEOAS Links

Responses include a `_links` object following the HAL (Hypertext Application Language) specification (draft-kelly-json-hal):

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "prod-web-01",
  "hostname": "192.168.1.100",
  "_links": {
    "self": {
      "href": "https://api.helixterminator.io/api/v1/hosts/
        550e8400-e29b-41d4-a716-446655440000"
    },
    "connections": {
```

```

    "href": "https://api.helixterminator.io/api/v1/hosts/
    550e8400-e29b-41d4-a716-446655440000/connections"
  },
  "group": {
    "href": "https://api.helixterminator.io/api/v1/groups/
    88e7f30c-1234-5678-abcd-ef0123456789"
  },
  "vault": {
    "href": "https://api.helixterminator.io/api/v1/vaults/
    aabbccdd-0011-2233-4455-66778899aabb"
  }
}
}

```

Collection responses include pagination links:

```

{
  "_links": {
    "self": { "href": "https://api.helixterminator.io/api/v1/
    hosts?cursor=abc&limit=25" },
    "next": { "href": "https://api.helixterminator.io/api/v1/
    hosts?cursor=def&limit=25" },
    "prev": { "href": "https://api.helixterminator.io/api/v1/
    hosts?cursor=xyz&limit=25&direction=prev" }
  }
}

```

1.10 Request ID and Tracing

Every request is assigned a unique trace ID: - If the client sends X-Request-ID: <uuid>, that value is used. - Otherwise, the server generates a UUID v4.

The trace ID is echoed back in every response:

X-Request-ID: 7f3a9b2c-1d4e-5f6a-8b9c-0d1e2f3a4b5c

X-Trace-ID: 7f3a9b2c-1d4e-5f6a-8b9c-0d1e2f3a4b5c

All log entries and error responses include this ID for correlation.

1.11 Content Negotiation

All API requests and responses use application/json unless stated otherwise.

Endpoint Type	Request Content-Type	Response Content-Type
Standard REST	application/json	application/json
File upload	multipart/form-data	application/json
File download	(none)	application/octet-stream or specific MIME
Error responses	—	application/problem+json
Session recording	(none)	application/x-asciicast (asciinema v2)
SSH config export	(none)	text/plain; charset=utf-8

1.12 OpenAPI 3.1 Specification Overview

The machine-readable OpenAPI 3.1 specification is available at:

```
GET /api/v1/openapi.json  - OpenAPI 3.1 document
GET /api/v1/openapi.yaml  - YAML version
GET /api/v1/docs          - Swagger UI
GET /api/v1/redoc         - ReDoc UI
```

The OpenAPI document uses the following component structure:

```
openapi: "3.1.0"
info:
  title: HelixTerminator API
  version: "1.0.0"
  contact:
    name: HelixTerminator Engineering
    email: api@helixterminator.io
  license:
    name: Proprietary

servers:
  - url: https://api.helixterminator.io/api/v1
    description: Production
  - url: https://staging-api.helixterminator.io/api/v1
    description: Staging
  - url: http://localhost:8080/api/v1
    description: Local development
```

```

components:
  securitySchemes:
    BearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT
    ApiKeyHeader:
      type: apiKey
      in: header
      name: X-API-Key

```

```

security:
  - BearerAuth: []

```

DEFERRED (next increment): Only the top-level OpenAPI document structure and security schemes are shown above. The full `components.schemas` request/response JSON Schema definitions for every endpoint in §2–§14 are not yet authored in this document; treat the per-endpoint JSON examples in this spec as illustrative, not as a substitute for a generated/validated schema.

1.13 Idempotency Keys

Mutating operations (POST, PATCH) that create resources or trigger actions support idempotency keys to prevent duplicate processing on retry:

Idempotency-Key: <client-generated-uuid>

The server caches the response for 24 hours keyed by `{user_id}`: `{idempotency_key}`. If the same key is replayed, the original response is returned without re-executing the operation. The response includes:

Idempotency-Key: 550e8400-e29b-41d4-a716-446655440000

Idempotency-Replayed: true

DEFERRED (next increment): “Support” above is described generically for all mutating operations; a definitive per-endpoint matrix stating which POST/PATCH operations *require* an Idempotency-Key (vs merely accept one) is not yet authored across §2–§14. Do not assume every mutating endpoint enforces it until that matrix exists.

1.14 Conditional Requests and ETags

GET responses include ETag and Last-Modified headers:

ETag: "33a64df551425fcc55e4d42a148795d9f25f89d4"

Last-Modified: Wed, 28 Jun 2026 10:00:00 GMT

Cache-Control: private, no-cache

Clients use conditional requests to avoid re-downloading unchanged data:

GET /api/v1/hosts/550e8400 HTTP/1.1

If-None-Match: "33a64df551425fcc55e4d42a148795d9f25f89d4"

Returns 304 Not Modified if unchanged.

For optimistic concurrency on updates:

PUT /api/v1/hosts/550e8400 HTTP/1.1

If-Match: "33a64df551425fcc55e4d42a148795d9f25f89d4"

Returns 412 Precondition Failed if the resource was modified since the ETag was retrieved.

1.15 CORS

Cross-Origin Resource Sharing is configured as follows:

Access-Control-Allow-Origin: https://app.helixterminator.io

Access-Control-Allow-Methods: GET, POST, PUT, PATCH, DELETE, OPTIONS

Access-Control-Allow-Headers: Authorization, Content-Type, X-Request-ID, Idempotency-Key, X-API-Key

Access-Control-Expose-Headers: X-Request-ID, X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset, ETag

Access-Control-Max-Age: 86400

Access-Control-Allow-Credentials: true

In development, Access-Control-Allow-Origin: * is permitted (credentials excluded).

2. Auth

Service

REST API

Base path: /

api/v1/auth

Service:
auth-service
(port 8081
internal, 443
external via
gateway)
Database:
auth_db
(PostgreSQL)

POST /api/v1/auth/register

Create a new user account.

Authentication: None required

Rate Limit: 5 requests per IP per 15 minutes

Request Headers:

Content-Type: application/json
X-Request-ID: <uuid> (optional)

Request Body:

```
{  
  "email": "alice@example.com",  
  "password": "SecureP@ssw0rd!",  
  "display_name": "Alice Smith",  
  "invite_code": "HELIX-INVITE-ABCD1234",  
  "accept_terms": true,  
  "locale": "en-US",  
  "timezone": "America/New_York"  
}
```

Field	Type	Required	Constraints
email	string	Yes	Valid email, max 255 chars
password	string	Yes	Min 12 chars, complexity requirements
display_name	string	Yes	2–100 chars, UTF-8
invite_code	string	No	

Field	Type	Required	Constraints
			Required if registration is invite-only
accept_terms	boolean	Yes	Must be true
locale	string	No	BCP 47 locale tag
timezone	string	No	IANA timezone name

Success Response — 201 Created:

```
{
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "alice@example.com",
    "display_name": "Alice Smith",
    "status": "active",
    "email_verified": false,
    "created_at": "2026-06-28T17:40:00Z"
  },
  "tokens": {
    "access_token": "eyJhbGciOiJIJZERTQSIInR5cCI6IkpXVCJ9...",
    "token_type": "Bearer",
    "expires_in": 900,
    "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
    "scope": "api:read api:write"
  },
  "requires_email_verification": true,
  "_links": {
    "self": { "href": "https://api.helixterminator.io/api/v1/users/me" },
    "verify_email": { "href": "https://api.helixterminator.io/api/v1/auth/email/verify" }
  }
}
```

Error Responses:

Status	Error Type	Condition
400	validation-error	Missing/malformed fields
409	conflict	Email already registered
422	validation-error	Password too weak, invalid email

Status	Error Type	Condition
429	rate-limit-exceeded	Too many registration attempts from IP

POST /api/v1/auth/login

Authenticate with email and password. Returns tokens or MFA challenge.

Authentication: None required

Rate Limit: 10 requests per IP per minute; 5 per account per minute

Request Headers:

Content-Type: application/json

User-Agent: HelixTerminator/1.0 (macOS 14.2)

X-Device-ID: <client-device-uuid> (optional, for trusted device tracking)

Request Body:

```
{
  "email": "alice@example.com",
  "password": "SecureP@ssw0rd!",
  "device_name": "Alice's MacBook Pro",
  "device_fingerprint": "sha256:abcdef1234567890...",
  "remember_me": true
}
```

Success Response (no MFA) — 200 OK:

```
{
  "status": "authenticated",
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "alice@example.com",
    "display_name": "Alice Smith",
    "avatar_url": "https://cdn.helixterminator.io/avatars/alice.jpg",
    "status": "active",
    "email_verified": true,
    "last_login_at": "2026-06-27T09:00:00Z",
    "mfa_enabled": false,
    "org_id": "org-550e8400-0000-0000-0000-000000000001"
  },
}
```

```

"tokens": {
  "access_token": "eyJhbGciOiJIJZERTQSI6InR5cCI6IkpXVCJ9... ",
  "token_type": "Bearer",
  "expires_in": 900,
  "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
  "scope": "api:read api:write"
},
"session": {
  "id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
  "device_id": "dev-550e8400-0000-0000-0000-112233445566",
  "created_at": "2026-06-28T17:40:00Z",
  "expires_at": "2026-07-28T17:40:00Z"
}
}

```

MFA Required Response — 200 OK:

```

{
  "status": "mfa_required",
  "mfa_challenge": {
    "challenge_id": "mfa-chal-550e8400-0000-0000-0000-aabbccdd0001",
    "methods": ["totp", "fido2"],
    "expires_at": "2026-06-28T17:45:00Z"
  }
}

```

Error Responses:

Status	Error Type	Condition
400	validation-error	Malformed request
401	authentication-required	Invalid credentials
403	forbidden	Account suspended or deleted
423	locked	Account temporarily locked after failed attempts
429	rate-limit-exceeded	Too many login attempts

POST /api/v1/auth/logout

Invalidate the current session and access token.

Authentication: Bearer token required

Request Body (optional):

```
{
  "all_sessions": false
}
```

Setting `all_sessions: true` invalidates all active sessions for the user.

Success Response — 204 No Content

Error Responses:

Status	Condition
401	Not authenticated

POST /api/v1/auth/refresh

Exchange a refresh token for a new access token.

Authentication: None required (refresh token in body or cookie)

Request Body:

```
{
  "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX"
}
```

If using cookie-based refresh tokens, the token is read from the `__Host-refresh_token` HttpOnly Secure cookie and the body can be empty.

Success Response — 200 OK:

```
{
  "access_token": "eyJhbGciOiJIJFZERTQSIzInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "expires_in": 900,
  "refresh_token": "rt_v1_yyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyyy",
  "scope": "api:read api:write"
}
```

The old refresh token is immediately invalidated (token rotation). The new refresh token extends the sliding window by 30 days.

Error Responses:

Status	Condition
401	Refresh token invalid, expired, or already used
403	Session revoked by admin

POST /api/v1/auth/mfa/totp/setup

Initiate TOTP (Time-based One-Time Password) setup. Returns the secret and provisioning URI.

Authentication: Bearer token required (access token, pre-MFA-verification)

Request Body: Empty {}

Success Response — 200 OK:

```
{
  "secret": "JBSWY3DPEHPK3PXP",
  "provisioning_uri": "otpauth://totp/HelixTerminator:alice%40example.com?secret=JBSWY3DPEHPK3PXP&issuer=HelixTerminator&algorithm=SHA1&digits=6&period=30",
  "qr_code_url": "https://api.helixterminator.io/api/v1/auth/mfa/totp/qr?token=setup-xyz",
  "backup_codes": [
    "AAAA-BBBB-CCCC",
    "DDDD-EEEE-FFFF",
    "GGGG-HHHH-IIII",
    "JJJJ-KKKK-LLLL",
    "MMMM-NNNN-OOOO",
    "PPPP-QQQQ-RRRR",
    "SSSS-TTTT-UUUU",
    "VVVV-WWWW-XXXX"
  ],
  "setup_token": "setup-token-expires-in-10-minutes",
  "expires_at": "2026-06-28T17:50:00Z"
}
```

The secret is not persisted until POST /api/v1/auth/mfa/totp/verify succeeds.

Error Responses:

Status	Condition
401	Not authenticated
409	TOTP already enabled for this account

POST /api/v1/auth/mfa/totp/verify

Verify a TOTP code to complete setup or authenticate.

Authentication: Bearer token required OR mfa_challenge_id in body

Request Body:

```
{
  "code": "123456",
  "setup_token": "setup-token-expires-in-10-minutes",
  "challenge_id": "mfa-chal-550e8400-0000-0000-0000-aabbccdd0001"
}
```

Provide setup_token when completing setup, challenge_id when authenticating.

Success Response (setup completion) — 200 OK:

```
{
  "enabled": true,
  "backup_codes_remaining": 8,
  "message": "TOTP successfully enabled."
}
```

Success Response (authentication) — 200 OK:

```
{
  "status": "authenticated",
  "tokens": {
    "access_token": "eyJhbGciOiJIJZERTQSI6InR5cCI6IkpXVCJ9...",
    "token_type": "Bearer",
    "expires_in": 900,
    "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
    "scope": "api:read api:write"
  }
}
```

Error Responses:

Status	Condition
400	Invalid code format
401	TOTP code incorrect or expired
410	Challenge expired
429	Too many TOTP attempts

POST /api/v1/auth/mfa/fido2/register/begin

Begin WebAuthn/FIDO2 credential registration. Returns a challenge for the authenticator.

Authentication: Bearer token required

Request Body:

```
{
  "authenticator_name": "YubiKey 5C",
  "authenticator_attachment": "cross-platform"
}
```

authenticator_attachment: "platform" (biometrics, TPM), "cross-platform" (security key), null (any).

Success Response — 200 OK:

```
{
  "registration_id": "reg-550e8400-0000-0000-0000-aabbccdd0002",
  "options": {
    "challenge": "dGhpcyBpcyBhIHRlc3QgY2hhbGxlbmdl",
    "rp": {
      "id": "helixterminator.io",
      "name": "HelixTerminator"
    },
    "user": {
      "id": "VVVlMDg0MDAtZTI5Yi00MWQ0LWE3MTYtNDQ2NjU1NDQwMDAw",
      "name": "alice@example.com",
      "displayName": "Alice Smith"
    },
    "pubKeyCredParams": [
      { "type": "public-key", "alg": -8 },
      { "type": "public-key", "alg": -7 },
      { "type": "public-key", "alg": -257 }
    ]
  }
}
```

```

    "timeout": 60000,
    "authenticatorSelection": {
      "authenticatorAttachment": "cross-platform",
      "requireResidentKey": false,
      "userVerification": "preferred"
    },
    "attestation": "indirect"
  },
  "expires_at": "2026-06-28T17:41:00Z"
}

```

Error Responses:

Status	Condition
401	Not authenticated

POST /api/v1/auth/mfa/fido2/register/complete

Complete WebAuthn/FIDO2 credential registration with authenticator response.

Authentication: Bearer token required

Request Body:

```

{
  "registration_id": "reg-550e8400-0000-0000-0000-aabbccdd0002",
  "authenticator_name": "YubiKey 5C",
  "credential": {
    "id": "credentialIdBase64url",
    "rawId": "credentialIdBase64url",
    "type": "public-key",
    "response": {
      "clientDataJSON":
        "eyJ0eXB1Ijoid2ViYXV0aG4uY3JlYXRlIiwiaY2hhbGxlbmdlIjoizEdocGN5Qn...",
      "attestationObject":
        "o2NmbXRmcGFja2VkZ2F0dFN0bXSiY2FsZyZjc2lnWEYwRAIgM..."
    }
  }
}

```

Success Response — 201 Created:

```

{
  "credential_id": "cred-550e8400-0000-0000-0000-aabbccdd0003",

```

```
"authenticator_name": "YubiKey 5C",
"credential_type": "public-key",
"created_at": "2026-06-28T17:40:30Z",
"aaguid": "2fc0579f-8113-47ea-b116-bb5a8db9202a",
"transports": ["usb", "nfc"]
}
```

POST /api/v1/auth/mfa/fido2/authenticate/begin

Begin FIDO2 authentication challenge.

Authentication: None (called after password verification, before full auth)

Request Body:

```
{
  "challenge_id": "mfa-chal-550e8400-0000-0000-0000-aabbccdd0001",
  "user_verification": "preferred"
}
```

Success Response — 200 OK:

```
{
  "fido2_challenge_id": "fido2-chal-550e8400-0000-0000-0000-aabbccdd0004",
  "options": {
    "challenge": "bGV0J3MgdGVzdCBGSURPMiBhdXRoZW50aWNhdGlvbg==",
    "timeout": 60000,
    "rpId": "helixterminator.io",
    "allowCredentials": [
      {
        "type": "public-key",
        "id": "credentialIdBase64url",
        "transports": ["usb", "nfc"]
      }
    ],
    "userVerification": "preferred"
  },
  "expires_at": "2026-06-28T17:41:00Z"
}
```

POST /api/v1/auth/mfa/fido2/authenticate/complete

Complete FIDO2 authentication.

Authentication: None (completes the MFA flow)

Request Body:

```
{
  "fido2_challenge_id": "fido2-chal-550e8400-0000-0000-0000-
    aabbccdd0004",
  "credential": {
    "id": "credentialIdBase64url",
    "rawId": "credentialIdBase64url",
    "type": "public-key",
    "response": {
      "clientDataJSON":
        "eyJ0eXB1Ijoid2ViYXV0aG4uZ2V0Iiwia2hhbGxlbmdlIjoieYkdWMEp5...",
      "authenticatorData": "SZYN5YgOjGh0NBcPZHgW4/
        krrmihjLHmVzzuoMdl2MBAAAABg==",
      "signature": "MEQCIBBgCLz3rRBOJb/
        mBLKIVHmRUKD9dR3JCZ0u08hPh8M5AiB4...",
      "userHandle": "VVVlMDg0MDAtZTI5Yi00MWQ0..."
    }
  }
}
```

Success Response — 200 OK:

```
{
  "status": "authenticated",
  "tokens": {
    "access_token": "eyJhbGciOiJIJZERTQSIiInR5cCI6IkpXVCJ9...",
    "token_type": "Bearer",
    "expires_in": 900,
    "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
    "scope": "api:read api:write"
  }
}
```

GET /api/v1/auth/devices

List all trusted devices for the authenticated user.

Authentication: Bearer token required

Query Parameters: | Parameter | Type | Description | |—|—|—| |
cursor | string | Pagination cursor | | limit | integer | Page size
(default 25) |

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "dev-550e8400-0000-0000-0000-112233445566",
      "name": "Alice's MacBook Pro",
      "fingerprint": "sha256:abcdef1234567890...",
      "platform": "macOS 14.2",
      "user_agent": "HelixTerminator/1.0 (macOS 14.2)",
      "trusted": true,
      "last_seen_at": "2026-06-28T17:40:00Z",
      "last_seen_ip": "192.168.1.10",
      "created_at": "2026-01-15T09:00:00Z",
      "is_current": true
    }
  ],
  "pagination": {
    "cursor_next": null,
    "has_next": false,
    "limit": 25,
    "total_count": 1
  }
}
```

DELETE /api/v1/auth/devices/{deviceId}

Revoke a trusted device. Future logins from this device will require full authentication.

Authentication: Bearer token required

Path Parameters: | Parameter | Type | Description | |—|—|—| |
deviceId | UUID | Device identifier |

Success Response — 204 No Content

Error Responses:

Status	Condition
404	Device not found
403	Device belongs to another user

POST /api/v1/auth/sso/{provider}/authorize

Initiate SSO OAuth2/OIDC authorization flow.

Authentication: None required

Path Parameters: | Parameter | Type | Description | |—|—|—| |
provider | string | SSO provider slug (e.g., github, google, azure, okta) |

Request Body:

```
{
  "redirect_uri": "https://app.helixterminator.io/auth/callback",
  "state": "client-generated-random-state",
  "org_slug": "mycompany"
}
```

Success Response — 200 OK:

```
{
  "authorization_url": "https://github.com/login/oauth/authorize?
    client_id=...&redirect_uri=...&state=...&scope=read%3Auser%20user%3Aemail",
  "state": "server-generated-state-token",
  "expires_at": "2026-06-28T17:50:00Z"
}
```

POST /api/v1/auth/sso/{provider}/callback

Handle SSO callback with authorization code.

Authentication: None required

Request Body:

```
{
  "code": "oauth2-authorization-code",
  "state": "server-generated-state-token",
  "redirect_uri": "https://app.helixterminator.io/auth/callback"
}
```

Success Response — 200 OK:

```
{
  "status": "authenticated",
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "alice@example.com",
    "display_name": "Alice Smith",
    "sso_provider": "github",
    "sso_subject": "12345678"
  },
  "tokens": {
    "access_token": "eyJhbGciOiJIJZERTQSIInR5cCI6IkpXVCJ9...",
    "token_type": "Bearer",
    "expires_in": 900,
    "refresh_token": "rt_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX"
  },
  "is_new_user": false
}
```

POST /api/v1/auth/api-keys

Create a new API key.

Authentication: Bearer token required

Request Body:

```
{
  "name": "CI/CD Pipeline Key",
  "scopes": ["hosts:read", "sessions:write", "snippets:execute"],
  "expires_at": "2027-06-28T00:00:00Z",
  "allowed_ips": ["10.0.0.0/8", "192.168.1.0/24"],
  "description": "Used by GitHub Actions for automated
    deployments"
}
```

Field	Type	Required	Description
name	string	Yes	Human-readable name (max 100 chars)
scopes	array	Yes	Permission scopes
expires_at	datetime	No	Expiration; omit for non-expiring

Field	Type	Required	Description
allowed_ips	array	No	IP allowlist (CIDR notation)
description	string	No	Usage description

Success Response — 201 Created:

```
{
  "id": "key-550e8400-0000-0000-0000-aabbccddeeff",
  "name": "CI/CD Pipeline Key",
  "key": "htk_v1_AbCdEfGhIjKlMnOpQrStUvWxYz0123456789ABCDE",
  "prefix": "htk_v1_AbCd",
  "scopes": ["hosts:read", "sessions:write", "snippets:execute"],
  "expires_at": "2027-06-28T00:00:00Z",
  "allowed_ips": ["10.0.0.0/8", "192.168.1.0/24"],
  "created_at": "2026-06-28T17:40:00Z",
  "last_used_at": null
}
```

IMPORTANT: The full key value is only returned once at creation. It cannot be retrieved again. Store it securely immediately.

GET /api/v1/auth/api-keys

List all API keys for the authenticated user.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "key-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "CI/CD Pipeline Key",
      "prefix": "htk_v1_AbCd",
      "scopes": ["hosts:read", "sessions:write",
        "snippets:execute"],
      "expires_at": "2027-06-28T00:00:00Z",
      "allowed_ips": ["10.0.0.0/8", "192.168.1.0/24"],
      "created_at": "2026-06-28T17:40:00Z",
      "last_used_at": "2026-06-28T12:00:00Z",
      "last_used_ip": "10.0.0.5",
      "is_expired": false
    }
  ]
}
```



```
],
"pagination": {
  "total_count": 1,
  "has_next": false
}
}
```

DELETE /api/v1/auth/api-keys/{keyId}

Revoke an API key. All requests using this key will immediately return 401.

Authentication: Bearer token required

Path Parameters:

Parameter	Type	Description
keyId	UUID	API key identifier

Success Response — 204 No Content

GET /api/v1/auth/sessions

List all active sessions for the authenticated user.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
      "device": {
        "id": "dev-550e8400-0000-0000-0000-112233445566",
        "name": "Alice's MacBook Pro",
        "platform": "macOS 14.2"
      },
      "ip_address": "192.168.1.10",
      "user_agent": "HelixTerminator/1.0 (macOS 14.2)",
      "created_at": "2026-06-28T09:00:00Z",
      "last_active_at": "2026-06-28T17:40:00Z",
      "expires_at": "2026-07-28T09:00:00Z",
      "is_current": true,
      "mfa_verified": true,
      "location": {
```

```
    "city": "San Francisco",
    "country": "US",
    "approximate": true
  }
],
"pagination": {
  "total_count": 1,
  "has_next": false
}
```

DELETE /api/v1/auth/sessions/{sessionId}

Terminate a specific session. The associated refresh token is revoked.

Authentication: Bearer token required

Path Parameters:

Parameter	Type	Description
sessionId	UUID	Session identifier

Success Response — 204 No Content

3. User Service REST API

Base path: /api/v1/users
Service: user-service (port 8082 internal)
Database: user_db (PostgreSQL)

GET /api/v1/users/me

Get the authenticated user's profile.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "email": "alice@example.com",
  "email_verified": true,
  "display_name": "Alice Smith",
```

```
"avatar_url": "https://cdn.helixterminator.io/avatars/
550e8400.jpg",
"status": "active",
"bio": "Infrastructure engineer at ExampleCorp",
"locale": "en-US",
"timezone": "America/New_York",
"created_at": "2026-01-15T09:00:00Z",
"updated_at": "2026-06-28T10:00:00Z",
"mfa": {
  "totp_enabled": true,
  "fido2_count": 2,
  "backup_codes_remaining": 6
},
"organization": {
  "id": "org-550e8400-0000-0000-0000-000000000001",
  "name": "ExampleCorp",
  "slug": "examplecorp",
  "role": "org_admin"
},
"_links": {
  "self": { "href": "https://api.helixterminator.io/api/v1/
users/me" },
  "preferences": { "href": "https://api.helixterminator.io/api/
v1/users/me/preferences" },
  "data_export": { "href": "https://api.helixterminator.io/api/
v1/users/me/data-export" }
}
}
```

PUT /api/v1/users/me

Update the authenticated user's profile.

Authentication: Bearer token required

Request Body:

```
{
  "display_name": "Alice J. Smith",
  "bio": "Senior Infrastructure Engineer at ExampleCorp",
  "locale": "en-GB",
  "timezone": "Europe/London"
}
```

All fields are optional; unincluded fields are not modified.

Success Response — 200 OK: Returns the updated user object (same schema as GET /api/v1/users/me).

PUT /api/v1/users/me/password

Change the authenticated user's password.

Authentication: Bearer token required

Rate Limit: 5 requests per user per 15 minutes

Request Body:

```
{
  "current_password": "OldSecureP@ssw0rd!",
  "new_password": "NewSecureP@ssw0rd!2026",
  "revoke_other_sessions": true
}
```

Success Response — 200 OK:

```
{
  "message": "Password changed successfully.",
  "sessions_revoked": 3
}
```

Error Responses:

Status	Condition
401	Current password incorrect
422	New password does not meet complexity requirements
422	New password matches a previously used password

PUT /api/v1/users/me/email

Initiate an email address change. Sends verification to both old and new addresses.

Authentication: Bearer token required

Request Body:

```
{
  "new_email": "alice.smith@newdomain.com",
}
```

```
"password": "SecureP@ssw0rd!"
}
```

Success Response — 200 OK:

```
{
  "message": "Verification emails sent to both addresses. Change
             will take effect once both are confirmed.",
  "pending_email": "alice.smith@newdomain.com",
  "expires_at": "2026-06-29T17:40:00Z"
}
```

GET /api/v1/users/me/preferences

Get the authenticated user's preferences.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "theme": "dark",
  "font_size": 14,
  "font_family": "JetBrains Mono",
  "terminal_color_scheme": "dracula",
  "cursor_style": "block",
  "cursor_blink": true,
  "scrollback_lines": 10000,
  "bell_sound": false,
  "notifications": {
    "email_on_new_login": true,
    "email_on_new_device": true,
    "push_session_started": false,
    "push_session_ended": false
  },
  "keyboard_shortcuts": {
    "new_tab": "Ctrl+T",
    "close_tab": "Ctrl+W",
    "split_horizontal": "Ctrl+Shift+H",
    "split_vertical": "Ctrl+Shift+V"
  },
  "default_vault_id": "vault-550e8400-0000-0000-0000-
                      aabbccddeeff",
  "startup_workspace_id": null,
  "sidebar_collapsed": false,
}
```

```
"updated_at": "2026-06-28T10:00:00Z"
}
```

PUT /api/v1/users/me/preferences

Update user preferences (partial update supported).

Authentication: Bearer token required

Request Body:

```
{
  "theme": "light",
  "font_size": 16,
  "notifications": {
    "email_on_new_login": false
  }
}
```

Success Response — 200 OK: Returns the full updated preferences object.

POST /api/v1/users/me/avatar

Upload a new profile avatar.

Authentication: Bearer token required

Content-Type: multipart/form-data

Limits: Max 5 MB; JPEG, PNG, GIF, WebP

Request Body (multipart):

Content-Disposition: form-data; name="avatar";
filename="photo.jpg"
Content-Type: image/jpeg

<binary data>

Success Response — 200 OK:

```
{
  "avatar_url": "https://cdn.helixterminator.io/avatars/550e8400-
    e29b-41d4-a716-446655440000.jpg",
}
```

```
"updated_at": "2026-06-28T17:45:00Z"
}
```

DELETE /api/v1/users/me/avatar

Remove the profile avatar and revert to default.

Authentication: Bearer token required

Success Response — 204 No Content

DELETE /api/v1/users/me

Request account deletion (GDPR Article 17 — Right to Erasure).

Authentication: Bearer token required

Rate Limit: 1 request per user per 24 hours

Request Body:

```
{
  "password": "SecureP@ssw0rd!",
  "reason": "No longer using the service",
  "confirm_deletion": true
}
```

confirm_deletion must be true.

Success Response — 202 Accepted:

```
{
  "message": "Account deletion scheduled. Your data will be
    permanently deleted within 30 days.",
  "deletion_scheduled_at": "2026-06-28T17:40:00Z",
  "permanent_deletion_at": "2026-07-28T17:40:00Z",
  "cancellation_url": "https://app.helixterminator.io/account/
    cancel-deletion?token=xxxx"
}
```

The account enters pending_deletion state. A cancellation link is valid for 14 days.

GET /api/v1/users/me/data-export

Request a GDPR data portability export (Article 20). Triggers an async job.

Authentication: Bearer token required

Rate Limit: 1 request per user per 7 days

Query Parameters: | Parameter | Type | Description | |—|—|—| |
format | string | json (default) or csv |

Success Response — 202 Accepted:

```
{
  "export_id": "exp-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "pending",
  "format": "json",
  "requested_at": "2026-06-28T17:40:00Z",
  "estimated_completion_at": "2026-06-28T18:10:00Z",
  "download_expires_at": null
}
```

When ready, the user receives an email with a signed download URL. The download link is valid for 48 hours.

GET /api/v1/users/me/data-export/{exportId}/status checks job status:

```
{
  "export_id": "exp-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "completed",
  "download_url": "https://cdn.helixterminator.io/exports/exp-xxx.json.zip?token=signed-url",
  "download_expires_at": "2026-06-30T18:10:00Z",
  "file_size_bytes": 2457600
}
```

4. Vault

Service REST

API

Base path: /

api/v1/vaults

Service: vault-

service (port

8083 internal)

Database:

vault_db
(PostgreSQL)
A **Vault** is an
end-to-end
encrypted
container for
secrets,
credentials,
host
configurations,
and SSH keys.
All vault item
data is
encrypted
client-side
before
transmission
using AES-256-
GCM with keys
derived from
the user's
master
password via
Argon2id. The
server never
sees plaintext
vault content.

GET /api/v1/vaults

List all vaults accessible to the authenticated user (owned + shared).

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Default	Description
cursor	string	—	Pagination cursor
limit	integer	25	Page size
owned_only	boolean	false	Only show vaults the user owns
sort	string	name:asc	Sort field

Success Response — 200 OK:

```
{  
  "data": [  
    {
```

```

    "id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
    "name": "Production Credentials",
    "description": "All production server credentials",
    "color": "#FF6B35",
    "icon": "server",
    "owner_id": "550e8400-e29b-41d4-a716-446655440000",
    "member_count": 5,
    "item_count": 247,
    "my_permission": "admin",
    "encrypted": true,
    "sync_enabled": true,
    "last_synced_at": "2026-06-28T17:00:00Z",
    "created_at": "2026-01-15T09:00:00Z",
    "updated_at": "2026-06-28T17:00:00Z",
    "_links": {
      "self": { "href": "https://api.helixterminator.io/api/v1/vaults/vault-550e8400-0000-0000-0000-aabbccddeeff" },
      "members": { "href": "https://api.helixterminator.io/api/v1/vaults/vault-550e8400-0000-0000-0000-aabbccddeeff/members" }
    }
  },
  ],
  "pagination": {
    "cursor_next": null,
    "has_next": false,
    "limit": 25,
    "total_count": 1
  }
}

```

POST /api/v1/vaults

Create a new vault.

Authentication: Bearer token required

Request Body:

```

{
  "name": "Staging Credentials",
  "description": "Staging environment hosts and keys",
  "color": "#4ECDC4",
  "icon": "database",
  "sync_enabled": true,

```

```
"encrypted_key_blob": "base64-encoded-encrypted-vault-key",
"kdf_params": {
  "algorithm": "argon2id",
  "memory_kib": 65536,
  "iterations": 3,
  "parallelism": 4,
  "salt": "base64-encoded-salt"
}
}
```

Success Response — 201 Created: Full vault object.

GET /api/v1/vaults/{vaultId}

Get a specific vault.

Authentication: Bearer token required

Path Parameters:

Parameter	Type	Description
vaultId	UUID	Vault identifier

Success Response — 200 OK: Full vault object.

Error Responses:

Status	Condition
403	User is not a member of this vault
404	Vault not found

PUT /api/v1/vaults/{vaultId}

Update vault metadata (name, description, color, icon, sync settings).

Authentication: Bearer token required; requires admin vault permission

Request Body:

```
{
  "name": "Staging Credentials v2",
  "color": "#FFE66D",
  "sync_enabled": false
}
```

Success Response — 200 OK: Updated vault object.

DELETE /api/v1/vaults/{vaultId}

Delete a vault permanently. All items are erased. Requires vault ownership.

Authentication: Bearer token required; must be vault owner

Request Body:

```
{
  "confirm_name": "Staging Credentials v2"
}
```

The confirm_name must match the vault name exactly.

Success Response — 204 No Content

POST /api/v1/vaults/{vaultId}/sync

Trigger a vault sync operation. Returns the server-side sync state for delta sync.

Authentication: Bearer token required; requires vault member access

Request Body:

```
{
  "client_cursor": "sync-cursor-opaque-value",
  "changes": [
    {
      "item_id": "item-550e8400-0000-0000-0000-aabbccddeeff",
      "operation": "upsert",
      "encrypted_data": "base64-encoded-encrypted-payload",
      "version": 5,
      "checksum": "sha256:aabbccdd..."
    }
  ]
}
```

Success Response — 200 OK:

```
{
  "server_cursor": "new-sync-cursor-opaque",

```

```

"applied_changes": 3,
"conflicts": [
  {
    "item_id": "item-550e8400-0000-0000-0000-aabbccddeeff",
    "conflict_type": "version_mismatch",
    "server_version": 7,
    "server_encrypted_data": "base64-encoded...",
    "resolution": "server_wins"
  }
],
"pending_changes": [
  {
    "item_id": "item-aabbccdd-0000-0000-0000-112233445566",
    "operation": "upsert",
    "encrypted_data": "base64-encoded-encrypted-payload",
    "version": 2
  }
]
}

```

GET /api/v1/vaults/{vaultId}/members

List vault members.

Authentication: Bearer token required; requires vault member access

Success Response — 200 OK:

```

{
  "data": [
    {
      "user_id": "550e8400-e29b-41d4-a716-446655440000",
      "email": "alice@example.com",
      "display_name": "Alice Smith",
      "avatar_url": "https://cdn.helixterminator.io/avatars/
        alice.jpg",
      "permission": "admin",
      "invited_by": null,
      "joined_at": "2026-01-15T09:00:00Z",
      "is_owner": true
    },
    {
      "user_id": "660e8400-e29b-41d4-a716-556655440000",
      "email": "bob@example.com",

```

```
    "display_name": "Bob Jones",
    "avatar_url": null,
    "permission": "write",
    "invited_by": "550e8400-e29b-41d4-a716-446655440000",
    "joined_at": "2026-02-01T10:00:00Z",
    "is_owner": false
  }
],
"pagination": {
  "total_count": 2,
  "has_next": false
}
}
```

POST /api/v1/vaults/{vaultId}/members

Add a member to a vault.

Authentication: Bearer token required; requires vault admin permission

Request Body:

```
{
  "user_id": "770e8400-e29b-41d4-a716-666655440000",
  "permission": "read",
  "encrypted_vault_key": "base64-encoded-vault-key-encrypted-for-
    new-member"
}
```

The `encrypted_vault_key` is the vault's symmetric key re-encrypted with the new member's public key. The server stores it as a blob and never decrypts it.

Success Response — 201 Created: New member object.

DELETE /api/v1/vaults/{vaultId}/members/{userId}

Remove a member from a vault.

Authentication: Bearer token required; requires vault admin permission (or self-removal)

Success Response — 204 No Content

PUT /api/v1/vaults/{vaultId}/members/{userId}/permissions

Update a vault member's permission level.

Authentication: Bearer token required; requires vault admin permission

Request Body:

```
{  
  "permission": "write"  
}
```

Valid permissions: read, write, admin.

Success Response — 200 OK: Updated member object.

5. Host Service REST API

Base path: /api/v1/hosts, /api/v1/groups
Service: host-service (port 8084 internal)
Database: host_db (PostgreSQL)

GET /api/v1/hosts

List hosts with filtering, sorting, and pagination.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Default	Description
cursor	string	—	Pagination cursor
limit	integer	25	Page size (max 100)
sort	string	name:asc	Sort field:direction
q	string	—	Full-text search
vault_id	UUID	—	Filter by vault
group_id	UUID	—	Filter by group
status	string	—	active, inactive, unreachable
os	string	—	OS filter (e.g., linux, macos, windows)
tag	string	—	Filter by tag (repeatable)
jump_enabled	boolean	—	Filter by jump host capability
last_connected_gte	datetime	—	Last connection after date
last_connected_lte	datetime	—	Last connection before date

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "host-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "prod-web-01",
      "hostname": "10.0.1.10",
      "port": 22,
      "username": "deploy",
      "auth_method": "key",
      "key_id": "key-550e8400-0000-0000-0000-112233445566",
      "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
      "group_id": "group-550e8400-0000-0000-0000-223344556677",
      "tags": ["production", "web", "nginx"],
      "os": "linux",
      "os_version": "Ubuntu 24.04 LTS",
      "arch": "amd64",
      "description": "Primary web server",
      "jump_host_id": null,
      "jump_enabled": false,
      "color": "#FF6B35",
      "icon": "server",
      "status": "active",
      "last_connected_at": "2026-06-28T16:00:00Z",
      "fingerprint_verified": true,
      "created_at": "2026-01-15T09:00:00Z",
      "updated_at": "2026-06-28T10:00:00Z",
      "_links": {
        "self": { "href": "https://api.helixterminator.io/api/v1/hosts/host-550e8400-0000-0000-0000-aabbccddeeff" },
        "connections": { "href": "https://api.helixterminator.io/api/v1/hosts/host-550e8400-0000-0000-0000-aabbccddeeff/connections" }
      }
    }
  ],
  "pagination": {
    "cursor_next": "eyJpZCI6Imhvc3Qtc...",
    "has_next": true,
    "limit": 25,
    "total_count": 347
  }
}
```

POST /api/v1/hosts

Create a new host.

Authentication: Bearer token required

Request Body:

```
{
  "name": "prod-db-01",
  "hostname": "10.0.2.10",
  "port": 22,
  "username": "ubuntu",
  "auth_method": "key",
  "key_id": "key-550e8400-0000-0000-0000-112233445566",
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
  "group_id": "group-550e8400-0000-0000-0000-223344556677",
  "tags": ["production", "database", "postgresql"],
  "description": "Primary PostgreSQL server",
  "jump_host_id": "host-550e8400-0000-0000-0000-334455667788",
  "color": "#4ECDC4",
  "icon": "database",
  "proxy_jump_command": null,
  "environment_variables": {
    "PGPASSWORD": "${secrets.PGPASSWORD}"
  },
  "startup_snippet_id": null,
  "connection_timeout_seconds": 30,
  "keepalive_interval_seconds": 60
}
```

Success Response — 201 Created: Full host object.

GET /api/v1/hosts/{hostId}

Get a specific host.

Authentication: Bearer token required

Path Parameters:

Parameter	Type	Description
hostId	UUID	Host identifier

Success Response — 200 OK: Full host object with additional computed fields:

```
{
  "id": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "name": "prod-web-01",
  "hostname": "10.0.1.10",
  "port": 22,
  "username": "deploy",
  "auth_method": "key",
  "key_id": "key-550e8400-0000-0000-0000-112233445566",
  "known_fingerprints": [
    {
      "algorithm": "SHA256",
      "fingerprint": "SHA256:abc123def456...",
      "added_at": "2026-01-15T09:05:00Z",
      "verified_by": "alice@example.com"
    }
  ],
  "jump_chain": [],
  "connection_stats": {
    "total_connections": 847,
    "total_duration_seconds": 284700,
    "last_connected_at": "2026-06-28T16:00:00Z",
    "average_session_seconds": 336
  }
}
```

PUT /api/v1/hosts/{hostId}

Replace a host configuration entirely.

Authentication: Bearer token required

Request Body: Same schema as POST.

Success Response — 200 OK: Updated host object.

DELETE /api/v1/hosts/{hostId}

Delete a host and all its connection history.

Authentication: Bearer token required

Success Response — 204 No Content

POST /api/v1/hosts/bulk

Create multiple hosts at once (batch operation).

Authentication: Bearer token required

Request Body:

```
{
  "hosts": [
    {
      "name": "prod-app-01",
      "hostname": "10.0.3.10",
      "port": 22,
      "username": "ubuntu",
      "auth_method": "key",
      "key_id": "key-550e8400-0000-0000-0000-112233445566",
      "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff"
    },
    {
      "name": "prod-app-02",
      "hostname": "10.0.3.11",
      "port": 22,
      "username": "ubuntu",
      "auth_method": "key",
      "key_id": "key-550e8400-0000-0000-0000-112233445566",
      "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff"
    }
  ],
  "group_id": "group-550e8400-0000-0000-0000-223344556677",
  "on_conflict": "skip"
}
```

on_conflict: skip (skip duplicates), update (update existing), error (fail on duplicate).

Success Response — 207 Multi-Status:

```
{
  "results": [
    { "index": 0, "status": 201, "id": "host-111", "name": "prod-app-01" },
    { "index": 1, "status": 409, "error": "hostname '10.0.3.11' already exists in vault" }
  ],
  "created": 1,
}
```

```
"skipped": 0,  
"errors": 1  
}
```

DELETE /api/v1/hosts/bulk

Delete multiple hosts by IDs.

Authentication: Bearer token required

Request Body:

```
{  
  "host_ids": [  
    "host-550e8400-0000-0000-0000-aabbccddeeff",  
    "host-660e8400-0000-0000-0000-bbccddeeaabb"  
  ]  
}
```

Success Response — 200 OK:

```
{  
  "deleted": 2,  
  "not_found": 0,  
  "forbidden": 0  
}
```

GET /api/v1/hosts/{hostId}/connections

Get connection history for a specific host.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
cursor	string	Pagination cursor
limit	integer	Page size
user_id	UUID	Filter by user
started_at_gte	datetime	Filter by start time

Success Response — 200 OK:

```
{  
  "data": [  
    {  
      "id": "conn-550e8400-0000-0000-0000-aabbccddeeff",  

```

```

    "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
    "user_id": "550e8400-e29b-41d4-a716-446655440000",
    "user_email": "alice@example.com",
    "session_id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
    "client_ip": "192.168.1.10",
    "started_at": "2026-06-28T16:00:00Z",
    "ended_at": "2026-06-28T16:15:30Z",
    "duration_seconds": 930,
    "bytes_sent": 48392,
    "bytes_received": 1204847,
    "exit_code": 0,
    "recording_available": true
  }
],
"pagination": {
  "cursor_next": null,
  "has_next": false,
  "total_count": 847
}
}

```

POST /api/v1/hosts/import

Import hosts from CSV, SSH config file, or Ansible inventory.

Authentication: Bearer token required

Content-Type: multipart/form-data

Request Body (multipart):

Content-Disposition: form-data; name="file"; filename="hosts.csv"

Content-Type: text/csv

```

name,hostname,port,username,auth_method
prod-web-01,10.0.1.10,22,ubuntu,key
prod-web-02,10.0.1.11,22,ubuntu,key

```

Query Parameters:

Parameter	Type	Description
format	string	csv, ssh_config, ansible
vault_id	UUID	Target vault
group_id	UUID	Target group
dry_run	boolean	Validate without importing

Success Response — 202 Accepted:

```
{
  "import_id": "imp-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "processing",
  "total_rows": 150,
  "dry_run": false
}
```

GET /api/v1/hosts/export

Export hosts to SSH config format.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
vault_id	UUID	Export from specific vault
group_id	UUID	Export from specific group
format	string	ssh_config (default), csv, json

Success Response — 200 OK:

Content-Type: text/plain; charset=utf-8

Content-Disposition: attachment; filename="helixterm-hosts.conf"

```
Host prod-web-01
  HostName 10.0.1.10
  Port 22
  User ubuntu
  IdentityFile ~/.ssh/helixterm_key

Host prod-db-01
  HostName 10.0.2.10
  Port 22
  User ubuntu
  ProxyJump prod-web-01
  IdentityFile ~/.ssh/helixterm_key
```

GET /api/v1/groups

List host groups.

Authentication: Bearer token required

Query Parameters: | Parameter | Type | Description | |—|—|—| |
vault_id | UUID | Filter by vault | | parent_id | UUID | Filter by
parent group | | q | string | Search query |

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "group-550e8400-0000-0000-0000-223344556677",
      "name": "Production",
      "description": "All production hosts",
      "parent_id": null,
      "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
      "color": "#FF6B35",
      "icon": "folder",
      "host_count": 47,
      "child_group_count": 3,
      "inherit_settings": true,
      "default_key_id":
        "key-550e8400-0000-0000-0000-112233445566",
      "default_username": "ubuntu",
      "created_at": "2026-01-15T09:00:00Z"
    }
  ],
  "pagination": {
    "total_count": 12,
    "has_next": false
  }
}
```

POST /api/v1/groups

Create a host group.

Authentication: Bearer token required

Request Body:

```
{
  "name": "Staging",
  "description": "Staging environment hosts",
  "parent_id": null,
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
  "color": "#4ECDC4",
```

```
"icon": "folder-open",
"inherit_settings": true,
"default_key_id": "key-550e8400-0000-0000-0000-112233445566",
"default_username": "ubuntu",
"default_port": 22
}
```

Success Response — 201 Created: Full group object.

GET /api/v1/groups/{groupId}

Get a specific group with its full configuration.

Authentication: Bearer token required

Success Response — 200 OK: Full group object.

PUT /api/v1/groups/{groupId}

Update a group.

Authentication: Bearer token required

Request Body: Same as POST, all fields optional.

Success Response — 200 OK: Updated group object.

DELETE /api/v1/groups/{groupId}

Delete a group. Hosts within the group are not deleted but become ungrouped.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
migrate_hosts_to	UUID	Move hosts to this group instead of ungrouping

Success Response — 204 No Content

POST /api/v1/groups/{groupId}/hosts

Add a host to a group.

Authentication: Bearer token required

Request Body:

```
{  
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff"  
}
```

Success Response — 204 No Content

DELETE /api/v1/groups/{groupId}/hosts/{hostId}

Remove a host from a group.

Authentication: Bearer token required

Success Response — 204 No Content

PUT /api/v1/groups/{groupId}/inherit

Configure inheritance settings for a group. Child groups and hosts inherit credentials, jump chains, and connection settings.

Authentication: Bearer token required

Request Body:

```
{  
  "inherit_from_parent": true,  
  "inherit_key": true,  
  "inherit_username": true,  
  "inherit_port": false,  
  "inherit_jump_host": true,  
  "inherit_environment_variables": true,  
  "inherit_startup_snippet": false  
}
```

Success Response — 200 OK: Updated inheritance settings.

6. SSH Proxy & Terminal Service REST API

Base path: /api/v1/sessions

Service: session-service (port 8085 internal), ssh-proxy-service (port 8086)

Database: session_db (PostgreSQL)

POST /api/v1/sessions/ssh

Initiate an SSH session. Returns a session token and WebSocket URL for terminal I/O.

Authentication: Bearer token required

Request Body:

```
{
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "terminal": {
    "cols": 220,
    "rows": 50,
    "term": "xterm-256color"
  },
  "recording_enabled": true,
  "collab_enabled": false,
  "read_only": false,
  "startup_snippet_id": null,
  "reason": "Investigating high CPU alert",
  "ticket_ref": "INC-20260628-001"
}
```

Field	Type	Required	Description
host_id	UUID	Yes	Target host
terminal.cols	integer	Yes	Terminal width in columns
terminal.rows	integer	Yes	Terminal height in rows
terminal.term	string	No	Terminal type (default xterm-256color)
recording_enabled	boolean	No	Enable session recording

Field	Type	Required	Description
			(default per org policy)
collab_enabled	boolean	No	Enable collaboration channel
read_only	boolean	No	Read-only mode (no input sent to host)
startup_snippet_id	UUID	No	Run snippet immediately on connect
reason	string	No	Reason for access (required by org policy)
ticket_ref	string	No	Incident/ticket reference

Success Response — 201 Created:

```
{
  "session_id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
  "session_token": "st_v1_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX",
  "websocket_url": "wss://proxy.helixterminator.io/api/v1/sessions/sess-550e8400/terminal",
  "collab_url": "wss://proxy.helixterminator.io/api/v1/sessions/sess-550e8400/collab",
  "status": "connecting",
  "host": {
    "id": "host-550e8400-0000-0000-0000-aabbccddeeff",
    "name": "prod-web-01",
    "hostname": "10.0.1.10",
    "port": 22
  },
  "recording_enabled": true,
  "expires_at": "2026-06-29T17:40:00Z",
  "created_at": "2026-06-28T17:40:00Z"
}
```

Error Responses:

Status	Condition
403	No permission to connect to this host
404	Host not found
409	Max concurrent sessions reached
422	Host unreachable, connection refused

GET /api/v1/sessions

List sessions (active and historical) for the authenticated user.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
status	string	active, closed, error Filter by status
host_id	UUID	Filter by host
cursor	string	Pagination cursor
limit	integer	Page size
started_at_gte	datetime	Filter by start time

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
      "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
      "host_name": "prod-web-01",
      "user_id": "550e8400-e29b-41d4-a716-446655440000",
      "status": "active",
      "recording_enabled": true,
      "collab_enabled": false,
      "started_at": "2026-06-28T17:40:00Z",
      "ended_at": null,
      "duration_seconds": null,
      "reason": "Investigating high CPU alert",
      "ticket_ref": "INC-20260628-001"
    }
  ],
  "pagination": {
    "total_count": 42,
    "has_next": false
  }
}
```

GET /api/v1/sessions/{sessionId}

Get details of a specific session.

Authentication: Bearer token required

Success Response — 200 OK: Full session object including connection metadata.

DELETE /api/v1/sessions/{sessionId}

Terminate an active SSH session.

Authentication: Bearer token required (session owner or org admin)

Success Response — 204 No Content

The SSH connection is terminated. If recording is enabled, the recording is finalized asynchronously.

POST /api/v1/sessions/{sessionId}/resize

Resize the terminal window for an active session.

Authentication: Bearer token required (must be session owner)

Request Body:

```
{  
  "cols": 240,  
  "rows": 60  
}
```

Success Response — 204 No Content

POST /api/v1/sessions/{sessionId}/broadcast

Broadcast input to multiple sessions simultaneously (requires org_admin permission).

Authentication: Bearer token required; requires org org_admin role

Request Body:

```
{
  "target_session_ids": [
    "sess-aabbccdd-0000-0000-0000-112233445566",
    "sess-bbccdde-0000-0000-0000-223344556677"
  ],
  "command": "sudo systemctl restart nginx\n",
  "require_confirmation": false
}
```

Success Response — 200 OK:

```
{
  "broadcast": 2,
  "failed": 0,
  "results": [
    { "session_id": "sess-aabbccdd", "status": "sent" },
    { "session_id": "sess-bbccdde", "status": "sent" }
  ]
}
```

GET /api/v1/sessions/{sessionId}/log

Download the full session log as plain text.

Authentication: Bearer token required (session owner or org admin)

Query Parameters: | Parameter | Type | Description | |—|—|—| |
format | string | text (default), ansi, html | | include_timestamps |
boolean | Prefix each line with timestamp |

Success Response — 200 OK:

Content-Type: text/plain; charset=utf-8

Content-Disposition: attachment; filename="session-
sess-550e8400-2026-06-28.log"

```
[2026-06-28 17:40:05] ubuntu@prod-web-01:~$ top
[2026-06-28 17:40:05] top - 17:40:05 up 42 days,  6:13,  1 user,
load average: 0.12, 0.08, 0.05
...
```

GET /api/v1/sessions/{sessionId}/recording

Download the session recording in asciinema v2 format.

Authentication: Bearer token required (session owner or org admin)

Query Parameters:

Parameter	Type	Description
format	string	asciicast (default), gif, mp4

Success Response — 200 OK:

Content-Type: application/x-asciicast

Content-Disposition: attachment; filename="session-sess-550e8400-2026-06-28.cast"

```
{"version": 2, "width": 220, "height": 50, "timestamp":  
1751125200, "title": "prod-web-01 session", "env": {"TERM":  
"xterm-256color"}}  
[0.0, "o", "\u001b[?1049h\u001b[22;0;0t"]  
[0.234, "o", "ubuntu@prod-web-01:~$ "  
...
```

7. SFTP Service REST API

Base path: /api/v1/sftp

Service: sftp-service (port 8087 internal)

Database: session_db (shared, sftp tables)

POST /api/v1/sftp/sessions

Open an SFTP session to a host.

Authentication: Bearer token required

Request Body:

```
{  
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",  
  "initial_path": "/var/www",  
  "transfer_mode": "binary"  
}
```

Success Response — 201 Created:

```
{
  "sftp_session_id": "sftp-550e8400-0000-0000-0000-aabbccddeeff",
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "host_name": "prod-web-01",
  "cwd": "/var/www",
  "status": "connected",
  "transfer_mode": "binary",
  "server_version": "SSH-2.0-OpenSSH_9.7",
  "created_at": "2026-06-28T17:40:00Z",
  "expires_at": "2026-06-28T23:40:00Z"
}
```

GET /api/v1/sftp/sessions/{sftpSessionId}/ls

List directory contents.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
path	string	Absolute path to list (default: /)
show_hidden	boolean	Include hidden files (default: false)

Success Response — 200 OK:

```
{
  "path": "/var/www",
  "entries": [
    {
      "name": "html",
      "path": "/var/www/html",
      "type": "directory",
      "size": 4096,
      "permissions": "drwxr-xr-x",
      "permissions_octal": "0755",
      "owner": "www-data",
      "group": "www-data",
      "modified_at": "2026-06-20T14:00:00Z",
      "is_symlink": false,
      "symlink_target": null
    },
    {
      "name": "index.html",
      "path": "/var/www/index.html",
      "type": "file",

```



```

    "size": 8192,
    "permissions": "-rw-r--r--",
    "permissions_octal": "0644",
    "owner": "www-data",
    "group": "www-data",
    "modified_at": "2026-06-28T10:00:00Z",
    "is_symlink": false,
    "symlink_target": null
  }
],
"total_entries": 2
}

```

POST /api/v1/sftp/sessions/{sftpSessionId}/upload

Upload a file to the remote host.

Authentication: Bearer token required

Content-Type: multipart/form-data

Max size: 10 GB per file (chunked)

Request Body (multipart):

Content-Disposition: form-data; name="file"; filename="app.tar.gz"

Content-Type: application/gzip

Query Parameters:

Parameter	Type	Description
path	string	Remote destination path
overwrite	boolean	Overwrite if exists (default: false)
chmod	string	Set permissions after upload (e.g., 0755)

Success Response — 201 Created:

```

{
  "transfer_id": "xfer-550e8400-0000-0000-0000-aabbccddeeff",
  "local_filename": "app.tar.gz",
  "remote_path": "/var/www/app.tar.gz",
  "bytes_transferred": 25600000,
  "checksum_sha256": "sha256:abcdef...",
  "duration_ms": 4800,
  "transferred_at": "2026-06-28T17:40:05Z"
}

```

GET /api/v1/sftp/sessions/{sftpSessionId}/download

Download a file from the remote host.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
path	string	Remote file path

Success Response — 200 OK:

Content-Type: application/octet-stream
Content-Disposition: attachment; filename="nginx.conf"
Content-Length: 4096
X-File-Permissions: 0644
X-File-Owner: root
X-File-Modified: Wed, 28 Jun 2026 10:00:00 GMT

<binary data>

POST /api/v1/sftp/sessions/{sftpSessionId}/mkdir

Create a remote directory.

Authentication: Bearer token required

Request Body:

```
{  
  "path": "/var/www/newdir",  
  "permissions": "0755",  
  "recursive": true  
}
```

Success Response — 201 Created:

```
{  
  "path": "/var/www/newdir",  
  "created_at": "2026-06-28T17:40:00Z"  
}
```

DELETE /api/v1/sftp/sessions/{sftpSessionId}/rm

Delete a remote file or directory.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
path	string	Remote path to delete
recursive	boolean	Delete directories recursively (default: false)

Success Response — 204 No Content

POST /api/v1/sftp/sessions/{sftpSessionId}/rename

Rename or move a remote file or directory.

Authentication: Bearer token required

Request Body:

```
{
  "source": "/var/www/old-name.html",
  "destination": "/var/www/new-name.html"
}
```

Success Response — 200 OK:

```
{
  "source": "/var/www/old-name.html",
  "destination": "/var/www/new-name.html",
  "renamed_at": "2026-06-28T17:40:00Z"
}
```

POST /api/v1/sftp/sessions/{sftpSessionId}/chmod

Change permissions on a remote file or directory.

Authentication: Bearer token required

Request Body:

```
{
  "path": "/var/www/script.sh",
  "permissions": "0755",
  "recursive": false
}
```

Success Response — 200 OK:

```
{
  "path": "/var/www/script.sh",
  "old_permissions": "0644",
  "new_permissions": "0755"
}
```

GET /api/v1/sftp/sessions/{sftpSessionId}/stat

Get file or directory metadata.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
path	string	Remote path
follow_symlinks	boolean	Resolve symlinks (default: true)

Success Response — 200 OK:

```
{
  "path": "/var/www/html/index.html",
  "type": "file",
  "size": 8192,
  "permissions": "-rw-r--r--",
  "permissions_octal": "0644",
  "owner": "www-data",
  "group": "www-data",
  "uid": 33,
  "gid": 33,
  "accessed_at": "2026-06-28T16:00:00Z",
  "modified_at": "2026-06-28T10:00:00Z",
  "changed_at": "2026-06-28T10:00:00Z",
  "is_symlink": false,
  "symlink_target": null
}
```

DELETE /api/v1/sftp/sessions/{sftpSessionId}

Close an SFTP session and release resources.

Authentication: Bearer token required

Success Response — 204 No Content

8. Port Forwarding Service REST API

Base path: /api/v1/port-forwards

Service: portforward-service (port 8088 internal)

Database: session_db (port forwarding tables)

GET /api/v1/port-forwards

List all port forwarding rules.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
host_id	UUID	Filter by host
type	string	local, remote, dynamic
status	string	active, inactive

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "pf-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "PostgreSQL Dev Access",
      "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
      "type": "local",
      "local_address": "127.0.0.1",
      "local_port": 15432,
      "remote_address": "localhost",
      "remote_port": 5432,
      "status": "active",
      "auto_start": true,
      "created_at": "2026-01-15T09:00:00Z"
    }
  ],
  "pagination": { "total_count": 5, "has_next": false }
}
```

POST /api/v1/port-forwards

Create a port forwarding rule.

Authentication: Bearer token required

Request Body:

```
{
  "name": "Redis Dev Access",
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "type": "local",
  "local_address": "127.0.0.1",
  "local_port": 16379,
  "remote_address": "localhost",
  "remote_port": 6379,
  "auto_start": true,
  "bind_address": "127.0.0.1",
  "description": "Forward remote Redis port to local dev"
}
```

type values: - local: Forward local_port on client to remote_address:remote_port on server - remote: Forward remote_port on server to local_address:local_port on client
- dynamic: SOCKS5 proxy on local_port

Success Response — 201 Created: Full rule object.

GET /api/v1/port-forwards/{ruleId}

Get a specific port forwarding rule.

Authentication: Bearer token required

Success Response — 200 OK: Full rule object.

PUT /api/v1/port-forwards/{ruleId}

Update a port forwarding rule. The rule must be stopped first.

Authentication: Bearer token required

Request Body: Same as POST, all fields optional.

Success Response — 200 OK: Updated rule object.

DELETE /api/v1/port-forwards/{ruleId}

Delete a port forwarding rule. Rule is automatically stopped first.

Authentication: Bearer token required

Success Response — 204 No Content

POST /api/v1/port-forwards/{ruleId}/start

Activate a port forwarding rule (establishes SSH tunnel).

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "id": "pf-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "active",
  "connection_id": "pfconn-550e8400-0000-0000-0000-aabbccddeeff",
  "started_at": "2026-06-28T17:40:00Z",
  "local_endpoint": "127.0.0.1:16379"
}
```

POST /api/v1/port-forwards/{ruleId}/stop

Deactivate a port forwarding rule (closes SSH tunnel).

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "id": "pf-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "inactive",
  "stopped_at": "2026-06-28T17:45:00Z",
  "bytes_transferred": 4096000
}
```

GET /api/v1/port-forwards/{ruleId}/status

Get real-time status of a port forwarding connection.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "id": "pf-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "active",
  "started_at": "2026-06-28T17:40:00Z",
  "bytes_sent": 102400,
  "bytes_received": 2048000,
  "active_connections": 3,
  "last_activity_at": "2026-06-28T17:44:55Z"
}
```

9.

Keychain

Service

REST API

Base path: /

api/v1/keys

Service:

keychain-

service

(port 8089

internal)

Database:

keychain_db

(PostgreSQL)

GET /api/v1/keys

List SSH keys and certificates.

Authentication: Bearer token required

Query Parameters: | Parameter | Type | Description | |—|—|—| |
vault_id | UUID | Filter by vault | | type | string | ssh_key,
certificate, pgp | | q | string | Search by name |

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "key-550e8400-0000-0000-0000-112233445566",
```



```

    "name": "Production Deploy Key",
    "type": "ssh_key",
    "algorithm": "ed25519",
    "bits": null,
    "fingerprint":
      "SHA256:abcdef1234567890abcdef1234567890abcdef12",
    "comment": "deploy@helixterm",
    "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
    "has_passphrase": false,
    "deployments_count": 47,
    "last_used_at": "2026-06-28T16:00:00Z",
    "expires_at": null,
    "created_at": "2026-01-15T09:00:00Z",
    "_links": {
      "self": { "href": "https://api.helixterminator.io/api/v1/
        keys/key-550e8400-0000-0000-0000-112233445566" },
      "public_key": { "href": "https://api.helixterminator.io/
        api/v1/keys/key-550e8400-0000-0000-0000-112233445566/
        public" }
    }
  },
],
"pagination": { "total_count": 12, "has_next": false }
}

```

POST /api/v1/keys

Create or register a new SSH key.

Authentication: Bearer token required

Request Body:

```

{
  "name": "Staging Deploy Key",
  "type": "ssh_key",
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
  "comment": "staging@helixterm",
  "encrypted_private_key": "base64-encoded-encrypted-pem",
  "public_key": "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIB...
    staging@helixterm",
  "passphrase_protected": false
}

```

The private key is always stored encrypted. The encrypted_private_key must be encrypted with the vault's key.

Success Response — 201 Created: Key object (without private key material).

GET /api/v1/keys/{keyId}

Get key metadata (never returns the private key).

Authentication: Bearer token required

Success Response — 200 OK: Full key object including deployment list.

PUT /api/v1/keys/{keyId}

Update key name, comment, or vault association.

Authentication: Bearer token required

Request Body:

```
{  
  "name": "Updated Key Name",  
  "comment": "new-comment@helixterm"  
}
```

Success Response — 200 OK: Updated key object.

DELETE /api/v1/keys/{keyId}

Delete a key. Hosts using this key will require re-association.

Authentication: Bearer token required

Success Response — 204 No Content

POST /api/v1/keys/generate

Generate a new SSH key pair server-side (or client-side via this endpoint's parameter guidance).

Authentication: Bearer token required

Request Body:

```
{
  "name": "Auto-Generated Key 2026",
  "algorithm": "ed25519",
  "comment": "generated@helixterm-2026-06-28",
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
  "bits": null
}
```

algorithm values: ed25519 (recommended), ecdsa, rsa4096.

bits: Only applicable for RSA (2048, 3072, 4096). Ed25519 ignores this.

Success Response — 201 Created:

```
{
  "id": "key-aabbccdd-0000-0000-0000-112233445566",
  "name": "Auto-Generated Key 2026",
  "algorithm": "ed25519",
  "fingerprint":
    "SHA256:newkeyfingerprinthere12345678901234567890",
  "public_key": "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIB2...
    generated@helixterm-2026-06-28",
  "encrypted_private_key": "base64-encoded-encrypted-private-key",
  "created_at": "2026-06-28T17:40:00Z"
}
```

The private key is returned once at generation, encrypted with the vault key. Store it securely.

GET /api/v1/keys/{keyId}/public

Get the public key in OpenSSH authorized_keys format.

Authentication: Bearer token required

Success Response — 200 OK:

Content-Type: text/plain; charset=utf-8

```
ssh-ed25519
AAAAC3NzaC1lZDI1NTE5AAAAIB2abcdef1234567890abcdef1234567890abcd
deploy@helixterm
```

POST /api/v1/keys/{keyId}/deploy

Deploy the public key to a host's `authorized_keys` file via SSH.

Authentication: Bearer token required

Request Body:

```
{
  "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "target_user": "ubuntu",
  "auth_key_id": "key-aabbccdd-0000-0000-0000-existing-key",
  "options": {
    "command": null,
    "restrict": false,
    "no_port_forwarding": false,
    "no_agent_forwarding": false
  }
}
```

`auth_key_id` is the key used to authenticate the deployment operation itself.

Success Response — 200 OK:

```
{
  "deployed_to": "host-550e8400-0000-0000-0000-aabbccddeeff",
  "host_name": "prod-web-01",
  "target_user": "ubuntu",
  "authorized_keys_path": "/home/ubuntu/.ssh/authorized_keys",
  "deployed_at": "2026-06-28T17:40:00Z",
  "was_already_present": false
}
```

POST /api/v1/keys/import

Import an SSH key from PEM, OpenSSH, or PKCS#8 format.

Authentication: Bearer token required

Request Body:

```
{
  "name": "Imported Legacy Key",
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
}
```

```
"private_key_pem": "-----BEGIN OPENSSH PRIVATE KEY-----
    \nb3BlbnNzaC1rZXkAAAAA...\n-----END OPENSSH PRIVATE
    KEY-----",
"passphrase": "optional-key-passphrase",
"comment": "legacy-server@old-domain"
}
```

Success Response — 201 Created: Key object.

10. Snippet Service REST API

Base path: /api/v1/snippets

Service: snippet-service (port 8090 internal)

Database: snippet_db (PostgreSQL)

GET /api/v1/snippets

List snippets accessible to the user.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
vault_id	UUID	Filter by vault
category_id	UUID	Filter by category
language	string	Filter by language (e.g., bash, python)
q	string	Full-text search

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "snip-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "Restart NGINX",
      "description": "Restart the NGINX web server and verify it
        started",
      "content": "sudo systemctl restart nginx && sudo systemctl
        status nginx",
      "language": "bash",
      "category_id": "cat-550e8400-0000-0000-0000-aabbccddeeff",
      "category_name": "System Administration",
      "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
      "tags": ["nginx", "systemctl", "webserver"],
      "shared": false,
      "parameters": [],
    }
  ]
}
```

```
    "executions_count": 142,
    "last_executed_at": "2026-06-28T15:00:00Z",
    "created_by": "550e8400-e29b-41d4-a716-446655440000",
    "created_at": "2026-01-15T09:00:00Z",
    "updated_at": "2026-06-20T10:00:00Z"
  }
],
"pagination": { "total_count": 87, "has_next": false }
}
```

POST /api/v1/snippets

Create a new snippet.

Authentication: Bearer token required

Request Body:

```
{
  "name": "Check Disk Usage",
  "description":
    "Show disk usage sorted by size for the specified
    directory",
  "content":
    "du -sh {{directory}}/* 2>/dev/null | sort -rh | head
    -20",
  "language": "bash",
  "category_id": "cat-550e8400-0000-0000-0000-aabbccddeeff",
  "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
  "tags": ["disk", "storage", "monitoring"],
  "shared": false,
  "parameters": [
    {
      "name": "directory",
      "description": "Directory to check",
      "type": "string",
      "default": "/var",
      "required": true
    }
  ]
}
```

Parameters in content use {{parameter_name}} syntax for variable substitution.

Success Response — 201 Created: Full snippet object.

GET /api/v1/snippets/{snippetId}

Get a specific snippet.

Authentication: Bearer token required

Success Response — 200 OK: Full snippet with execution history summary.

PUT /api/v1/snippets/{snippetId}

Update a snippet.

Authentication: Bearer token required

Request Body: Same as POST, all fields optional.

Success Response — 200 OK: Updated snippet.

DELETE /api/v1/snippets/{snippetId}

Delete a snippet.

Authentication: Bearer token required

Success Response — 204 No Content

POST /api/v1/snippets/{snippetId}/execute

Execute a snippet on one or more hosts.

Authentication: Bearer token required

Request Body:

```
{
  "host_ids": [
    "host-550e8400-0000-0000-0000-aabbccddeeff",
    "host-660e8400-0000-0000-0000-bbccddeeaabb"
  ],
  "parameters": {
```

```

    "directory": "/home"
  },
  "execution_mode": "parallel",
  "timeout_seconds": 60,
  "record": false
}

```

execution_mode: parallel (all hosts simultaneously) or sequential (one at a time).

Success Response — 202 Accepted:

```

{
  "execution_id": "exec-550e8400-0000-0000-0000-aabbccddeeff",
  "snippet_id": "snip-550e8400-0000-0000-0000-aabbccddeeff",
  "status": "running",
  "host_count": 2,
  "started_at": "2026-06-28T17:40:00Z"
}

```

Poll GET /api/v1/snippets/executions/{executionId} for results.

GET /api/v1/snippets/search

Full-text search across snippet names, descriptions, and content.

Authentication: Bearer token required

Query Parameters:

Parameter	Type	Description
q	string	Search query (required)
limit	integer	Max results (default 20, max 50)

Success Response — 200 OK:

```

{
  "query": "nginx restart",
  "results": [
    {
      "id": "snip-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "Restart NGINX",
      "_score": 0.97,
      "_highlights": {
        "name": "<mark>Restart NGINX</mark>",
        "content": "sudo systemctl <mark>restart nginx</mark> && ..."
      }
    }
  ]
}

```



```
    }  
  ],  
  "total_results": 3  
}
```

11. Workspace Service REST API

Base path: /api/v1/workspaces

Service: workspace-service (port 8091 internal)

Database: workspace_db (PostgreSQL)

A **Workspace** is a saved layout of terminal tabs, panels, and connections — analogous to a saved IDE project layout.

GET /api/v1/workspaces

List all workspaces.

Authentication: Bearer token required

Success Response — 200 OK:

```
{  
  "data": [  
    {  
      "id": "ws-550e8400-0000-0000-0000-aabbccddeeff",  
      "name": "Production Debug Session",  
      "description": "4-pane layout for production debugging",  
      "thumbnail_url": "https://cdn.helixterminator.io/ws-  
        thumbnails/ws-550e8400.png",  
      "layout": {  
        "type": "grid",  
        "panels": 4,  
        "arrangement": "2x2"  
      },  
      "session_count": 4,  
      "is_template": false,  
      "template_id": null,  
      "pinned": true,  
      "last_opened_at": "2026-06-28T16:00:00Z",  
      "created_at": "2026-01-15T09:00:00Z"  
    },  
  ],  
}
```

```
"pagination": { "total_count": 8, "has_next": false }
}
```

POST /api/v1/workspaces

Create a new workspace.

Authentication: Bearer token required

Request Body:

```
{
  "name": "Database Monitoring",
  "description": "Monitor all PostgreSQL instances",
  "layout": {
    "type": "split",
    "direction": "horizontal",
    "panes": [
      {
        "id": "pane-1",
        "type": "terminal",
        "host_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
        "startup_snippet_id": "snip-
aabbccdd-0000-0000-0000-112233445566",
        "size_percent": 50
      },
      {
        "id": "pane-2",
        "type": "terminal",
        "host_id": "host-660e8400-0000-0000-0000-bbccddeeaabb",
        "startup_snippet_id": null,
        "size_percent": 50
      }
    ]
  },
  "auto_connect": true,
  "template_id": null
}
```

Success Response — 201 Created: Full workspace object.

GET /api/v1/workspaces/{workspaceId}

Get a specific workspace.

Authentication: Bearer token required

Success Response — 200 OK: Full workspace object.

PUT /api/v1/workspaces/{workspaceId}

Update a workspace layout or metadata.

Authentication: Bearer token required

Success Response — 200 OK: Updated workspace.

DELETE /api/v1/workspaces/{workspaceId}

Delete a workspace.

Authentication: Bearer token required

Success Response — 204 No Content

POST /api/v1/workspaces/{workspaceId}/restore

Restore a workspace to a previous saved state.

Authentication: Bearer token required

Request Body:

```
{  
  "snapshot_id": "snap-550e8400-0000-0000-0000-aabbccddeeff"  
}
```

Success Response — 200 OK: Restored workspace.

GET /api/v1/workspace-templates

List available workspace templates.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "tpl-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "4-Pane Server Monitor",
      "description": "Connect to 4 servers in a 2x2 grid",
      "category": "monitoring",
      "pane_count": 4,
      "preview_url": "https://cdn.helixterminator.io/templates/4pane-preview.png",
      "usage_count": 1247,
      "created_by": "system",
      "created_at": "2026-01-01T00:00:00Z"
    }
  ]
}
```

POST /api/v1/workspace-templates

Create a workspace template from an existing workspace.

Authentication: Bearer token required

Request Body:

```
{
  "workspace_id": "ws-550e8400-0000-0000-0000-aabbccddeeff",
  "name": "My Monitoring Template",
  "description": "4-pane monitoring template",
  "public": false
}
```

Success Response — 201 Created: Template object.

12. Organization & Team Service REST API

Base path: /api/v1/orgs

Service: org-service (port 8092 internal)

Database: org_db (PostgreSQL)

GET /api/v1/orgs/me

Get the authenticated user's organization.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "id": "org-550e8400-0000-0000-0000-000000000001",
  "name": "ExampleCorp",
  "slug": "examplecorp",
  "plan": "enterprise",
  "member_count": 87,
  "max_members": 500,
  "domain": "examplecorp.com",
  "domain_verified": true,
  "sso_enabled": true,
  "sso_provider": "azure",
  "enforce_mfa": true,
  "session_recording_required": true,
  "vault_count": 12,
  "host_count": 347,
  "created_at": "2025-06-01T00:00:00Z"
}
```

GET /api/v1/orgs/me/members

List organization members.

Authentication: Bearer token required; requires org org_admin or member role

Query Parameters:

Parameter	Type	Description
role	string	Filter by role: org_admin, team_admin, member, auditor,
api_user	string	Filter by team membership
team_id	UUID	Filter by team membership
q	string	Search by name or email
cursor	string	Pagination cursor
limit	integer	Page size

Success Response — 200 OK:

```
{
  "data": [
    {
```

```
    "id": "mem-550e8400-0000-0000-0000-aabbccddeeff",
    "user_id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "alice@example.com",
    "display_name": "Alice Smith",
    "role": "org_admin",
    "teams": ["platform", "sre"],
    "invited_by": null,
    "joined_at": "2025-06-01T00:00:00Z",
    "last_active_at": "2026-06-28T17:00:00Z",
    "status": "active"
  }
],
"pagination": { "total_count": 87, "has_next": true }
}
```

POST /api/v1/orgs/me/invitations

Invite a new member to the organization.

Authentication: Bearer token required; requires org_admin role

Request Body:

```
{
  "email": "newmember@examplecorp.com",
  "role": "member",
  "team_ids": ["team-550e8400-0000-0000-0000-aabbccddeeff"],
  "message": "Welcome to HelixTerminator!"
}
```

Success Response — 201 Created:

```
{
  "invitation_id": "inv-550e8400-0000-0000-0000-aabbccddeeff",
  "email": "newmember@examplecorp.com",
  "role": "member",
  "invited_by": "alice@example.com",
  "expires_at": "2026-07-05T17:40:00Z",
  "status": "pending"
}
```

GET /api/v1/orgs/me/teams

List teams in the organization.

Authentication: Bearer token required

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "team-550e8400-0000-0000-0000-aabbccddeeff",
      "name": "Platform Engineering",
      "slug": "platform",
      "description": "Infrastructure and platform team",
      "member_count": 12,
      "vault_access": [
        {
          "vault_id": "vault-550e8400-0000-0000-0000-aabbccddeeff",
          "vault_name": "Production Credentials",
          "permission": "write"
        }
      ],
      "created_at": "2025-06-01T00:00:00Z"
    }
  ],
  "pagination": { "total_count": 5, "has_next": false }
}
```

POST /api/v1/orgs/me/teams

Create a new team.

Authentication: Bearer token required; requires org_admin role

Request Body:

```
{
  "name": "Security Engineering",
  "slug": "security",
  "description": "Security and compliance team"
}
```

Success Response — 201 Created: Full team object.

GET /api/v1/orgs/me/teams/{teamId}

Get a specific team.

Authentication: Bearer token required

Success Response — 200 OK: Full team with members.

PUT /api/v1/orgs/me/teams/{teamId}

Update a team.

Authentication: Bearer token required; requires org_admin role

Success Response — 200 OK: Updated team.

DELETE /api/v1/orgs/me/teams/{teamId}

Delete a team.

Authentication: Bearer token required; requires org_admin role

Success Response — 204 No Content

POST /api/v1/orgs/me/teams/{teamId}/members

Add a member to a team.

Authentication: Bearer token required; requires org_admin role

Request Body:

```
{  
  "user_id": "660e8400-e29b-41d4-a716-556655440000",  
  "role": "member"  
}
```

Success Response — 201 Created: Team member object.

DELETE /api/v1/orgs/me/teams/{teamId}/members/{userId}

Remove a member from a team.

Authentication: Bearer token required; requires org_admin role

Success Response — 204 No Content

13. Audit Service REST API

Base path: /api/v1/audit

Service: audit-service (port 8093 internal)

Database: audit_db (PostgreSQL, append-only, partitioned)

GET /api/v1/audit/events

Query audit events with comprehensive filtering.

Authentication: Bearer token required; requires org org_admin role

Query Parameters:

Parameter	Type	Description
cursor	string	Pagination cursor
limit	integer	Page size (max 100)
user_id	UUID	Filter by user
event_type	string	Filter by event type (repeatable)
resource_type	string	host, vault, session, key, etc.
resource_id	UUID	Filter by specific resource
outcome	string	success, failure
ip_address	string	Filter by source IP
created_at_gte	datetime	After date
created_at_lte	datetime	Before date
sort	string	created_at:desc (default)

Success Response — 200 OK:

```
{
  "data": [
    {
      "id": "audit-550e8400-0000-0000-0000-aabbccddeeff",
      "event_type": "session.ssh.started",
      "user_id": "550e8400-e29b-41d4-a716-446655440000",
      "user_email": "alice@example.com",
      "resource_type": "host",
      "resource_id": "host-550e8400-0000-0000-0000-aabbccddeeff",
      "resource_name": "prod-web-01",
      "outcome": "success",
    }
  ]
}
```

```

    "ip_address": "192.168.1.10",
    "user_agent": "HelixTerminator/1.0",
    "session_id": "sess-550e8400-0000-0000-0000-aabbccddeeff",
    "metadata": {
      "host_name": "prod-web-01",
      "jump_chain": [],
      "recording_enabled": true,
      "ticket_ref": "INC-20260628-001"
    },
    "hash": "sha256:event-integrity-hash",
    "prev_hash": "sha256:previous-event-hash",
    "created_at": "2026-06-28T17:40:00Z"
  }
],
"pagination": {
  "cursor_next": "eyJpZCI6...",
  "has_next": true,
  "total_count": 48291
}
}

```

GET /api/v1/audit/events/{eventId}

Get a specific audit event by ID.

Authentication: Bearer token required; requires org org_admin role

Success Response — 200 OK: Full audit event object.

GET /api/v1/audit/export

Export audit events to a file.

Authentication: Bearer token required; requires org org_admin role

Query Parameters:

Parameter	Type	Description
format	string	json, csv, syslog
created_at_gte	datetime	Start date (required)
created_at_lte	datetime	End date (required)
event_type	string	Filter by event type

Success Response — 202 Accepted:

```

{
  "export_id": "aexp-550e8400-0000-0000-0000-aabbccddeeff",

```

```
"status": "processing",
"estimated_rows": 48291,
"format": "json",
"requested_at": "2026-06-28T17:40:00Z"
}
```

14. AI Service REST API

Base path: /api/v1/ai

Service: ai-service (port 8094 internal)

Rate Limit: 100 requests per user per hour

POST /api/v1/ai/complete

AI-powered command completion in the terminal context.

Authentication: Bearer token required

Request Body:

```
{
  "context": {
    "cwd": "/var/www/html",
    "hostname": "prod-web-01",
    "os": "Ubuntu 24.04",
    "shell": "bash",
    "history": [
      "ls -la",
      "cat nginx.conf",
      "sudo systemctl status nginx"
    ]
  },
  "partial_command": "sudo journalctl -u nginx",
  "max_suggestions": 5
}
```

Success Response — 200 OK:

```
{
  "suggestions": [
    {
```

```
"completion":
  "sudo journalctl -u nginx --since '1 hour ago' | tail
  -100",
"description": "View nginx logs from the last hour",
"confidence": 0.95
},
{
  "completion": "sudo journalctl -u nginx -n 50 --no-pager",
  "description": "Show last 50 log entries",
  "confidence": 0.88
}
],
"model": "helixterm-cmd-v1",
"latency_ms": 124
}
```

POST /api/v1/ai/explain

Explain a command in plain English.

Authentication: Bearer token required

Request Body:

```
{
  "command": "find /var/log -name '*.log' -mtime +7 -exec gzip {}
  \\\;",
  "context": {
    "os": "Ubuntu 24.04",
    "shell": "bash"
  },
  "detail_level": "detailed"
}
```

detail_level: brief, standard, detailed.

Success Response — 200 OK:

```
{
  "command": "find /var/log -name '*.log' -mtime +7 -exec gzip {}
  \\\;",
  "explanation": {
    "summary":
      "Compresses all .log files in /var/log that haven't been
      modified in more than 7 days.",
    "parts": [
      {
```

```

    "token": "find /var/log",
    "description": "Start searching from the /var/log
    directory"
  },
  {
    "token": "-name '*.log'",
    "description": "Match only files ending in .log"
  },
  {
    "token": "-mtime +7",
    "description": "Modified more than 7 days ago"
  },
  {
    "token": "-exec gzip {} \\;",
    "description": "Execute gzip on each matching file"
  }
],
"risk_level": "low",
"side_effects": ["Creates .log.gz files", "Deletes
  original .log files (gzip default)"],
"man_page_refs": ["find(1)", "gzip(1)"]
},
"model": "helixterm-explain-v1",
"latency_ms": 89
}

```

POST /api/v1/ai/suggest

Get AI suggestions for the next action based on terminal context.

Authentication: Bearer token required

Request Body:

```

{
  "context": {
    "cwd": "/var/www",
    "hostname": "prod-web-01",
    "os": "Ubuntu 24.04",
    "shell": "bash",
    "recent_output": "nginx: [warn] could not build optimal
      variables_hash...\nnginx: the configuration file /etc/
      nginx/nginx.conf syntax is ok",
    "history": [
      "sudo nginx -t",

```

```
    "sudo cat /etc/nginx/nginx.conf"
  ],
  "goal": "Fix the nginx configuration warning"
}
```

Success Response — 200 OK:

```
{
  "suggestions": [
    {
      "type": "command",
      "command": "sudo sed -i 's/variables_hash_bucket_size.*/variables_hash_bucket_size 128;/' /etc/nginx/nginx.conf",
      "description": "Increase variables_hash_bucket_size to resolve the warning",
      "confidence": 0.91,
      "explanation": "The warning indicates variables_hash_bucket_size needs to be increased. The default is 64 bytes; setting it to 128 typically resolves this."
    },
    {
      "type": "documentation",
      "url": "https://nginx.org/en/docs/hash.html",
      "description": "NGINX hash configuration documentation"
    }
  ],
  "model": "helixterm-suggest-v1",
  "latency_ms": 156
}
```

15. WebSocket API

HelixTerminator uses WebSocket connections for real-time terminal I/O, collaboration, and vault sync.

Authentication

All WebSocket connections require authentication via: 1. **Query**

parameter: wss://proxy.helixterminator.io/...?

token=<session_token> 2. **First message:** Send

`{"type": "auth", "token": "<session_token>"}` within 5 seconds of connecting.

WS /api/v1/sessions/{sessionId}/terminal

Bidirectional terminal I/O stream. Protocol: Binary and text frames.

Connection URL:

wss://proxy.helixterminator.io/api/v1/sessions/sess-550e8400/
terminal?token=st_v1_xxx

Client → Server messages:

Input data (terminal keystrokes):

```
{"type": "input", "data": "bHMgLWxhCg=="}
```

data is base64-encoded raw bytes.

Terminal resize:

```
{"type": "resize", "cols": 240, "rows": 60}
```

Ping/keepalive:

```
{"type": "ping"}
```

Server → Client messages:

Output data:

```
{"type": "output", "data": "dG90YWwgNDgK..."}
```

Session state:

```
{"type": "state", "status": "connected", "host": "prod-web-01",  
  "latency_ms": 12}
```

Error:

```
{"type": "error", "code": "host_unreachable", "message":  
  "Connection to host timed out after 30 seconds"}
```

Close:

```
{"type": "close", "code": 0, "message": "Session ended by user"}
```

WebSocket close codes:

Code	Meaning
4000	Normal close (session ended)
4001	Authentication failed
4002	Session expired
4003	Host connection lost
4004	Server error
4005	Rate limit exceeded
4006	Idle timeout (5 minutes of no input)

WS /api/v1/sessions/{sessionId}/collab

Collaboration channel for shared terminal sessions.

Connection URL:

```
wss://proxy.helixterminator.io/api/v1/sessions/sess-550e8400/  
collab?token=st_v1_xxx
```

Server → Client messages:

Participant joined:

```
{"type": "participant_joined", "user_id": "660e8400-...",  
  "display_name": "Bob Jones", "role": "viewer"}
```

Participant left:

```
{"type": "participant_left", "user_id": "660e8400-..."}
```

Cursor position (other participants' cursors):

```
{"type": "cursor", "user_id": "660e8400-...", "position": {"row":  
  5, "col": 12}, "color": "#FF6B35"}
```

Chat message:

```
{"type": "chat", "user_id": "660e8400-...", "display_name": "Bob  
  Jones", "message": "Check line 42", "timestamp":  
  "2026-06-28T17:45:00Z"}
```


Client → Server messages:

Send chat:

```
{"type": "chat", "message": "I see the issue!"}
```

Request control:

```
{"type": "request_control"}
```

WS /api/v1/sync

Vault sync channel for real-time vault synchronization across clients.

Connection URL:

```
wss://api.helixterminator.io/api/v1/sync?  
token=<access_token>&vault_id=<vault_id>
```

Client → Server messages:

Sync push:

```
{  
  "type": "push",  
  "vault_id": "vault-550e8400-...",  
  "changes": [  
    {  
      "item_id": "item-550e8400-...",  
      "operation": "upsert",  
      "encrypted_data": "base64...",  
      "version": 6,  
      "checksum": "sha256:..."  
    }  
  ],  
  "client_cursor": "cursor-value"  
}
```

Server → Client messages:

Sync pull (server-initiated changes):

```
{  
  "type": "pull",  
  "vault_id": "vault-550e8400-...",  
  "changes": [...],  
}
```

```
    "server_cursor": "new-cursor"
}
```

Conflict notification:

```
{
  "type": "conflict",
  "item_id": "item-550e8400-...",
  "your_version": 5,
  "server_version": 7
}
```

16. gRPC Service Definitions

Internal gRPC services communicate over mTLS on the internal Kubernetes network. All proto files live at `internal/proto/` in the monorepo.

Base configuration:

```
syntax = "proto3";
package helixterm.v1;
option go_package = "helixterminator.io/core/internal/
    proto;proto";
```

auth.proto

```
syntax = "proto3";
package helixterm.v1;
option go_package = "helixterminator.io/core/internal/
    proto;proto";

import "google/protobuf/timestamp.proto";
import "google/protobuf/empty.proto";

// AuthService provides internal authentication operations called
// by other microservices.
service AuthService {
  // ValidateToken validates a JWT access token and returns the
  // claims.
  rpc ValidateToken(ValidateTokenRequest) returns
    (ValidateTokenResponse);

  // ValidateApiKey validates an API key and returns the
  // associated user and scopes.
}
```

```

rpc ValidateApiKey(ValidateApiKeyRequest) returns
    (ValidateApiKeyResponse);

// GetUserContext retrieves user context (user ID, org, roles)
// for an authenticated request.
rpc GetUserContext(GetUserContextRequest) returns (UserContext);

// RevokeToken adds a token to the blocklist.
rpc RevokeToken(RevokeTokenRequest) returns
    (google.protobuf.Empty);

// GetSession retrieves session details by session ID.
rpc GetSession(GetSessionRequest) returns (Session);

// InvalidateSession terminates a session.
rpc InvalidateSession(InvalidateSessionRequest) returns
    (google.protobuf.Empty);

// CheckPermission verifies a user has a specific permission on
// a resource.
rpc CheckPermission(CheckPermissionRequest) returns
    (CheckPermissionResponse);

// BulkCheckPermission checks multiple permissions at once.
rpc BulkCheckPermission(BulkCheckPermissionRequest) returns
    (BulkCheckPermissionResponse);
}

message ValidateTokenRequest {
    string token = 1;
    repeated string required_scopes = 2;
    string audience = 3;
}

message ValidateTokenResponse {
    bool valid = 1;
    string user_id = 2;
    string session_id = 3;
    string org_id = 4;
    repeated string scopes = 5;
    bool mfa_verified = 6;
    google.protobuf.Timestamp expires_at = 7;
    string error_code = 8;
    string error_message = 9;
}

message ValidateApiKeyRequest {

```

```
    string api_key = 1;
    repeated string required_scopes = 2;
}
```

```
message ValidateApiKeyResponse {
    bool valid = 1;
    string user_id = 2;
    string key_id = 3;
    string org_id = 4;
    repeated string scopes = 5;
    repeated string allowed_ips = 6;
    string error_code = 7;
}
```

```
message GetUserContextRequest {
    string user_id = 1;
}
```

```
message UserContext {
    string user_id = 1;
    string email = 2;
    string display_name = 3;
    string org_id = 4;
    string org_slug = 5;
    string org_role = 6;
    repeated string team_ids = 7;
    repeated string permissions = 8;
    bool mfa_enabled = 9;
    string locale = 10;
    string timezone = 11;
    string status = 12;
}
```

```
message RevokeTokenRequest {
    string token_jti = 1;
    google.protobuf.Timestamp expires_at = 2;
    string reason = 3;
}
```

```
message GetSessionRequest {
    string session_id = 1;
}
```

```
message Session {
```

```

    string id = 1;
    string user_id = 2;
    string device_id = 3;
    string ip_address = 4;
    string user_agent = 5;
    bool mfa_verified = 6;
    google.protobuf.Timestamp created_at = 7;
    google.protobuf.Timestamp last_active_at = 8;
    google.protobuf.Timestamp expires_at = 9;
    string status = 10;
}

message InvalidateSessionRequest {
    string session_id = 1;
    string reason = 2;
}

message CheckPermissionRequest {
    string user_id = 1;
    string resource_type = 2;
    string resource_id = 3;
    string action = 4;
    string org_id = 5;
}

message CheckPermissionResponse {
    bool allowed = 1;
    string reason = 2;
}

message BulkCheckPermissionRequest {
    repeated CheckPermissionRequest checks = 1;
}

message BulkCheckPermissionResponse {
    repeated CheckPermissionResponse results = 1;
}

```

vault.proto

```

syntax = "proto3";
package helixterm.v1;

```

```

option go_package = "helixterminator.io/core/internal/
    proto;proto";

import "google/protobuf/timestamp.proto";
import "google/protobuf/empty.proto";

// VaultService provides internal vault operations for other
// microservices.
service VaultService {
    // GetVault retrieves vault metadata (not encrypted contents).
    rpc GetVault(GetVaultRequest) returns (Vault);

    // CheckVaultAccess verifies a user's access to a vault.
    rpc CheckVaultAccess(CheckVaultAccessRequest) returns
        (CheckVaultAccessResponse);

    // GetVaultMember retrieves vault membership for a specific
    // user.
    rpc GetVaultMember(GetVaultMemberRequest) returns (VaultMember);

    // RecordVaultEvent records an event in vault audit log.
    rpc RecordVaultEvent(RecordVaultEventRequest) returns
        (google.protobuf.Empty);

    // GetVaultSyncState gets the current sync cursor for a vault.
    rpc GetVaultSyncState(GetVaultSyncStateRequest) returns
        (VaultSyncState);

    // UpdateVaultSyncState updates the sync cursor after successful
    // sync.
    rpc UpdateVaultSyncState(UpdateVaultSyncStateRequest) returns
        (google.protobuf.Empty);

    // GetEncryptedVaultKey retrieves the vault key blob encrypted
    // for a specific user.
    rpc GetEncryptedVaultKey(GetEncryptedVaultKeyRequest) returns
        (GetEncryptedVaultKeyResponse);

    // ListVaultMembers lists all members of a vault.
    rpc ListVaultMembers(ListVaultMembersRequest) returns
        (ListVaultMembersResponse);
}

message GetVaultRequest {
    string vault_id = 1;
}

message Vault {

```

```
    string id = 1;
    string name = 2;
    string owner_id = 3;
    string org_id = 4;
    bool sync_enabled = 5;
    bool encrypted = 6;
    google.protobuf.Timestamp created_at = 7;
    google.protobuf.Timestamp updated_at = 8;
}
```

```
message CheckVaultAccessRequest {
    string vault_id = 1;
    string user_id = 2;
    string required_permission = 3;
}
```

```
message CheckVaultAccessResponse {
    bool allowed = 1;
    string permission = 2;
    string reason = 3;
}
```

```
message GetVaultMemberRequest {
    string vault_id = 1;
    string user_id = 2;
}
```

```
message VaultMember {
    string vault_id = 1;
    string user_id = 2;
    string permission = 3;
    bytes encrypted_vault_key = 4;
    google.protobuf.Timestamp joined_at = 5;
}
```

```
message RecordVaultEventRequest {
    string vault_id = 1;
    string user_id = 2;
    string event_type = 3;
    string resource_type = 4;
    string resource_id = 5;
    bytes metadata_json = 6;
    string ip_address = 7;
}
```

```

message GetVaultSyncStateRequest {
    string vault_id = 1;
    string client_id = 2;
}

message VaultSyncState {
    string vault_id = 1;
    string client_id = 2;
    string cursor = 3;
    google.protobuf.Timestamp last_synced_at = 4;
    int64 server_version = 5;
}

message UpdateVaultSyncStateRequest {
    string vault_id = 1;
    string client_id = 2;
    string cursor = 3;
    int64 server_version = 4;
}

message GetEncryptedVaultKeyRequest {
    string vault_id = 1;
    string user_id = 2;
}

message GetEncryptedVaultKeyResponse {
    bytes encrypted_key_blob = 1;
    bytes kdf_params_json = 2;
}

message ListVaultMembersRequest {
    string vault_id = 1;
}

message ListVaultMembersResponse {
    repeated VaultMember members = 1;
}

```

pki.proto

```

syntax = "proto3";
package helixterm.v1;

```



```

option go_package = "helixterminator.io/core/internal/
    proto;proto";

import "google/protobuf/timestamp.proto";
import "google/protobuf/duration.proto";
import "google/protobuf/empty.proto";

// PKIService provides SSH certificate issuance and management.
// HelixTerminator supports SSH certificate-based authentication
// as an alternative to authorized_keys.
service PKIService {
    // SignUserCertificate signs a user's public key with the CA.
    rpc SignUserCertificate(SignUserCertificateRequest) returns
        (SignUserCertificateResponse);

    // SignHostCertificate signs a host public key with the CA.
    rpc SignHostCertificate(SignHostCertificateRequest) returns
        (SignHostCertificateResponse);

    // GetCACertificate returns the CA public key for trust
    // distribution.
    rpc GetCACertificate(GetCACertificateRequest) returns
        (GetCACertificateResponse);

    // RevokeCertificate revokes a previously issued certificate.
    rpc RevokeCertificate(RevokeCertificateRequest) returns
        (google.protobuf.Empty);

    // GetCRL returns the current certificate revocation list.
    rpc GetCRL(GetCRLRequest) returns (GetCRLResponse);

    // CheckCertificate validates whether a certificate is valid and
    // not revoked.
    rpc CheckCertificate(CheckCertificateRequest) returns
        (CheckCertificateResponse);

    // ListCertificates lists issued certificates for a user or
    // host.
    rpc ListCertificates(ListCertificatesRequest) returns
        (ListCertificatesResponse);

    // RotateCA rotates the CA key (requires admin authority).
    rpc RotateCA(RotateCARequest) returns (RotateCAResponse);
}

message SignUserCertificateRequest {
    string public_key_openssh = 1;
    string user_id = 2;

```

```

    string username = 3;
    repeated string principals = 4;
    repeated CertExtension extensions = 5;
    google.protobuf.Duration validity = 6;
    string source_address = 7;
    bool force_command = 8;
    string forced_command = 9;
}

message SignUserCertificateResponse {
    string certificate_openssh = 1;
    string certificate_id = 2;
    google.protobuf.Timestamp valid_after = 3;
    google.protobuf.Timestamp valid_before = 4;
    string serial = 5;
}

message SignHostCertificateRequest {
    string public_key_openssh = 1;
    string host_id = 2;
    repeated string hostnames = 3;
    google.protobuf.Duration validity = 4;
}

message SignHostCertificateResponse {
    string certificate_openssh = 1;
    string certificate_id = 2;
    google.protobuf.Timestamp valid_before = 3;
    string serial = 4;
}

message CertExtension {
    string name = 1;
    string value = 2;
    bool critical = 3;
}

message GetCACertificateRequest {
    string ca_type = 1; // "user" or "host"
}

message GetCACertificateResponse {
    string public_key_openssh = 1;
    string fingerprint = 2;
}

```

```
    google.protobuf.Timestamp created_at = 3;
    google.protobuf.Timestamp rotated_at = 4;
    string version = 5;
}

message RevokeCertificateRequest {
    string certificate_id = 1;
    string reason = 2;
    string revoked_by = 3;
}

message GetCRLRequest {
    string ca_type = 1;
}

message GetCRLResponse {
    repeated string revoked_serials = 1;
    string krl_binary_base64 = 2;
    google.protobuf.Timestamp generated_at = 3;
}

message CheckCertificateRequest {
    string certificate_openssh = 1;
}

message CheckCertificateResponse {
    bool valid = 1;
    bool revoked = 2;
    bool expired = 3;
    string certificate_id = 4;
    string reason = 5;
    google.protobuf.Timestamp expires_at = 6;
}

message ListCertificatesRequest {
    string entity_type = 1; // "user" or "host"
    string entity_id = 2;
    bool include_revoked = 3;
    bool include_expired = 4;
}

message Certificate {
    string id = 1;
    string entity_type = 2;
```

```

    string entity_id = 3;
    string serial = 4;
    string fingerprint = 5;
    repeated string principals = 6;
    google.protobuf.Timestamp valid_after = 7;
    google.protobuf.Timestamp valid_before = 8;
    bool revoked = 9;
    google.protobuf.Timestamp revoked_at = 10;
    string revocation_reason = 11;
    google.protobuf.Timestamp created_at = 12;
}

message ListCertificatesResponse {
    repeated Certificate certificates = 1;
    int32 total_count = 2;
}

message RotateCARequest {
    string ca_type = 1;
    string reason = 2;
    string rotated_by = 3;
}

message RotateCAResponse {
    string old_public_key = 1;
    string new_public_key = 2;
    string new_fingerprint = 3;
    google.protobuf.Timestamp rotated_at = 4;
}

```

session.proto

```

syntax = "proto3";
package helixterm.v1;
option go_package = "helixterminator.io/core/internal/
    proto;proto";

import "google/protobuf/timestamp.proto";
import "google/protobuf/empty.proto";

// SessionService manages SSH session lifecycle for internal
// service communication.
service SessionService {
    // CreateSession creates a new session record.

```

```

rpc CreateSession(CreateSessionRequest) returns
    (CreateSessionResponse);

// GetSession retrieves session details.
rpc GetSession(GetSessionRequest) returns (SSHSession);

// UpdateSessionStatus updates the status of a session.
rpc UpdateSessionStatus(UpdateSessionStatusRequest) returns
    (google.protobuf.Empty);

// TerminateSession forcibly terminates an active session.
rpc TerminateSession(TerminateSessionRequest) returns
    (google.protobuf.Empty);

// ListActiveSessions lists currently active sessions,
// optionally filtered.
rpc ListActiveSessions(ListActiveSessionsRequest) returns
    (ListActiveSessionsResponse);

// RecordSessionEvent appends an event to the session event log.
rpc RecordSessionEvent(RecordSessionEventRequest) returns
    (google.protobuf.Empty);

// GetSessionStats returns aggregate statistics for a session.
rpc GetSessionStats(GetSessionStatsRequest) returns
    (SessionStats);

// FinalizeRecording marks a session recording as complete and
// triggers processing.
rpc FinalizeRecording(FinalizeRecordingRequest) returns
    (google.protobuf.Empty);

// BroadcastToSession sends a command to one or more sessions.
rpc BroadcastToSession(BroadcastRequest) returns
    (BroadcastResponse);

// GetPortForward retrieves port forwarding rule details.
rpc GetPortForward(GetPortForwardRequest) returns
    (PortForwardRule);

// UpdatePortForwardStatus updates the status of a port
// forwarding connection.
rpc UpdatePortForwardStatus(UpdatePortForwardStatusRequest)
    returns (google.protobuf.Empty);
}

message CreateSessionRequest {
    string user_id = 1;
    string host_id = 2;

```

```

    string vault_id = 3;
    string client_ip = 4;
    string user_agent = 5;
    int32 terminal_cols = 6;
    int32 terminal_rows = 7;
    string terminal_type = 8;
    bool recording_enabled = 9;
    bool collab_enabled = 10;
    bool read_only = 11;
    string reason = 12;
    string ticket_ref = 13;
    string startup_snippet_id = 14;
}

message CreateSessionResponse {
    string session_id = 1;
    string session_token = 2;
    google.protobuf.Timestamp expires_at = 3;
}

message GetSessionRequest {
    string session_id = 1;
}

message SSHSession {
    string id = 1;
    string user_id = 2;
    string host_id = 3;
    string vault_id = 4;
    string status = 5;
    string client_ip = 6;
    int32 terminal_cols = 7;
    int32 terminal_rows = 8;
    bool recording_enabled = 9;
    bool collab_enabled = 10;
    bool read_only = 11;
    string reason = 12;
    string ticket_ref = 13;
    google.protobuf.Timestamp started_at = 14;
    google.protobuf.Timestamp ended_at = 15;
    int64 bytes_sent = 16;
    int64 bytes_received = 17;
}

```

```

message UpdateSessionStatusRequest {
    string session_id = 1;
    string status = 2;
    string error_message = 3;
    int32 exit_code = 4;
}

message TerminateSessionRequest {
    string session_id = 1;
    string terminated_by = 2;
    string reason = 3;
}

message ListActiveSessionsRequest {
    string user_id = 1;
    string host_id = 2;
    string org_id = 3;
}

message ListActiveSessionsResponse {
    repeated SSHSession sessions = 1;
}

message RecordSessionEventRequest {
    string session_id = 1;
    string event_type = 2;
    google.protobuf.Timestamp occurred_at = 3;
    bytes data = 4;
    string direction = 5; // "i" (input) or "o" (output)
}

message GetSessionStatsRequest {
    string session_id = 1;
}

message SessionStats {
    string session_id = 1;
    int64 bytes_sent = 2;
    int64 bytes_received = 3;
    int64 events_count = 4;
    google.protobuf.Timestamp duration = 5;
    int32 resize_count = 6;
}

```

```
message FinalizeRecordingRequest {
    string session_id = 1;
    string recording_path = 2;
    int64 file_size_bytes = 3;
}

message BroadcastRequest {
    repeated string session_ids = 1;
    bytes data = 2;
    bool require_confirmation = 3;
}

message BroadcastResponse {
    int32 sent = 1;
    int32 failed = 2;
    repeated string failed_session_ids = 3;
}

message GetPortForwardRequest {
    string rule_id = 1;
}

message PortForwardRule {
    string id = 1;
    string user_id = 2;
    string host_id = 3;
    string name = 4;
    string type = 5;
    string local_address = 6;
    int32 local_port = 7;
    string remote_address = 8;
    int32 remote_port = 9;
    bool auto_start = 10;
    string status = 11;
}

message UpdatePortForwardStatusRequest {
    string rule_id = 1;
    string connection_id = 2;
    string status = 3;
    int64 bytes_sent = 4;
    int64 bytes_received = 5;
}
```

audit.proto

```
syntax = "proto3";
package helixterm.v1;
option go_package = "helixterminator.io/core/internal/
    proto;proto";

import "google/protobuf/timestamp.proto";
import "google/protobuf/empty.proto";

// AuditService receives and stores audit events from all
// microservices.
// It maintains a cryptographic hash chain for tamper evidence.
service AuditService {
    // RecordEvent appends an audit event to the immutable log.
    rpc RecordEvent(RecordEventRequest) returns
        (RecordEventResponse);

    // RecordEventBatch appends multiple events in a single RPC
    // (efficient bulk logging).
    rpc RecordEventBatch(RecordEventBatchRequest) returns
        (RecordEventBatchResponse);

    // QueryEvents queries the audit log with filters (admin only).
    rpc QueryEvents(QueryEventsRequest) returns
        (QueryEventsResponse);

    // GetEvent retrieves a single audit event.
    rpc GetEvent(GetEventRequest) returns (AuditEvent);

    // VerifyIntegrity verifies the hash chain for a given time
    // range.
    rpc VerifyIntegrity(VerifyIntegrityRequest) returns
        (VerifyIntegrityResponse);

    // ExportEvents triggers an async export job.
    rpc ExportEvents(ExportEventsRequest) returns
        (ExportEventsResponse);
}

message RecordEventRequest {
    string event_type = 1;
    string user_id = 2;
    string org_id = 3;
    string resource_type = 4;
    string resource_id = 5;
    string resource_name = 6;
```

```

    string outcome = 7;
    string ip_address = 8;
    string user_agent = 9;
    string session_id = 10;
    bytes metadata_json = 11;
    google.protobuf.Timestamp occurred_at = 12;
    string source_service = 13;
}

message RecordEventResponse {
    string event_id = 1;
    string hash = 2;
    google.protobuf.Timestamp recorded_at = 3;
}

message RecordEventBatchRequest {
    repeated RecordEventRequest events = 1;
}

message RecordEventBatchResponse {
    int32 recorded = 1;
    int32 failed = 2;
    repeated RecordEventResponse results = 3;
}

message QueryEventsRequest {
    string org_id = 1;
    string user_id = 2;
    string resource_type = 3;
    string resource_id = 4;
    string event_type = 5;
    string outcome = 6;
    google.protobuf.Timestamp created_at_gte = 7;
    google.protobuf.Timestamp created_at_lte = 8;
    string cursor = 9;
    int32 limit = 10;
    string sort_direction = 11;
}

message QueryEventsResponse {
    repeated AuditEvent events = 1;
    string next_cursor = 2;
    bool has_next = 3;
    int64 total_count = 4;
}

```

```
}
```

```
message GetEventRequest {  
    string event_id = 1;  
}
```

```
message AuditEvent {  
    string id = 1;  
    string event_type = 2;  
    string user_id = 3;  
    string org_id = 4;  
    string resource_type = 5;  
    string resource_id = 6;  
    string resource_name = 7;  
    string outcome = 8;  
    string ip_address = 9;  
    string user_agent = 10;  
    string session_id = 11;  
    bytes metadata_json = 12;  
    string hash = 13;  
    string prev_hash = 14;  
    string source_service = 15;  
    google.protobuf.Timestamp occurred_at = 16;  
    google.protobuf.Timestamp recorded_at = 17;  
}
```

```
message VerifyIntegrityRequest {  
    string org_id = 1;  
    google.protobuf.Timestamp from = 2;  
    google.protobuf.Timestamp to = 3;  
}
```

```
message VerifyIntegrityResponse {  
    bool valid = 1;  
    int64 events_checked = 2;  
    string first_broken_event_id = 3;  
    string error_description = 4;  
}
```

```
message ExportEventsRequest {  
    string org_id = 1;  
    string format = 2;  
    google.protobuf.Timestamp created_at_gte = 3;  
    google.protobuf.Timestamp created_at_lte = 4;
```

```

    string requested_by = 5;
}

message ExportEventsResponse {
    string export_id = 1;
    string status = 2;
}

```

17. PostgreSQL Database Schemas

Each microservice owns its own PostgreSQL 17.2 database. In production, each runs on a dedicated cluster or schema. In development, all schemas live in a single PostgreSQL instance with separate databases.

> **DEFERRED**

(next

increment):

Multi-tenant isolation below is

app-layer WHERE

org_id = ... only

— no > table in

§17 has

ROW LEVEL

SECURITY enabled

or a CREATE

POLICY defined. A

missing/incorrect

WHERE > clause in

application code

is a cross-tenant

data leak (IDOR)

today; do not read the schemas below as > DB-enforced isolation. Adding RLS policies (mirroring the intended audit_events pattern) to every > multi-tenant table in §17 is deferred to the next increment. > **DEFERRED (next increment):** No backup/restore or Point-In-Time-Recovery (PITR) procedure is specified > for any of the per-service PostgreSQL databases below (or for Redis in §18). Authoring RPO/RTO targets > and a concrete backup/PITR runbook is deferred; until then, assume no tested recovery path exists.

Conventions: - All primary keys are UUID using gen_random_uuid() (pgcrypto / pg 13+). - All timestamps are `TIMESTAMP WITH`

TIME ZONE NOT
NULL DEFAULT
NOW(). - Soft
delete:
deleted_at
TIMESTAMP WITH
TIME ZONE — NULL
means not
deleted. - Partial
indexes on
deleted_at IS
NULL for all soft-
delete tables. - All
ENUM-like status
columns use
VARCHAR with
CHECK constraints
(easier migrations
than SQL
ENUMs). - JSONB
for flexible
metadata fields
with GIN indexes.
- BRIN indexes on
all created_at /
occurred_at for
append-heavy
tables.

17.1 auth_db

```
-- Enable required extensions
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
CREATE EXTENSION IF NOT EXISTS "citext";

-- =====
-- TABLE: users
-- Core user identity. Owns authentication credentials.
-- =====
CREATE TABLE users (
    id                UUID PRIMARY KEY DEFAULT
                    gen_random_uuid(),
    email             CITEXT NOT NULL,
```

```

email_verified_at      TIMESTAMP WITH TIME ZONE,
email_pending          CITEXT,
email_pending_token    VARCHAR(255),
email_pending_expires_at  TIMESTAMP WITH TIME ZONE,
password_hash          VARCHAR(255),
display_name           VARCHAR(100) NOT NULL,
avatar_url             TEXT,
bio                    TEXT,
locale                 VARCHAR(20) NOT NULL DEFAULT 'en-US',
timezone               VARCHAR(100) NOT NULL DEFAULT 'UTC',
status                 VARCHAR(20) NOT NULL DEFAULT 'active'
                        CHECK (status IN ('active',
                        'suspended', 'pending_deletion', 'deleted')),
failed_login_attempts  INTEGER NOT NULL DEFAULT 0,
locked_until           TIMESTAMP WITH TIME ZONE,
last_login_at          TIMESTAMP WITH TIME ZONE,
last_login_ip          INET,
password_changed_at    TIMESTAMP WITH TIME ZONE,
terms_accepted_at      TIMESTAMP WITH TIME ZONE,
terms_version           VARCHAR(20),
deletion_requested_at  TIMESTAMP WITH TIME ZONE,
deletion_scheduled_at  TIMESTAMP WITH TIME ZONE,
created_at             TIMESTAMP WITH TIME ZONE NOT NULL
                        DEFAULT NOW(),
updated_at             TIMESTAMP WITH TIME ZONE NOT NULL
                        DEFAULT NOW(),
deleted_at             TIMESTAMP WITH TIME ZONE
);

```

```

CREATE UNIQUE INDEX idx_users_email ON users(email) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_users_email_pending ON users(email_pending)
    WHERE email_pending IS NOT NULL;
CREATE INDEX idx_users_status ON users(status) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_users_created_at ON users USING BRIN
    (created_at);
CREATE INDEX idx_users_deletion_scheduled ON
    users(deletion_scheduled_at)
    WHERE deletion_scheduled_at IS NOT NULL;

```

```

-- Trigger: update updated_at on row modification
CREATE OR REPLACE FUNCTION trigger_set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = NOW();
    RETURN NEW;

```

```

END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_users_updated_at
    BEFORE UPDATE ON users
    FOR EACH ROW EXECUTE FUNCTION trigger_set_updated_at();

-- =====
-- TABLE: user_sessions
-- Active authentication sessions (login sessions).
-- =====

CREATE TABLE user_sessions (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL REFERENCES users(id) ON DELETE
        CASCADE,
    device_id         UUID,
    ip_address        INET NOT NULL,
    user_agent        TEXT,
    location_city     VARCHAR(100),
    location_country  VARCHAR(10),
    mfa_verified      BOOLEAN NOT NULL DEFAULT FALSE,
    mfa_method        VARCHAR(20) CHECK (mfa_method IN ('totp',
        'fido2', 'backup_code', NULL)),
    status            VARCHAR(20) NOT NULL DEFAULT 'active'
        CHECK (status IN ('active', 'expired',
        'revoked'))),
    revoked_reason    VARCHAR(100),
    last_active_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    expires_at        TIMESTAMP WITH TIME ZONE NOT NULL
);

CREATE INDEX idx_user_sessions_user_id ON user_sessions(user_id);
CREATE INDEX idx_user_sessions_status ON user_sessions(status,
    expires_at)
    WHERE status = 'active';
CREATE INDEX idx_user_sessions_expires_at ON user_sessions USING
    BRIN (expires_at);

-- =====
-- TABLE: refresh_tokens
-- Refresh token storage for token rotation.
-- =====

```



```

CREATE TABLE refresh_tokens (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id  UUID NOT NULL REFERENCES user_sessions(id) ON
        DELETE CASCADE,
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
        CASCADE,
    token_hash  VARCHAR(255) NOT NULL UNIQUE,
    family      UUID NOT NULL,
    generation  INTEGER NOT NULL DEFAULT 1,
    ip_address  INET,
    user_agent  TEXT,
    used_at     TIMESTAMP WITH TIME ZONE,
    revoked     BOOLEAN NOT NULL DEFAULT FALSE,
    revoked_at  TIMESTAMP WITH TIME ZONE,
    revoked_reason VARCHAR(100),
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    expires_at  TIMESTAMP WITH TIME ZONE NOT NULL
);

```

```

CREATE INDEX idx_refresh_tokens_token_hash ON
    refresh_tokens(token_hash);
CREATE INDEX idx_refresh_tokens_session_id ON
    refresh_tokens(session_id);
CREATE INDEX idx_refresh_tokens_family ON refresh_tokens(family);
CREATE INDEX idx_refresh_tokens_expires_at ON refresh_tokens
    USING BRIN (expires_at);

```

```

-- =====
-- TABLE: device_tokens
-- Trusted device registrations.
-- =====

```

```

CREATE TABLE device_tokens (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
        CASCADE,
    name        VARCHAR(255) NOT NULL,
    fingerprint VARCHAR(512) NOT NULL,
    platform    VARCHAR(255),
    user_agent  TEXT,
    trusted     BOOLEAN NOT NULL DEFAULT TRUE,
    last_seen_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    last_seen_ip INET,
    revoked     BOOLEAN NOT NULL DEFAULT FALSE,
    revoked_at  TIMESTAMP WITH TIME ZONE,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```
);
```

```
CREATE UNIQUE INDEX idx_device_tokens_user_fingerprint ON
    device_tokens(user_id, fingerprint)
    WHERE revoked = FALSE;
CREATE INDEX idx_device_tokens_user_id ON device_tokens(user_id);
```

```
-- =====
-- TABLE: mfa_totp_credentials
-- TOTP (authenticator app) MFA credentials.
-- =====
```

```
CREATE TABLE mfa_totp_credentials (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
                CASCADE,
    encrypted_secret TEXT NOT NULL,
    issuer      VARCHAR(100) NOT NULL DEFAULT
                'HelixTerminator',
    algorithm   VARCHAR(20) NOT NULL DEFAULT 'SHA1',
    digits      INTEGER NOT NULL DEFAULT 6,
    period      INTEGER NOT NULL DEFAULT 30,
    enabled     BOOLEAN NOT NULL DEFAULT TRUE,
    last_used_at  TIMESTAMP WITH TIME ZONE,
    last_used_code VARCHAR(10),
    created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                NOW(),
    updated_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);
```

```
CREATE UNIQUE INDEX idx_mfa_totp_user_enabled ON
    mfa_totp_credentials(user_id)
    WHERE enabled = TRUE;
```

```
-- =====
-- TABLE: mfa_totp_backup_codes
-- One-time backup codes for TOTP recovery.
-- =====
```

```
CREATE TABLE mfa_totp_backup_codes (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
                CASCADE,
    code_hash   VARCHAR(255) NOT NULL,
    used        BOOLEAN NOT NULL DEFAULT FALSE,
    used_at     TIMESTAMP WITH TIME ZONE,
```

```

        used_ip      INET,
        created_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
    );

    CREATE INDEX idx_backup_codes_user_id ON
        mfa_totp_backup_codes(user_id)
    WHERE used = FALSE;

-- =====
-- TABLE: mfa_fido2_credentials
-- WebAuthn/FIDO2 credential registrations.
-- =====

CREATE TABLE mfa_fido2_credentials (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL REFERENCES users(id) ON
        DELETE CASCADE,
    credential_id      BYTEA NOT NULL UNIQUE,
    credential_id_b64  TEXT NOT NULL,
    name              VARCHAR(255) NOT NULL,
    public_key_cbor    BYTEA NOT NULL,
    aaguid            UUID,
    sign_count         BIGINT NOT NULL DEFAULT 0,
    transports         TEXT[] DEFAULT '{}',
    backup_eligible    BOOLEAN NOT NULL DEFAULT FALSE,
    backup_state       BOOLEAN NOT NULL DEFAULT FALSE,
    attestation_type   VARCHAR(50),
    attestation_data   JSONB,
    last_used_at       TIMESTAMP WITH TIME ZONE,
    enabled            BOOLEAN NOT NULL DEFAULT TRUE,
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    updated_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW()
);

CREATE INDEX idx_fido2_user_id ON mfa_fido2_credentials(user_id)
    WHERE enabled = TRUE;
CREATE INDEX idx_fido2_credential_id ON
    mfa_fido2_credentials(credential_id_b64);

-- =====
-- TABLE: api_keys
-- API key management.
-- =====

```

```

CREATE TABLE api_keys (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
              CASCADE,
  org_id      UUID,
  name        VARCHAR(100) NOT NULL,
  description  TEXT,
  key_hash    VARCHAR(255) NOT NULL UNIQUE,
  key_prefix  VARCHAR(20) NOT NULL,
  scopes      TEXT[] NOT NULL DEFAULT '{}',
  allowed_ips CIDR[] DEFAULT '{}',
  last_used_at TIMESTAMP WITH TIME ZONE,
  last_used_ip INET,
  revoked     BOOLEAN NOT NULL DEFAULT FALSE,
  revoked_at  TIMESTAMP WITH TIME ZONE,
  revoked_reason VARCHAR(255),
  expires_at  TIMESTAMP WITH TIME ZONE,
  created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
              NOW(),
  updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_api_keys_user_id ON api_keys(user_id) WHERE
  revoked = FALSE;

```

```

CREATE INDEX idx_api_keys_key_hash ON api_keys(key_hash);

```

```

-- =====
-- TABLE: login_history
-- Immutable login event log.
-- =====

```

```

CREATE TABLE login_history (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
              CASCADE,
  event_type   VARCHAR(50) NOT NULL
              CHECK (event_type IN (
                  'login_success', 'login_failure',
                  'logout',
                  'mfa_success', 'mfa_failure',
                  'token_refresh',
                  'api_key_used', 'sso_login',
                  'password_reset'
              )),
  ip_address   INET,
  user_agent   TEXT,
  device_id    UUID,

```

```

        session_id          UUID,
        failure_reason       VARCHAR(255),
        metadata             JSONB DEFAULT '{}',
        occurred_at          TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                               NOW()
    );

CREATE INDEX idx_login_history_user_id ON login_history(user_id);
CREATE INDEX idx_login_history_occurred_at ON login_history USING
    BRIN (occurred_at);
CREATE INDEX idx_login_history_event_type ON
    login_history(event_type, occurred_at DESC);

-- =====
-- TABLE: password_history
-- Stores hashes of previous passwords (prevents reuse).
-- =====
CREATE TABLE password_history (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
                CASCADE,
    password_hash VARCHAR(255) NOT NULL,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_password_history_user_id ON
    password_history(user_id);

-- =====
-- TABLE: sso_providers
-- Configured SSO provider integrations per organization.
-- =====
CREATE TABLE sso_providers (
    id          UUID PRIMARY KEY DEFAULT
                gen_random_uuid(),
    org_id      UUID NOT NULL,
    provider     VARCHAR(50) NOT NULL
                CHECK (provider IN ('github', 'google',
                                   'azure', 'okta', 'saml', 'oidc')),
    slug        VARCHAR(100) NOT NULL,
    display_name VARCHAR(255),
    client_id    VARCHAR(512),
    encrypted_client_secret TEXT,
    discovery_url TEXT,
    authorization_url TEXT,

```

```

token_url          TEXT,
userinfo_url       TEXT,
jwks_uri          TEXT,
scopes            TEXT[] DEFAULT ARRAY['openid', 'email',
    'profile'],
attribute_mapping  JSONB DEFAULT '{}',
enabled           BOOLEAN NOT NULL DEFAULT TRUE,
enforce_for_domain VARCHAR(255),
created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
    NOW(),
updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
    NOW()
);

CREATE UNIQUE INDEX idx_sso_providers_org_provider ON
    sso_providers(org_id, provider)
    WHERE enabled = TRUE;
CREATE INDEX idx_sso_providers_slug ON sso_providers(slug);

-- =====
-- TABLE: sso_identities
-- Links a local user to a remote SSO identity.
-- =====
CREATE TABLE sso_identities (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
        CASCADE,
    provider_id UUID NOT NULL REFERENCES sso_providers(id) ON
        DELETE CASCADE,
    subject     VARCHAR(512) NOT NULL,
    access_token TEXT,
    refresh_token TEXT,
    token_expires_at TIMESTAMP WITH TIME ZONE,
    profile_data JSONB DEFAULT '{}',
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE UNIQUE INDEX idx_sso_identities_provider_subject ON
    sso_identities(provider_id, subject);
CREATE INDEX idx_sso_identities_user_id ON
    sso_identities(user_id);

-- =====
-- TABLE: jwt_blocklist

```

```
-- Blocklisted JTIs (revoked tokens before expiry).
-- =====
CREATE TABLE jwt_blocklist (
    jti          VARCHAR(255) PRIMARY KEY,
    user_id      UUID NOT NULL,
    revoked_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    expires_at   TIMESTAMP WITH TIME ZONE NOT NULL,
    reason       VARCHAR(100)
);

CREATE INDEX idx_jwt_blocklist_expires_at ON jwt_blocklist USING
    BRIN (expires_at);
-- Periodically delete expired entries: DELETE FROM jwt_blocklist
-- WHERE expires_at < NOW();
```

17.2 vault_db

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

-- =====
-- TABLE: vaults
-- Vault containers (E2E encrypted).
-- =====
CREATE TABLE vaults (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id        UUID NOT NULL,
    owner_id      UUID NOT NULL,
    name          VARCHAR(255) NOT NULL,
    description    TEXT,
    color         VARCHAR(20),
    icon          VARCHAR(50),
    encrypted     BOOLEAN NOT NULL DEFAULT TRUE,
    sync_enabled  BOOLEAN NOT NULL DEFAULT TRUE,
    item_count    INTEGER NOT NULL DEFAULT 0,
    storage_bytes BIGINT NOT NULL DEFAULT 0,
    version       BIGINT NOT NULL DEFAULT 1,
    created_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    deleted_at    TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_vaults_org_id ON vaults(org_id) WHERE deleted_at
    IS NULL;
```

```

CREATE INDEX idx_vaults_owner_id ON vaults(owner_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_vaults_name_trgm ON vaults USING GIN (name
    gin_trgm_ops) WHERE deleted_at IS NULL;

-- =====
-- TABLE: vault_members
-- Users with access to each vault.
-- =====

CREATE TABLE vault_members (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vault_id          UUID NOT NULL REFERENCES vaults(id) ON
        DELETE CASCADE,
    user_id           UUID NOT NULL,
    permission        VARCHAR(20) NOT NULL CHECK (permission IN
        ('read', 'write', 'admin')),
    is_owner          BOOLEAN NOT NULL DEFAULT FALSE,
    invited_by        UUID,
    encrypted_vault_key BYTEA NOT NULL,
    kdf_params        JSONB NOT NULL DEFAULT '{}',
    joined_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW()
);

CREATE UNIQUE INDEX idx_vault_members_vault_user ON
    vault_members(vault_id, user_id);
CREATE INDEX idx_vault_members_user_id ON vault_members(user_id);

-- =====
-- TABLE: vault_items
-- Individual encrypted items within a vault.
-- =====

CREATE TABLE vault_items (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vault_id          UUID NOT NULL REFERENCES vaults(id) ON DELETE
        CASCADE,
    item_type         VARCHAR(50) NOT NULL
        CHECK (item_type IN (
            'host', 'ssh_key', 'password', 'note',
            'certificate', 'totp_secret',
            'api_credential', 'file'
        )),
    encrypted_data    BYTEA NOT NULL,

```



```

checksum      VARCHAR(128) NOT NULL,
iv            BYTEA NOT NULL,
version       INTEGER NOT NULL DEFAULT 1,
is_deleted    BOOLEAN NOT NULL DEFAULT FALSE,
deleted_at    TIMESTAMP WITH TIME ZONE,
created_by    UUID NOT NULL,
updated_by    UUID,
created_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
updated_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_vault_items_vault_id ON vault_items(vault_id)
    WHERE is_deleted = FALSE;
CREATE INDEX idx_vault_items_updated_at ON vault_items USING BRIN
    (updated_at);

```

```

-- =====
-- TABLE: vault_item_versions
-- Version history for vault items (for conflict resolution).
-- =====

```

```

CREATE TABLE vault_item_versions (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    item_id     UUID NOT NULL REFERENCES vault_items(id) ON
        DELETE CASCADE,
    vault_id    UUID NOT NULL,
    version     INTEGER NOT NULL,
    encrypted_data BYTEA NOT NULL,
    checksum    VARCHAR(128) NOT NULL,
    iv          BYTEA NOT NULL,
    changed_by  UUID NOT NULL,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_vault_item_versions_item_id ON
    vault_item_versions(item_id, version DESC);
CREATE INDEX idx_vault_item_versions_vault_id ON
    vault_item_versions(vault_id);

```

```

-- =====
-- TABLE: vault_sync_states
-- Per-client sync cursors for delta synchronization.
-- =====

```

```

CREATE TABLE vault_sync_states (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```

```

vault_id      UUID NOT NULL REFERENCES vaults(id) ON DELETE
               CASCADE,
user_id       UUID NOT NULL,
client_id     VARCHAR(255) NOT NULL,
cursor       TEXT NOT NULL DEFAULT '',
server_version BIGINT NOT NULL DEFAULT 0,
last_synced_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
client_platform VARCHAR(100),
app_version   VARCHAR(50)
);

```

```

CREATE UNIQUE INDEX idx_vault_sync_states_vault_client ON
    vault_sync_states(vault_id, client_id);
CREATE INDEX idx_vault_sync_states_user_id ON
    vault_sync_states(user_id);

```

```

-- =====
-- TABLE: vault_audit_events
-- Vault-level audit log (operations on vault items).
-- =====

```

```

CREATE TABLE vault_audit_events (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vault_id    UUID NOT NULL,
    item_id     UUID,
    user_id     UUID NOT NULL,
    event_type  VARCHAR(100) NOT NULL,
    ip_address  INET,
    metadata    JSONB DEFAULT '{}',
    occurred_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
) PARTITION BY RANGE (occurred_at);

```

```

CREATE INDEX idx_vault_audit_vault_id ON
    vault_audit_events(vault_id);
CREATE INDEX idx_vault_audit_user_id ON
    vault_audit_events(user_id);
CREATE INDEX idx_vault_audit_occurred_at ON vault_audit_events
    USING BRIN (occurred_at);

```

```

-- Create quarterly partitions
CREATE TABLE vault_audit_events_2026_q2 PARTITION OF
    vault_audit_events
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE vault_audit_events_2026_q3 PARTITION OF
    vault_audit_events
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

```

```
CREATE TABLE vault_audit_events_2026_q4 PARTITION OF
    vault_audit_events
    FOR VALUES FROM ('2026-10-01') TO ('2027-01-01');
```

17.3 host_db

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
```

```
-- =====
-- TABLE: hosts
-- SSH host definitions.
-- =====

CREATE TABLE hosts (
    id                UUID PRIMARY KEY DEFAULT
                      gen_random_uuid(),
    vault_id          UUID NOT NULL,
    group_id          UUID,
    org_id            UUID NOT NULL,
    created_by        UUID NOT NULL,
    name              VARCHAR(255) NOT NULL,
    hostname          VARCHAR(512) NOT NULL,
    port              INTEGER NOT NULL DEFAULT 22
                      CHECK (port BETWEEN 1 AND 65535),
    username          VARCHAR(255),
    auth_method       VARCHAR(20) NOT NULL DEFAULT 'key'
                      CHECK (auth_method IN (
                        'key', 'password',
                        'certificate',
                        'interactive', 'agent', 'pgp'
                      )),
    key_id            UUID,
    encrypted_password BYTEA,
    certificate_id     UUID,
    os                VARCHAR(50),
    os_version        VARCHAR(100),
    arch              VARCHAR(20),
    description        TEXT,
    color             VARCHAR(20),
    icon              VARCHAR(50),
    tags              TEXT[] NOT NULL DEFAULT '{}',
    jump_host_id      UUID,
    proxy_command      TEXT,
    connection_timeout_seconds INTEGER NOT NULL DEFAULT 30,
```

```

keepalive_interval_seconds    INTEGER NOT NULL DEFAULT 60,
keepalive_count_max           INTEGER NOT NULL DEFAULT 3,
server_alive_interval         INTEGER NOT NULL DEFAULT 0,
compression                   BOOLEAN NOT NULL DEFAULT FALSE,
cipher_suite                  TEXT,
macs                           TEXT,
kex_algorithms                TEXT,
host_key_algorithms           TEXT,
environment_variables         JSONB NOT NULL DEFAULT '{}',
startup_snippet_id            UUID,
status                         VARCHAR(20) NOT NULL DEFAULT
    'active'
                                CHECK (status IN ('active',
    'inactive', 'unreachable', 'archived')),
last_connected_at             TIMESTAMP WITH TIME ZONE,
last_connection_status         VARCHAR(20),
fingerprint_verified          BOOLEAN NOT NULL DEFAULT FALSE,
custom_fields                 JSONB NOT NULL DEFAULT '{}',
sort_order                    INTEGER NOT NULL DEFAULT 0,
created_at                    TIMESTAMP WITH TIME ZONE NOT NULL
    DEFAULT NOW(),
updated_at                    TIMESTAMP WITH TIME ZONE NOT NULL
    DEFAULT NOW(),
deleted_at                    TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_hosts_vault_id ON hosts(vault_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_hosts_group_id ON hosts(group_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_hosts_org_id ON hosts(org_id) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_hosts_status ON hosts(status) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_hosts_name_trgm ON hosts USING GIN (name
    gin_trgm_ops) WHERE deleted_at IS NULL;
CREATE INDEX idx_hosts_hostname_trgm ON hosts USING GIN (hostname
    gin_trgm_ops) WHERE deleted_at IS NULL;
CREATE INDEX idx_hosts_tags ON hosts USING GIN (tags);
CREATE INDEX idx_hosts_last_connected_at ON
    hosts(last_connected_at DESC NULLS LAST) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_hosts_jump_host_id ON hosts(jump_host_id) WHERE
    jump_host_id IS NOT NULL;

-- =====
-- TABLE: host_groups
-- Hierarchical host grouping.

```

```

-- =====
CREATE TABLE host_groups (
    id UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    vault_id UUID NOT NULL,
    org_id UUID NOT NULL,
    parent_id UUID REFERENCES host_groups(id)
        ON DELETE SET NULL,
    name VARCHAR(255) NOT NULL,
    description TEXT,
    color VARCHAR(20),
    icon VARCHAR(50),
    default_key_id UUID,
    default_username VARCHAR(255),
    default_port INTEGER CHECK (default_port
        BETWEEN 1 AND 65535),
    default_jump_host_id UUID,
    default_connection_timeout INTEGER,
    default_keepalive_interval INTEGER,
    inherit_from_parent BOOLEAN NOT NULL DEFAULT TRUE,
    inherit_key BOOLEAN NOT NULL DEFAULT TRUE,
    inherit_username BOOLEAN NOT NULL DEFAULT TRUE,
    inherit_port BOOLEAN NOT NULL DEFAULT FALSE,
    inherit_jump_host BOOLEAN NOT NULL DEFAULT TRUE,
    inherit_environment_variables BOOLEAN NOT NULL DEFAULT TRUE,
    inherit_startup_snippet BOOLEAN NOT NULL DEFAULT FALSE,
    sort_order INTEGER NOT NULL DEFAULT 0,
    path TEXT NOT NULL DEFAULT '',
    depth INTEGER NOT NULL DEFAULT 0,
    created_at TIMESTAMP WITH TIME ZONE NOT
        NULL DEFAULT NOW(),
    updated_at TIMESTAMP WITH TIME ZONE NOT
        NULL DEFAULT NOW(),
    deleted_at TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_host_groups_vault_id ON host_groups(vault_id)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_host_groups_parent_id ON host_groups(parent_id)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_host_groups_org_id ON host_groups(org_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_host_groups_path ON host_groups(path) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_host_groups_name_trgm ON host_groups USING GIN
    (name gin_trgm_ops) WHERE deleted_at IS NULL;

```

```

-- =====
-- TABLE: host_group_members
-- Many-to-many hosts→groups (a host can be in multiple groups).
-- =====

CREATE TABLE host_group_members (
    host_id      UUID NOT NULL REFERENCES hosts(id) ON DELETE
                  CASCADE,
    group_id     UUID NOT NULL REFERENCES host_groups(id) ON DELETE
                  CASCADE,
    added_by     UUID NOT NULL,
    added_at     TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    PRIMARY KEY (host_id, group_id)
);

CREATE INDEX idx_host_group_members_group_id ON
    host_group_members(group_id);

-- =====
-- TABLE: host_labels
-- Flexible key-value label system for hosts.
-- =====

CREATE TABLE host_labels (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    host_id      UUID NOT NULL REFERENCES hosts(id) ON DELETE CASCADE,
    key          VARCHAR(100) NOT NULL,
    value        VARCHAR(500) NOT NULL,
    created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE UNIQUE INDEX idx_host_labels_host_key ON
    host_labels(host_id, key);
CREATE INDEX idx_host_labels_key_value ON host_labels(key, value);

-- =====
-- TABLE: host_known_fingerprints
-- SSH host key fingerprints for TOFU (Trust on First Use).
-- =====

CREATE TABLE host_known_fingerprints (
    id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    host_id      UUID NOT NULL REFERENCES hosts(id) ON DELETE
                  CASCADE,
    algorithm     VARCHAR(20) NOT NULL,
    fingerprint   VARCHAR(512) NOT NULL,
    raw_key       TEXT NOT NULL,

```

```

verified          BOOLEAN NOT NULL DEFAULT FALSE,
verified_by       UUID,
verified_at       TIMESTAMP WITH TIME ZONE,
first_seen_at     TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
last_seen_at      TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
revoked           BOOLEAN NOT NULL DEFAULT FALSE,
revoked_at        TIMESTAMP WITH TIME ZONE,
revoke_reason     VARCHAR(255)
);

```

```

CREATE UNIQUE INDEX idx_known_fingerprints_host_algo ON
    host_known_fingerprints(host_id, algorithm)
    WHERE revoked = FALSE;
CREATE INDEX idx_known_fingerprints_fingerprint ON
    host_known_fingerprints(fingerprint);

```

```

-- =====
-- TABLE: host_connection_history
-- Log of every SSH connection attempt.
-- =====

```

```

CREATE TABLE host_connection_history (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    host_id     UUID NOT NULL,
    user_id     UUID NOT NULL,
    org_id      UUID NOT NULL,
    session_id  UUID,
    client_ip   INET NOT NULL,
    auth_method VARCHAR(20) NOT NULL,
    key_id      UUID,
    started_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    ended_at    TIMESTAMP WITH TIME ZONE,
    duration_seconds INTEGER,
    bytes_sent  BIGINT NOT NULL DEFAULT 0,
    bytes_received BIGINT NOT NULL DEFAULT 0,
    exit_code   INTEGER,
    disconnect_reason VARCHAR(255),
    recording_path TEXT,
    jump_chain  JSONB DEFAULT '[]',
    metadata    JSONB DEFAULT '{}')
) PARTITION BY RANGE (started_at);

```

```

CREATE INDEX idx_host_conn_history_host_id ON
    host_connection_history(host_id, started_at DESC);
CREATE INDEX idx_host_conn_history_user_id ON
    host_connection_history(user_id, started_at DESC);

```

```

CREATE INDEX idx_host_conn_history_started_at ON
    host_connection_history USING BRIN (started_at);

CREATE TABLE host_connection_history_2026_q2 PARTITION OF
    host_connection_history
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE host_connection_history_2026_q3 PARTITION OF
    host_connection_history
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');
CREATE TABLE host_connection_history_2026_q4 PARTITION OF
    host_connection_history
    FOR VALUES FROM ('2026-10-01') TO ('2027-01-01');

```

```

-- =====
-- TABLE: jump_host_chains
-- Saved multi-hop jump host configurations.
-- =====

CREATE TABLE jump_host_chains (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vault_id    UUID NOT NULL,
    org_id      UUID NOT NULL,
    name        VARCHAR(255) NOT NULL,
    description TEXT,
    hops        JSONB NOT NULL DEFAULT '[]',
    created_by  UUID NOT NULL,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_jump_chains_vault_id ON
    jump_host_chains(vault_id);

```

17.4 keychain_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

```

```

-- =====
-- TABLE: ssh_keys
-- SSH key metadata (private key stored encrypted).
-- =====

CREATE TABLE ssh_keys (
    id          UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),

```



```

vault_id          UUID NOT NULL,
org_id            UUID NOT NULL,
user_id           UUID NOT NULL,
name              VARCHAR(255) NOT NULL,
type              VARCHAR(20) NOT NULL DEFAULT 'ssh_key'
                  CHECK (type IN ('ssh_key',
                                'certificate', 'pgp', 'gpg')),
algorithm         VARCHAR(20) NOT NULL
                  CHECK (algorithm IN ('ed25519',
                                'ecdsa', 'rsa', 'dsa', 'ecdsa-sk', 'ed25519-sk')),
bits              INTEGER,
comment           VARCHAR(512),
fingerprint       VARCHAR(512) NOT NULL,
public_key_openssh TEXT NOT NULL,
encrypted_private_key BYTEA,
private_key_iv    BYTEA,
has_passphrase    BOOLEAN NOT NULL DEFAULT FALSE,
passphrase_protected BOOLEAN NOT NULL DEFAULT FALSE,
is_agent_forwarding BOOLEAN NOT NULL DEFAULT FALSE,
source            VARCHAR(20) NOT NULL DEFAULT 'generated'
                  CHECK (source IN ('generated',
                                'imported', 'agent')),
expires_at        TIMESTAMP WITH TIME ZONE,
created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                  NOW(),
updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                  NOW(),
deleted_at        TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_ssh_keys_vault_id ON ssh_keys(vault_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_ssh_keys_user_id ON ssh_keys(user_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_ssh_keys_fingerprint ON ssh_keys(fingerprint);
CREATE INDEX idx_ssh_keys_name_trgm ON ssh_keys USING GIN (name
    gin_trgm_ops) WHERE deleted_at IS NULL;

-- =====
-- TABLE: key_deployments
-- Record of public key deployments to hosts.
-- =====

CREATE TABLE key_deployments (
    id          UUID PRIMARY KEY DEFAULT
                gen_random_uuid(),
    key_id      UUID NOT NULL REFERENCES ssh_keys(id) ON
                DELETE CASCADE,

```

```

host_id          UUID NOT NULL,
host_name        VARCHAR(255) NOT NULL,
target_user      VARCHAR(255) NOT NULL,
auth_key_options JSONB DEFAULT '{}',
deployed_by      UUID NOT NULL,
deployed_at      TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                  NOW(),
revoked          BOOLEAN NOT NULL DEFAULT FALSE,
revoked_at       TIMESTAMP WITH TIME ZONE,
revoked_by       UUID,
revoke_reason    VARCHAR(255),
status           VARCHAR(20) NOT NULL DEFAULT 'active'
                  CHECK (status IN ('active', 'revoked',
                                     'expired', 'error'))
);

CREATE INDEX idx_key_deployments_key_id ON
  key_deployments(key_id);
CREATE INDEX idx_key_deployments_host_id ON
  key_deployments(host_id);

-- =====
-- TABLE: key_usage_log
-- Immutable log of key usage events.
-- =====

CREATE TABLE key_usage_log (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  key_id      UUID NOT NULL,
  host_id     UUID,
  user_id     UUID NOT NULL,
  session_id  UUID,
  event_type  VARCHAR(50) NOT NULL
              CHECK (event_type IN ('auth_success',
                                     'auth_failure', 'sign', 'deploy', 'revoke')),
  ip_address  INET,
  occurred_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
) PARTITION BY RANGE (occurred_at);

CREATE INDEX idx_key_usage_key_id ON key_usage_log(key_id);
CREATE INDEX idx_key_usage_occurred_at ON key_usage_log
  USING BRIN (occurred_at);

CREATE TABLE key_usage_log_2026_q2 PARTITION OF key_usage_log
  FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE key_usage_log_2026_q3 PARTITION OF key_usage_log

```

```

    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');
CREATE TABLE key_usage_log_2026_q4 PARTITION OF key_usage_log
    FOR VALUES FROM ('2026-10-01') TO ('2027-01-01');

-- =====
-- TABLE: certificate_store
-- SSH certificates issued or stored for use.
-- =====

CREATE TABLE certificate_store (
    id                UUID PRIMARY KEY DEFAULT
                    gen_random_uuid(),
    key_id            UUID REFERENCES ssh_keys(id) ON DELETE
                    SET NULL,
    vault_id          UUID NOT NULL,
    user_id           UUID NOT NULL,
    certificate_type   VARCHAR(20) NOT NULL CHECK
                    (certificate_type IN ('user', 'host')),
    certificate_openssh TEXT NOT NULL,
    serial            BIGINT NOT NULL,
    fingerprint       VARCHAR(512) NOT NULL,
    principals        TEXT[] NOT NULL DEFAULT '{}',
    extensions        JSONB DEFAULT '{}',
    critical_options   JSONB DEFAULT '{}',
    valid_after        TIMESTAMP WITH TIME ZONE NOT NULL,
    valid_before       TIMESTAMP WITH TIME ZONE NOT NULL,
    signed_by_ca_id    UUID,
    revoked            BOOLEAN NOT NULL DEFAULT FALSE,
    revoked_at         TIMESTAMP WITH TIME ZONE,
    revoke_reason      VARCHAR(255),
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW()
);

CREATE UNIQUE INDEX idx_cert_store_serial ON
    certificate_store(serial);
CREATE INDEX idx_cert_store_key_id ON certificate_store(key_id);
CREATE INDEX idx_cert_store_user_id ON certificate_store(user_id);
CREATE INDEX idx_cert_store_valid_before ON
    certificate_store(valid_before)
    WHERE revoked = FALSE;

```

17.5 session_db

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- =====
-- TABLE: ssh_sessions
-- SSH terminal session records.
-- =====

CREATE TABLE ssh_sessions (
  id                UUID PRIMARY KEY DEFAULT
                    gen_random_uuid(),
  user_id           UUID NOT NULL,
  host_id           UUID NOT NULL,
  vault_id          UUID NOT NULL,
  org_id            UUID NOT NULL,
  client_ip         INET NOT NULL,
  user_agent        TEXT,
  terminal_cols     SMALLINT NOT NULL DEFAULT 80,
  terminal_rows     SMALLINT NOT NULL DEFAULT 24,
  terminal_type     VARCHAR(50) NOT NULL DEFAULT
                    'xterm-256color',
  auth_method       VARCHAR(20),
  key_id            UUID,
  recording_enabled BOOLEAN NOT NULL DEFAULT FALSE,
  recording_path    TEXT,
  recording_size_bytes BIGINT DEFAULT 0,
  collab_enabled    BOOLEAN NOT NULL DEFAULT FALSE,
  read_only         BOOLEAN NOT NULL DEFAULT FALSE,
  status            VARCHAR(20) NOT NULL DEFAULT 'connecting'
                    CHECK (status IN (
                        'connecting', 'connected',
                        'disconnected',
                        'error', 'terminated'
                    )),
  reason            TEXT,
  ticket_ref        VARCHAR(255),
  startup_snippet_id UUID,
  jump_chain        JSONB DEFAULT '[]',
  exit_code         INTEGER,
  disconnect_reason TEXT,
  bytes_sent        BIGINT NOT NULL DEFAULT 0,
  bytes_received    BIGINT NOT NULL DEFAULT 0,
  commands_count    INTEGER NOT NULL DEFAULT 0,
  resize_count      INTEGER NOT NULL DEFAULT 0,
  started_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW(),
```

```

        connected_at          TIMESTAMP WITH TIME ZONE,
        ended_at              TIMESTAMP WITH TIME ZONE,
        duration_seconds       INTEGER
    ) PARTITION BY RANGE (started_at);

CREATE INDEX idx_ssh_sessions_user_id ON ssh_sessions(user_id,
    started_at DESC);
CREATE INDEX idx_ssh_sessions_host_id ON ssh_sessions(host_id,
    started_at DESC);
CREATE INDEX idx_ssh_sessions_org_id ON ssh_sessions(org_id,
    started_at DESC);
CREATE INDEX idx_ssh_sessions_status ON ssh_sessions(status,
    started_at DESC)
    WHERE status IN ('connecting', 'connected');
CREATE INDEX idx_ssh_sessions_started_at ON ssh_sessions USING
    BRIN (started_at);

CREATE TABLE ssh_sessions_2026_q2 PARTITION OF ssh_sessions
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE ssh_sessions_2026_q3 PARTITION OF ssh_sessions
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');
CREATE TABLE ssh_sessions_2026_q4 PARTITION OF ssh_sessions
    FOR VALUES FROM ('2026-10-01') TO ('2027-01-01');

-- =====
-- TABLE: session_events
-- Per-event recording of terminal I/O (asciinema-compatible).
-- =====

CREATE TABLE session_events (
    id          BIGSERIAL,
    session_id  UUID NOT NULL,
    occurred_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    elapsed_ms  BIGINT NOT NULL,
    direction   CHAR(1) NOT NULL CHECK (direction IN ('i', 'o')),
    data        BYTEA NOT NULL,
    event_type  VARCHAR(20) NOT NULL DEFAULT 'data'
                CHECK (event_type IN ('data', 'resize',
                    'marker', 'metadata'))
) PARTITION BY RANGE (occurred_at);

CREATE INDEX idx_session_events_session_id ON
    session_events(session_id, occurred_at);
CREATE INDEX idx_session_events_occurred_at ON session_events
    USING BRIN (occurred_at);

CREATE TABLE session_events_2026_q2 PARTITION OF session_events

```

```

    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE session_events_2026_q3 PARTITION OF session_events
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

-- =====
-- TABLE: session_recordings
-- Metadata for session recording files (asciinema v2).
-- =====
CREATE TABLE session_recordings (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id        UUID NOT NULL,
    storage_path       TEXT NOT NULL,
    storage_backend    VARCHAR(20) NOT NULL DEFAULT 's3'
                        CHECK (storage_backend IN ('s3', 'gcs',
                        'azure_blob', 'local')),
    file_size_bytes    BIGINT NOT NULL DEFAULT 0,
    duration_seconds   INTEGER,
    format             VARCHAR(20) NOT NULL DEFAULT 'asciicast_v2',
    terminal_cols       SMALLINT NOT NULL,
    terminal_rows       SMALLINT NOT NULL,
    checksum_sha256     VARCHAR(64),
    compressed         BOOLEAN NOT NULL DEFAULT TRUE,
    encryption_key_id  UUID,
    processed          BOOLEAN NOT NULL DEFAULT FALSE,
    processed_at        TIMESTAMP WITH TIME ZONE,
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                        NOW()
);

CREATE UNIQUE INDEX idx_session_recordings_session_id ON
    session_recordings(session_id);
CREATE INDEX idx_session_recordings_created_at ON
    session_recordings USING BRIN (created_at);

-- =====
-- TABLE: sftp_sessions
-- SFTP session records.
-- =====
CREATE TABLE sftp_sessions (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL,
    host_id           UUID NOT NULL,
    org_id            UUID NOT NULL,
    client_ip         INET NOT NULL,

```

```

cwd                TEXT NOT NULL DEFAULT '/',
transfer_mode      VARCHAR(10) NOT NULL DEFAULT 'binary'
                  CHECK (transfer_mode IN ('binary',
                  'ascii')),
server_version     VARCHAR(100),
status             VARCHAR(20) NOT NULL DEFAULT 'connected'
                  CHECK (status IN ('connected', 'closed',
                  'error', 'expired')),
files_uploaded     INTEGER NOT NULL DEFAULT 0,
files_downloaded   INTEGER NOT NULL DEFAULT 0,
bytes_uploaded     BIGINT NOT NULL DEFAULT 0,
bytes_downloaded   BIGINT NOT NULL DEFAULT 0,
started_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                  NOW(),
ended_at           TIMESTAMP WITH TIME ZONE,
expires_at        TIMESTAMP WITH TIME ZONE NOT NULL
);

CREATE INDEX idx_sftp_sessions_user_id ON sftp_sessions(user_id);
CREATE INDEX idx_sftp_sessions_host_id ON sftp_sessions(host_id);
CREATE INDEX idx_sftp_sessions_started_at ON sftp_sessions USING
BRIN (started_at);

-- =====
-- TABLE: sftp_transfers
-- Individual SFTP file transfer records.
-- =====
CREATE TABLE sftp_transfers (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  sftp_session_id   UUID NOT NULL REFERENCES sftp_sessions(id) ON
                  DELETE CASCADE,
  user_id           UUID NOT NULL,
  host_id           UUID NOT NULL,
  direction         VARCHAR(10) NOT NULL CHECK (direction IN
                  ('upload', 'download')),
  local_filename    VARCHAR(1024),
  remote_path       TEXT NOT NULL,
  file_size_bytes   BIGINT NOT NULL DEFAULT 0,
  bytes_transferred BIGINT NOT NULL DEFAULT 0,
  checksum_sha256   VARCHAR(64),
  status            VARCHAR(20) NOT NULL DEFAULT 'completed'
                  CHECK (status IN ('pending', 'in_progress',
                  'completed', 'failed', 'cancelled')),
  error_message     TEXT,
  duration_ms       INTEGER,

```

```

        transferred_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                           NOW()
    ) PARTITION BY RANGE (transferred_at);

CREATE INDEX idx_sftp_transfers_session_id ON
    sftp_transfers(sftp_session_id);
CREATE INDEX idx_sftp_transfers_user_id
    ON sftp_transfers(user_id, transferred_at DESC);
CREATE INDEX idx_sftp_transfers_transferred_at ON sftp_transfers
    USING BRIN (transferred_at);

CREATE TABLE sftp_transfers_2026_q2 PARTITION OF sftp_transfers
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE sftp_transfers_2026_q3 PARTITION OF sftp_transfers
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

-- =====
-- TABLE: port_forward_rules
-- Port forwarding rule definitions.
-- =====

CREATE TABLE port_forward_rules (
    id                UUID PRIMARY KEY DEFAULT
                      gen_random_uuid(),
    user_id           UUID NOT NULL,
    host_id           UUID NOT NULL,
    vault_id          UUID NOT NULL,
    org_id            UUID NOT NULL,
    name              VARCHAR(255) NOT NULL,
    description        TEXT,
    type              VARCHAR(20) NOT NULL
                      CHECK (type IN ('local', 'remote',
                                      'dynamic')),
    local_address      VARCHAR(255) NOT NULL DEFAULT '127.0.0.1',
    local_port         INTEGER NOT NULL CHECK (local_port
                      BETWEEN 1 AND 65535),
    remote_address     VARCHAR(255),
    remote_port        INTEGER CHECK (remote_port BETWEEN 1 AND
                      65535),
    bind_address       VARCHAR(255),
    auto_start         BOOLEAN NOT NULL DEFAULT FALSE,
    status             VARCHAR(20) NOT NULL DEFAULT 'inactive'
                      CHECK (status IN ('active', 'inactive',
                                      'error')),
    sort_order         INTEGER NOT NULL DEFAULT 0,
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                      NOW(),

```



```

        updated_at          TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                             NOW(),
        deleted_at          TIMESTAMP WITH TIME ZONE
    );

CREATE INDEX idx_pf_rules_user_id ON port_forward_rules(user_id)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_pf_rules_host_id ON port_forward_rules(host_id)
    WHERE deleted_at IS NULL;

-- =====
-- TABLE: port_forward_connections
-- Active and historical port forwarding connections.
-- =====

CREATE TABLE port_forward_connections (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    rule_id           UUID NOT NULL REFERENCES port_forward_rules(id)
                     ON DELETE CASCADE,
    user_id           UUID NOT NULL,
    host_id           UUID NOT NULL,
    ssh_session_id    UUID,
    status             VARCHAR(20) NOT NULL DEFAULT 'active'
                     CHECK (status IN ('active', 'closed',
                                     'error')),
    bytes_sent        BIGINT NOT NULL DEFAULT 0,
    bytes_received    BIGINT NOT NULL DEFAULT 0,
    started_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    ended_at          TIMESTAMP WITH TIME ZONE,
    error_message      TEXT
);

CREATE INDEX idx_pf_connections_rule_id ON
    port_forward_connections(rule_id);
CREATE INDEX idx_pf_connections_started_at ON
    port_forward_connections USING BRIN (started_at);

```

17.6 snippet_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

-- =====
-- TABLE: snippet_categories
-- Hierarchical snippet categorization.
-- =====

```

```

CREATE TABLE snippet_categories (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  vault_id    UUID NOT NULL,
  org_id      UUID NOT NULL,
  parent_id   UUID REFERENCES snippet_categories(id) ON DELETE
              SET NULL,
  name        VARCHAR(255) NOT NULL,
  description TEXT,
  color       VARCHAR(20),
  icon        VARCHAR(50),
  sort_order  INTEGER NOT NULL DEFAULT 0,
  created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
  updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_snippet_categories_vault_id ON
  snippet_categories(vault_id);
CREATE INDEX idx_snippet_categories_parent_id ON
  snippet_categories(parent_id);

```

```

-- =====
-- TABLE: snippets
-- Command snippets / scripts.
-- =====

```

```

CREATE TABLE snippets (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  vault_id    UUID NOT NULL,
  org_id      UUID NOT NULL,
  created_by  UUID NOT NULL,
  category_id UUID REFERENCES snippet_categories(id) ON
              DELETE SET NULL,
  name        VARCHAR(255) NOT NULL,
  description TEXT,
  content     TEXT NOT NULL,
  language    VARCHAR(50) NOT NULL DEFAULT 'bash',
  interpreter VARCHAR(255),
  shebang     VARCHAR(255),
  tags        TEXT[] NOT NULL DEFAULT '{}',
  parameters  JSONB NOT NULL DEFAULT '[]',
  shared      BOOLEAN NOT NULL DEFAULT FALSE,
  pinned      BOOLEAN NOT NULL DEFAULT FALSE,
  executions_count BIGINT NOT NULL DEFAULT 0,
  last_executed_at  TIMESTAMP WITH TIME ZONE,
  fts_vector   TSVECTOR,

```

```

        created_at          TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                               NOW(),
        updated_at          TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                               NOW(),
        deleted_at          TIMESTAMP WITH TIME ZONE
    );

    CREATE INDEX idx_snippets_vault_id ON snippets(vault_id) WHERE
        deleted_at IS NULL;
    CREATE INDEX idx_snippets_category_id ON snippets(category_id)
        WHERE deleted_at IS NULL;
    CREATE INDEX idx_snippets_tags ON snippets USING GIN (tags) WHERE
        deleted_at IS NULL;
    CREATE INDEX idx_snippets_name_trgm ON snippets USING GIN (name
        gin_trgm_ops) WHERE deleted_at IS NULL;
    CREATE INDEX idx_snippets_content_trgm ON snippets USING GIN
        (content gin_trgm_ops) WHERE deleted_at IS NULL;
    CREATE INDEX idx_snippets_fts ON snippets USING GIN (fts_vector)
        WHERE deleted_at IS NULL;

-- Trigger to maintain FTS vector
CREATE OR REPLACE FUNCTION snippets_fts_update()
RETURNS TRIGGER AS $$
BEGIN
    NEW.fts_vector :=
        setweight(to_tsvector('english', coalesce(NEW.name, '')),
            'A') ||
        setweight(to_tsvector('english', coalesce(NEW.description,
            '')), 'B') ||
        setweight(to_tsvector('english', coalesce(NEW.content, '')),
            'C');
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_snippets_fts
    BEFORE INSERT OR UPDATE ON snippets
    FOR EACH ROW EXECUTE FUNCTION snippets_fts_update();

-- =====
-- TABLE: snippet_executions
-- Log of snippet executions.
-- =====
CREATE TABLE snippet_executions (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    snippet_id        UUID NOT NULL,
    user_id           UUID NOT NULL,

```

```

org_id          UUID NOT NULL,
execution_mode   VARCHAR(20) NOT NULL DEFAULT 'parallel'
                CHECK (execution_mode IN ('parallel',
                'sequential')),
host_count       INTEGER NOT NULL DEFAULT 0,
parameters       JSONB DEFAULT '{}',
status           VARCHAR(20) NOT NULL DEFAULT 'pending'
                CHECK (status IN ('pending', 'running',
                'completed', 'failed', 'cancelled')),
completed_count  INTEGER NOT NULL DEFAULT 0,
failed_count     INTEGER NOT NULL DEFAULT 0,
timeout_seconds  INTEGER NOT NULL DEFAULT 60,
started_at       TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                NOW(),
completed_at     TIMESTAMP WITH TIME ZONE
) PARTITION BY RANGE (started_at);

```

```

CREATE INDEX idx_snippet_executions_snippet_id ON
    snippet_executions(snippet_id);
CREATE INDEX idx_snippet_executions_user_id ON
    snippet_executions(user_id, started_at DESC);
CREATE INDEX idx_snippet_executions_started_at ON
    snippet_executions USING BRIN (started_at);

```

```

CREATE TABLE snippet_executions_2026_q2 PARTITION OF
    snippet_executions
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE snippet_executions_2026_q3 PARTITION OF
    snippet_executions
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

```

```

-- =====
-- TABLE: snippet_execution_results
-- Per-host results for each execution.
-- =====

CREATE TABLE snippet_execution_results (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    execution_id  UUID NOT NULL,
    host_id      UUID NOT NULL,
    host_name     VARCHAR(255) NOT NULL,
    session_id   UUID,
    status        VARCHAR(20) NOT NULL DEFAULT 'pending'
                CHECK (status IN ('pending', 'running',
                'success', 'failure', 'timeout')),
    exit_code     INTEGER,
    stdout        TEXT,

```

```

    stderr          TEXT,
    error_message    TEXT,
    started_at       TIMESTAMP WITH TIME ZONE,
    completed_at     TIMESTAMP WITH TIME ZONE,
    duration_ms      INTEGER
);

CREATE INDEX idx_exec_results_execution_id ON
    snippet_execution_results(execution_id);

```

17.7 workspace_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- =====
-- TABLE: workspaces
-- Saved terminal layout configurations.
-- =====

CREATE TABLE workspaces (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL,
    org_id            UUID NOT NULL,
    name              VARCHAR(255) NOT NULL,
    description        TEXT,
    layout            JSONB NOT NULL DEFAULT '{}',
    thumbnail_url     TEXT,
    is_template        BOOLEAN NOT NULL DEFAULT FALSE,
    template_id       UUID,
    pinned            BOOLEAN NOT NULL DEFAULT FALSE,
    auto_connect       BOOLEAN NOT NULL DEFAULT TRUE,
    last_opened_at    TIMESTAMP WITH TIME ZONE,
    open_count        INTEGER NOT NULL DEFAULT 0,
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    deleted_at        TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_workspaces_user_id ON workspaces(user_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_workspaces_org_id ON workspaces(org_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_workspaces_layout ON workspaces USING GIN
    (layout) WHERE deleted_at IS NULL;

```

```

-- =====
-- TABLE: workspace_snapshots
-- Point-in-time snapshots of workspace layouts.
-- =====

CREATE TABLE workspace_snapshots (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  workspace_id UUID NOT NULL REFERENCES workspaces(id) ON DELETE
    CASCADE,
  layout      JSONB NOT NULL,
  snapshot_name VARCHAR(255),
  created_by  UUID NOT NULL,
  created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_workspace_snapshots_workspace_id ON
  workspace_snapshots(workspace_id, created_at DESC);

-- =====
-- TABLE: workspace_sessions
-- Maps workspace to the sessions opened within it.
-- =====

CREATE TABLE workspace_sessions (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  workspace_id UUID NOT NULL REFERENCES workspaces(id) ON DELETE
    CASCADE,
  session_id  UUID NOT NULL,
  pane_id     VARCHAR(50) NOT NULL,
  user_id     UUID NOT NULL,
  created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_workspace_sessions_workspace_id ON
  workspace_sessions(workspace_id);
CREATE INDEX idx_workspace_sessions_session_id ON
  workspace_sessions(session_id);

-- =====
-- TABLE: workspace_templates
-- Reusable workspace layout templates.
-- =====

CREATE TABLE workspace_templates (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  created_by  UUID NOT NULL,
  org_id      UUID,

```

```

name          VARCHAR(255) NOT NULL,
description   TEXT,
category      VARCHAR(100),
layout        JSONB NOT NULL DEFAULT '{}',
pane_count    SMALLINT NOT NULL DEFAULT 1,
preview_url   TEXT,
public        BOOLEAN NOT NULL DEFAULT FALSE,
usage_count   INTEGER NOT NULL DEFAULT 0,
created_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
updated_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_workspace_templates_org_id ON
    workspace_templates(org_id);
CREATE INDEX idx_workspace_templates_public ON
    workspace_templates(public, usage_count DESC)
    WHERE public = TRUE;

```

17.8 collab_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";

```

```

-- =====
-- TABLE: collaboration_sessions
-- Collaborative terminal session metadata.
-- =====

```

```

CREATE TABLE collaboration_sessions (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    ssh_session_id  UUID NOT NULL UNIQUE,
    org_id       UUID NOT NULL,
    owner_id     UUID NOT NULL,
    title        VARCHAR(255),
    max_participants SMALLINT NOT NULL DEFAULT 10,
    allow_input   BOOLEAN NOT NULL DEFAULT FALSE,
    status       VARCHAR(20) NOT NULL DEFAULT 'active'
                CHECK (status IN ('active', 'ended')),
    started_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW(),
    ended_at      TIMESTAMP WITH TIME ZONE
);

```

```

CREATE INDEX idx_collab_sessions_ssh_session ON
    collaboration_sessions(ssh_session_id);
CREATE INDEX idx_collab_sessions_org_id ON
    collaboration_sessions(org_id, started_at DESC);

```

```

-- =====
-- TABLE: collaboration_participants
-- Participants in a collaboration session.
-- =====

CREATE TABLE collaboration_participants (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    collab_id     UUID NOT NULL REFERENCES
        collaboration_sessions(id) ON DELETE CASCADE,
    user_id       UUID NOT NULL,
    display_name  VARCHAR(100) NOT NULL,
    role          VARCHAR(20) NOT NULL DEFAULT 'viewer'
        CHECK (role IN ('owner', 'contributor',
            'viewer')),
    cursor_color  VARCHAR(20),
    connected_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    disconnected_at TIMESTAMP WITH TIME ZONE,
    is_active     BOOLEAN NOT NULL DEFAULT TRUE
);

CREATE INDEX idx_collab_participants_collab_id ON
    collaboration_participants(collab_id)
    WHERE is_active = TRUE;
CREATE INDEX idx_collab_participants_user_id ON
    collaboration_participants(user_id);

-- =====
-- TABLE: collaboration_events
-- Event log for collaboration sessions (chat, cursor, control).
-- =====

CREATE TABLE collaboration_events (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    collab_id     UUID NOT NULL,
    user_id       UUID NOT NULL,
    event_type     VARCHAR(50) NOT NULL
        CHECK (event_type IN ('chat', 'cursor', 'join',
            'leave', 'control_request', 'control_granted')),
    payload        JSONB DEFAULT '{}',
    occurred_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
) PARTITION BY RANGE (occurred_at);

CREATE INDEX idx_collab_events_collab_id ON
    collaboration_events(collab_id, occurred_at);
CREATE INDEX idx_collab_events_occurred_at ON
    collaboration_events USING BRIN (occurred_at);

```



```

CREATE TABLE collaboration_events_2026_q2 PARTITION OF
    collaboration_events
    FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');
CREATE TABLE collaboration_events_2026_q3 PARTITION OF
    collaboration_events
    FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

```

17.9 org_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "citext";

```

```

-- =====
-- TABLE: organizations
-- Organizations (tenants).
-- =====

CREATE TABLE organizations (
    id                                UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    name                             VARCHAR(255) NOT NULL,
    slug                             CITEXT NOT NULL UNIQUE,
    domain                           CITEXT,
    domain_verified                   BOOLEAN NOT NULL DEFAULT FALSE,
    domain_verified_at               TIMESTAMP WITH TIME ZONE,
    domain_verification_token         VARCHAR(255),
    plan                              VARCHAR(50) NOT NULL DEFAULT
        'free'
        CHECK (plan IN ('free',
            'pro', 'team', 'enterprise')),
    plan_expires_at                  TIMESTAMP WITH TIME ZONE,
    max_members                       INTEGER NOT NULL DEFAULT 5,
    max_vaults                        INTEGER NOT NULL DEFAULT 3,
    max_hosts                         INTEGER NOT NULL DEFAULT 50,
    max_sessions_concurrent           INTEGER NOT NULL DEFAULT 5,
    enforce_mfa                       BOOLEAN NOT NULL DEFAULT FALSE,
    session_recording_required        BOOLEAN NOT NULL DEFAULT FALSE,
    session_recording_retention_days  INTEGER NOT NULL DEFAULT 90,
    ip_allowlist                      CIDR[] DEFAULT '{}',
    sso_required                      BOOLEAN NOT NULL DEFAULT FALSE,
    audit_log_retention_days           INTEGER NOT NULL DEFAULT 365,
    owner_id                          UUID NOT NULL,
    billing_email                     CITEXT,
    stripe_customer_id                VARCHAR(255),

```

```

status          VARCHAR(20) NOT NULL DEFAULT
                'active'
                CHECK (status IN ('active',
                'suspended', 'trial', 'cancelled')),
settings        JSONB NOT NULL DEFAULT '{}',
created_at      TIMESTAMP WITH TIME ZONE NOT
                NULL DEFAULT NOW(),
updated_at      TIMESTAMP WITH TIME ZONE NOT
                NULL DEFAULT NOW(),
deleted_at      TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_organizations_slug ON organizations(slug) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_organizations_domain ON organizations(domain)
    WHERE deleted_at IS NULL AND domain IS NOT NULL;
CREATE INDEX idx_organizations_owner_id ON
    organizations(owner_id);

-- =====
-- TABLE: org_members
-- Members of organizations.
-- =====

CREATE TABLE org_members (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL REFERENCES organizations(id) ON
                DELETE CASCADE,
    user_id     UUID NOT NULL,
    -- Canonical 6-role vocabulary (CD-8; single RBAC schema,
    -- reconciles the prior
    -- owner/admin/member/viewer/billing set). 'billing' is a
    -- permission, not a
    -- role – granted via `roles` / `role_assignments`, not this
    -- column.
    role        VARCHAR(20) NOT NULL DEFAULT 'member'
                CHECK (role IN ('super_admin', 'org_admin',
                'team_admin', 'member', 'auditor', 'api_user')),
    invited_by  UUID,
    joined_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    status      VARCHAR(20) NOT NULL DEFAULT 'active'
                CHECK (status IN ('active', 'suspended',
                'pending')),
    last_active_at TIMESTAMP WITH TIME ZONE
);

CREATE UNIQUE INDEX idx_org_members_org_user ON
    org_members(org_id, user_id)
    WHERE status != 'suspended';

```

```

CREATE INDEX idx_org_members_user_id ON org_members(user_id);
CREATE INDEX idx_org_members_role ON org_members(org_id, role);

-- =====
-- TABLE: teams
-- Teams within an organization.
-- =====

CREATE TABLE teams (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id      UUID NOT NULL REFERENCES organizations(id) ON
              DELETE CASCADE,
  name        VARCHAR(255) NOT NULL,
  slug        CITEXT NOT NULL,
  description  TEXT,
  settings    JSONB DEFAULT '{}',
  created_by  UUID NOT NULL,
  created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
  updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
  deleted_at  TIMESTAMP WITH TIME ZONE
);

CREATE UNIQUE INDEX idx_teams_org_slug ON teams(org_id, slug)
  WHERE deleted_at IS NULL;
CREATE INDEX idx_teams_org_id ON teams(org_id) WHERE deleted_at
  IS NULL;

-- =====
-- TABLE: team_members
-- Members of teams.
-- =====

CREATE TABLE team_members (
  team_id     UUID NOT NULL REFERENCES teams(id) ON DELETE
              CASCADE,
  user_id     UUID NOT NULL,
  -- Canonical 6-role vocabulary (CD-8): 'lead' reconciled to
  -- 'team_admin'.
  role        VARCHAR(20) NOT NULL DEFAULT 'member'
              CHECK (role IN ('team_admin', 'member')),
  added_by    UUID NOT NULL,
  added_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
  PRIMARY KEY (team_id, user_id)
);

CREATE INDEX idx_team_members_user_id ON team_members(user_id);

```

```

-- =====
-- TABLE: roles
-- Custom RBAC roles.
-- =====

CREATE TABLE roles (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL REFERENCES organizations(id) ON
        DELETE CASCADE,
    name        VARCHAR(100) NOT NULL,
    description  TEXT,
    is_system    BOOLEAN NOT NULL DEFAULT FALSE,
    permissions  JSONB NOT NULL DEFAULT '[]',
    created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE UNIQUE INDEX idx_roles_org_name ON roles(org_id, name);
CREATE INDEX idx_roles_permissions ON roles USING GIN
    (permissions);

-- =====
-- TABLE: role_assignments
-- Assigns roles to users within an org (for custom RBAC).
-- =====

CREATE TABLE role_assignments (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL,
    user_id     UUID NOT NULL,
    role_id     UUID NOT NULL REFERENCES roles(id) ON DELETE
        CASCADE,
    resource_type VARCHAR(50),
    resource_id  UUID,
    granted_by   UUID NOT NULL,
    granted_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    expires_at   TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_role_assignments_org_user ON
    role_assignments(org_id, user_id);
CREATE INDEX idx_role_assignments_resource ON
    role_assignments(resource_type, resource_id)
    WHERE resource_type IS NOT NULL;

```

```

-- =====
-- TABLE: invitations
-- Pending organization invitations.
-- =====

CREATE TABLE invitations (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL REFERENCES organizations(id) ON
                DELETE CASCADE,
    email       CITEXT NOT NULL,
    role        VARCHAR(20) NOT NULL DEFAULT 'member',
    team_ids    UUID[] DEFAULT '{}',
    invited_by  UUID NOT NULL,
    invitation_token VARCHAR(255) NOT NULL UNIQUE,
    message     TEXT,
    status      VARCHAR(20) NOT NULL DEFAULT 'pending'
                CHECK (status IN ('pending', 'accepted',
                                'declined', 'expired', 'cancelled')),
    accepted_at TIMESTAMP WITH TIME ZONE,
    declined_at TIMESTAMP WITH TIME ZONE,
    expires_at  TIMESTAMP WITH TIME ZONE NOT NULL,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_invitations_org_email ON invitations(org_id,
    email)
    WHERE status = 'pending';
CREATE INDEX idx_invitations_token ON
    invitations(invitation_token);
CREATE INDEX idx_invitations_expires_at ON invitations(expires_at)
    WHERE status = 'pending';

```

17.10 audit_db

DEFERRED (next increment): The hash/prev_hash chain below is an in-table integrity check only — it has no external/independent anchoring (e.g. S3 Object Lock compliance mode, HSM signature, or off-DB notarization). A DB principal with write access to audit_db can rewrite the chain, including the genesis row, undetected. Do not read “Immutable” in the comment below as tamper-proof; it is tamper-evident against non-privileged writers only. External WORM anchoring is deferred.

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
```

```

-- =====
-- TABLE: audit_events
-- Immutable, cryptographically-chained audit log.
-- Partitioned by month for efficient time-range queries.
-- =====

CREATE TABLE audit_events (
    id                UUID NOT NULL DEFAULT gen_random_uuid(),
    seq               BIGSERIAL,
    org_id            UUID NOT NULL,
    event_type        VARCHAR(100) NOT NULL,
    user_id           UUID,
    resource_type     VARCHAR(50),
    resource_id       UUID,
    resource_name     VARCHAR(512),
    outcome           VARCHAR(20) NOT NULL DEFAULT 'success'
                    CHECK (outcome IN ('success', 'failure',
                    'partial')),
    ip_address        INET,
    user_agent        TEXT,
    session_id        UUID,
    source_service    VARCHAR(50) NOT NULL,
    metadata          JSONB NOT NULL DEFAULT '{}',
    hash              VARCHAR(128) NOT NULL,
    prev_hash         VARCHAR(128),
    occurred_at       TIMESTAMP WITH TIME ZONE NOT NULL,
    recorded_at       TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    PRIMARY KEY (id, occurred_at)
) PARTITION BY RANGE (occurred_at);

CREATE INDEX idx_audit_events_org_id ON audit_events(org_id,
    occurred_at DESC);
CREATE INDEX idx_audit_events_user_id ON audit_events(user_id,
    occurred_at DESC)
    WHERE user_id IS NOT NULL;
CREATE INDEX idx_audit_events_event_type ON
    audit_events(event_type, occurred_at DESC);
CREATE INDEX idx_audit_events_resource ON
    audit_events(resource_type, resource_id, occurred_at DESC)
    WHERE resource_type IS NOT NULL;
CREATE INDEX idx_audit_events_occurred_at ON audit_events USING
    BRIN (occurred_at);
CREATE INDEX idx_audit_events_metadata ON audit_events USING GIN
    (metadata);

CREATE TABLE audit_events_2026_01 PARTITION OF audit_events
    FOR VALUES FROM ('2026-01-01') TO ('2026-02-01');
CREATE TABLE audit_events_2026_02 PARTITION OF audit_events

```

```

    FOR VALUES FROM ('2026-02-01') TO ('2026-03-01');
CREATE TABLE audit_events_2026_03 PARTITION OF audit_events
    FOR VALUES FROM ('2026-03-01') TO ('2026-04-01');
CREATE TABLE audit_events_2026_04 PARTITION OF audit_events
    FOR VALUES FROM ('2026-04-01') TO ('2026-05-01');
CREATE TABLE audit_events_2026_05 PARTITION OF audit_events
    FOR VALUES FROM ('2026-05-01') TO ('2026-06-01');
CREATE TABLE audit_events_2026_06 PARTITION OF audit_events
    FOR VALUES FROM ('2026-06-01') TO ('2026-07-01');
CREATE TABLE audit_events_2026_07 PARTITION OF audit_events
    FOR VALUES FROM ('2026-07-01') TO ('2026-08-01');
CREATE TABLE audit_events_2026_08 PARTITION OF audit_events
    FOR VALUES FROM ('2026-08-01') TO ('2026-09-01');
CREATE TABLE audit_events_2026_09 PARTITION OF audit_events
    FOR VALUES FROM ('2026-09-01') TO ('2026-10-01');
CREATE TABLE audit_events_2026_10 PARTITION OF audit_events
    FOR VALUES FROM ('2026-10-01') TO ('2026-11-01');
CREATE TABLE audit_events_2026_11 PARTITION OF audit_events
    FOR VALUES FROM ('2026-11-01') TO ('2026-12-01');
CREATE TABLE audit_events_2026_12 PARTITION OF audit_events
    FOR VALUES FROM ('2026-12-01') TO ('2027-01-01');

-- =====
-- TABLE: audit_event_hash_chain
-- Tracks the chain head for tamper detection.
-- =====
CREATE TABLE audit_event_hash_chain (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL UNIQUE,
    last_event_id UUID NOT NULL,
    last_hash   VARCHAR(128) NOT NULL,
    chain_length BIGINT NOT NULL DEFAULT 0,
    updated_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

-- =====
-- TABLE: audit_exports
-- Records of audit log export jobs.
-- =====
CREATE TABLE audit_exports (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL,

```

```

requested_by      UUID NOT NULL,
format            VARCHAR(20) NOT NULL CHECK (format IN
                ('json', 'csv', 'syslog')),
filter_json       JSONB NOT NULL DEFAULT '{}',
status            VARCHAR(20) NOT NULL DEFAULT 'pending'
                CHECK (status IN ('pending', 'processing',
                'completed', 'failed')),
event_count       BIGINT DEFAULT 0,
file_size_bytes   BIGINT DEFAULT 0,
storage_path      TEXT,
download_token     VARCHAR(255),
download_expires_at TIMESTAMP WITH TIME ZONE,
error_message     TEXT,
created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                NOW(),
completed_at      TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_audit_exports_org_id ON audit_exports(org_id,
                created_at DESC);

```

17.11 pki_db

```

CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- =====
-- TABLE: ca_keys
-- Certificate Authority key configurations.
-- =====

CREATE TABLE ca_keys (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id        UUID NOT NULL,
    ca_type       VARCHAR(20) NOT NULL CHECK (ca_type IN ('user',
                'host')),
    version       INTEGER NOT NULL DEFAULT 1,
    public_key_openssh TEXT NOT NULL,
    fingerprint   VARCHAR(512) NOT NULL,
    algorithm     VARCHAR(20) NOT NULL DEFAULT 'ed25519',
    encrypted_private_key_ref TEXT NOT NULL,
    key_manager    VARCHAR(20) NOT NULL DEFAULT 'vault'
                CHECK (key_manager IN ('vault', 'aws_kms',
                'gcp_kms', 'hsm')),
    active        BOOLEAN NOT NULL DEFAULT TRUE,
    last_used_at  TIMESTAMP WITH TIME ZONE,
    rotated_at    TIMESTAMP WITH TIME ZONE,

```



```
        created_at      TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
    );
```

```
CREATE UNIQUE INDEX idx_ca_keys_org_type_active
    ON ca_keys(org_id, ca_type)
    WHERE active = TRUE;
```

```
-- =====
-- TABLE: certificates
-- Issued SSH certificates.
-- =====
```

```
CREATE TABLE certificates (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id            UUID NOT NULL,
    ca_key_id         UUID NOT NULL REFERENCES ca_keys(id),
    cert_type         VARCHAR(10) NOT NULL CHECK (cert_type IN
        ('user', 'host')),
    entity_type       VARCHAR(20) NOT NULL,
    entity_id         UUID NOT NULL,
    serial            BIGINT NOT NULL,
    fingerprint       VARCHAR(512) NOT NULL,
    certificate_openssh TEXT NOT NULL,
    principals        TEXT[] NOT NULL DEFAULT '{}',
    extensions        JSONB DEFAULT '{}',
    critical_options  JSONB DEFAULT '{}',
    valid_after       TIMESTAMP WITH TIME ZONE NOT NULL,
    valid_before      TIMESTAMP WITH TIME ZONE NOT NULL,
    revoked           BOOLEAN NOT NULL DEFAULT FALSE,
    revoked_at        TIMESTAMP WITH TIME ZONE,
    revoke_reason     VARCHAR(255),
    revoked_by        UUID,
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);
```

```
CREATE UNIQUE INDEX idx_certificates_serial ON
    certificates(org_id, serial);
CREATE INDEX idx_certificates_entity ON certificates(entity_type,
    entity_id);
CREATE INDEX idx_certificates_valid_before ON
    certificates(valid_before)
    WHERE revoked = FALSE;
CREATE INDEX idx_certificates_fingerprint ON
    certificates(fingerprint);
```

```

-- =====
-- TABLE: certificate_revocations
-- Revocation records with reason codes.
-- =====

CREATE TABLE certificate_revocations (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    certificate_id UUID NOT NULL REFERENCES certificates(id) ON
        DELETE CASCADE,
    org_id      UUID NOT NULL,
    serial      BIGINT NOT NULL,
    reason      VARCHAR(50) NOT NULL
        CHECK (reason IN (
            'unspecified', 'key_compromise',
            'ca_compromise',
            'affiliation_changed', 'superseded',
            'cessation',
            'certificate_hold', 'remove_from_crl',
            'privilege_withdrawn',
            'aa_compromise'
        )),
    revoked_by  UUID NOT NULL,
    revoked_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

CREATE INDEX idx_cert_revocations_org_id ON
    certificate_revocations(org_id);
CREATE INDEX idx_cert_revocations_serial ON
    certificate_revocations(serial);

-- =====
-- TABLE: crt_entries
-- Certificate Revocation List (CRL / KRL) entries for fast
-- lookup.
-- =====

CREATE TABLE crt_entries (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id      UUID NOT NULL,
    ca_key_id   UUID NOT NULL REFERENCES ca_keys(id),
    serial      BIGINT NOT NULL,
    revoked_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    reason      VARCHAR(50) NOT NULL
);

CREATE UNIQUE INDEX idx_crt_entries_org_serial ON
    crt_entries(org_id, serial);
CREATE INDEX idx_crt_entries_ca_key_id ON crt_entries(ca_key_id);

```

18. Redis

Data

Structures

Redis 8 is used for ephemeral, high-throughput data: session state, rate limiting, distributed locks, caching, pub/sub, and real-time

presence. All

keys use a

{service}:

{type}:

{identifier}

namespace

convention

to enable

Redis

Cluster key

sharding

where

appropriate.

Keys are

prefixed by

service to

enable

logical

separation.

TTLs are

always set;

no key

should be

persisted

indefinitely

without

explicit
review.

18.1 Session Storage

Active SSH Session State

Key: session:{sessionId}:state

Type: Hash

TTL: 24 hours (refreshed on activity)

Purpose: Store the current state of an active SSH session for the proxy service.

```
HSET session:sess-550e8400:state
  user_id          "550e8400-e29b-41d4-a716-446655440000"
  host_id          "host-550e8400-0000-0000-0000-aabbccddeeff"
  org_id           "org-550e8400-0000-0000-0000-000000000001"
  status           "connected"
  cols             "220"
  rows             "50"
  terminal_type     "xterm-256color"
  recording         "true"
  collab           "false"
  proxy_node        "proxy-node-2"
  started_at        "1751120400"
  last_activity     "1751125200"
```

```
EXPIRE session:sess-550e8400:state 86400
```

Session Token → Session ID Mapping

Key: session_token:{tokenHash}:session

Type: String

TTL: 24 hours

Purpose: Validate WebSocket session tokens quickly without database lookup.

```
SET session_token:sha256abc123:session
"sess-550e8400-0000-0000-0000-aabbccddeeff"
EX 86400
```

User Active Sessions Index

Key: user:{userId}:active_sessions

Type: Set

TTL: 24 hours (refreshed on session activity)

Purpose: Track all active session IDs for a user (for GET /api/v1/sessions fast path and session limits).

```
SADD user:550e8400:active_sessions "sess-550e8400-aaa"
"sess-550e8400-bbb"
EXPIRE user:550e8400:active_sessions 86400
```

18.2 Authentication Cache

JWT Blocklist

Key: jwt:blocklist:{jti}

Type: String (value: reason)

TTL: Set to token's remaining lifetime

Purpose: O(1) lookup for revoked JWT tokens. Backed by jwt_blocklist table for persistence.

```
SET jwt:blocklist:jti-550e8400-unique "user_logout"
EXAT 1751120400
```

Auth Rate Limit Counters (Sliding Window)

Key: rl:auth:{ipAddress}:{windowMinute}

Type: String (integer counter)

TTL: 2 minutes (covers current + previous window)

Purpose: Rate limit login/register endpoints per IP.

```
INCR rl:auth:192.168.1.10:29185440
EXPIRE rl:auth:192.168.1.10:29185440 120
```

```
windowMinute = floor(unix_timestamp / 60)
```

General Rate Limit Counter (User)

Key: rl:user:{userId}:{windowMinute}

Type: String (integer counter)

TTL: 2 minutes

Purpose: Rate limit per authenticated user.

```
INCR rl:user:550e8400:29185440
```

```
EXPIRE rl:user:550e8400:29185440 120
```

AI Endpoint Rate Limit (Hourly)

Key: rl:ai:{userId}:{windowHour}

Type: String (integer counter)

TTL: 2 hours

Purpose: Rate limit AI service usage (100 req/hour per user).

```
INCR rl:ai:550e8400:485757
```

```
EXPIRE rl:ai:550e8400:485757 7200
```

```
windowHour = floor(unix_timestamp / 3600)
```

MFA Challenge State

Key: mfa:challenge:{challengeId}

Type: Hash

TTL: 5 minutes

Purpose: Store MFA challenge context during the login flow.

```
HSET mfa:challenge:mfa-chal-550e8400
```

```
  user_id      "550e8400-e29b-41d4-a716-446655440000"
```

```
  method      "totp"
```

```
  stage       "pending"
```

```
  session_id  "temp-session-id"
```

```
  ip          "192.168.1.10"
```

```
  attempts    "0"
```

```
  created_at  "1751125200"
```

```
EXPIRE mfa:challenge:mfa-chal-550e8400 300
```

FIDO2 Authentication Challenge

Key: fido2:challenge:{challengeId}

Type: Hash

TTL: 60 seconds

Purpose: Store FIDO2 WebAuthn challenge bytes and expected state.

```
HSET fido2:challenge:fido2-chal-550e8400
  challenge_b64    "bGV0J3MgdGVzdA=="
  user_id          "550e8400-e29b-41d4-a716-446655440000"
  rp_id            "helixterminator.io"
  user_verification "preferred"
  mfa_challenge_id "mfa-chal-550e8400"
```

```
EXPIRE fido2:challenge:fido2-chal-550e8400 60
```

TOTP Setup Token

Key: totp:setup:{setupToken}

Type: Hash

TTL: 10 minutes

Purpose: Temporarily store TOTP secret during setup (not persisted until verified).

```
HSET totp:setup:setup-token-xyz
  user_id          "550e8400-e29b-41d4-a716-446655440000"
  encrypted_secret "AES256GCM:base64..."
  algorithm        "SHA1"
  digits           "6"
  period           "30"
```

```
EXPIRE totp:setup:setup-token-xyz 600
```

18.3 Vault Sync Cursors

Vault Sync Cursor Cache

Key: vault:{vaultId}:sync:{clientId}:cursor

Type: String

TTL: 7 days

Purpose: Fast lookup of client sync position without hitting PostgreSQL.

```
SET vault:vault-550e8400:sync:client-abc123:cursor
"eyJzZXEiOjE0Mjh9"
EX 604800
```

Vault Version Counter

Key: vault:{vaultId}:version

Type: String (integer)

TTL: None (persistent until vault deleted)

Purpose: Incrementing server version for optimistic concurrency control.

```
INCR vault:vault-550e8400:version
```

Vault Change Notification Stream

Key: vault:{vaultId}:changes

Type: Stream

TTL: 48 hours (maxlen 10000)

Purpose: Notify connected WebSocket clients of vault changes for real-time sync.

```
XADD vault:vault-550e8400:changes MAXLEN ~ 10000 *
  item_id    "item-550e8400-aaa"
  operation  "upsert"
  version    "6"
  changed_by "550e8400-user"
  timestamp  "1751125200"
```

18.4 SSH Connection State

SSH Session Lock

Key: lock:session:{sessionId}

Type: String (lock value = node ID)

TTL: 30 seconds (refreshed by lock holder)

Purpose: Distributed mutex to prevent concurrent operations on the same session (e.g., resize while terminating).

```
SET lock:session:sess-550e8400 "proxy-node-2:1751125200" NX EX 30
```


Renewal: SET lock:session:sess-550e8400 "proxy-node-2:1751125200"
XX EX 30

SSH Proxy Node Registration

Key: proxy:node:{nodeId}:info

Type: Hash

TTL: 60 seconds (heartbeat)

Purpose: Register proxy nodes for load balancing and session routing.

```
HSET proxy:node:proxy-node-2:info
  address      "10.0.10.2:8086"
  active_sessions "47"
  max_sessions  "200"
  cpu_percent   "23"
  region       "us-east-1"
  version      "1.2.3"
```

```
EXPIRE proxy:node:proxy-node-2:info 60
```

Active Proxy Nodes Index

Key: proxy:nodes

Type: Sorted Set (score = last heartbeat timestamp)

TTL: None (managed by heartbeat)

Purpose: Discover available proxy nodes for session routing.

```
ZADD proxy:nodes 1751125200 "proxy-node-2"
ZADD proxy:nodes 1751125195 "proxy-node-1"
```

Remove stale nodes: ZREMRANGEBYSCORE proxy:nodes -inf (unix_now - 90)

Port Forward Active Connection

Key: portfwd:{ruleId}:connection

Type: Hash

TTL: Duration of connection + 300 seconds

Purpose: Track active port forwarding connection metadata.

```
HSET portfwd:pf-550e8400:connection
  connection_id  "pfconn-550e8400-aaa"
  ssh_session_id "sess-550e8400-bbb"
```

proxy_node	"proxy-node-2"
local_port	"16379"
started_at	"1751125200"
bytes_sent	"0"
bytes_received	"0"

18.5 User Presence

Online User Presence

Key: presence:org:{orgId}:online

Type: Sorted Set (score = last seen timestamp)

TTL: None (managed by activity updates)

Purpose: Track which users are currently online for org-level presence indicators.

```
ZADD presence:org:org-550e8400:online 1751125200 "user:550e8400-alice"
```

```
ZADD presence:org:org-550e8400:online 1751125190 "user:660e8400-bob"
```

```
Mark offline: ZREM presence:org:org-550e8400:online "user:550e8400-alice"
```

```
Get online users: ZRANGEBYSCORE presence:org:org-550e8400:online  
(unix_now - 300) +inf
```

User Activity Heartbeat

Key: presence:user:{userId}:heartbeat

Type: String (value = ISO8601 timestamp)

TTL: 5 minutes

Purpose: Per-user heartbeat; expiry = user went offline.

```
SET presence:user:550e8400:heartbeat "2026-06-28T17:40:00Z"  
EX 300
```

18.6 Notification Queues

User Notification Queue

Key: notifications:user:{userId}:queue

Type: List (LPUSH producer, BRPOP consumer)

TTL: 7 days

Purpose: Queue notifications for delivery to connected WebSocket clients.

```
LPUSH notifications:user:550e8400:queue
```

```
'{"id":"notif-abc","type":"session_started","data":  
{ "host":"prod-web-01"}, "ts":1751125200}{'
```

```
EXPIRE notifications:user:550e8400:queue 604800
```

Broadcast Message Queue

Key: broadcast:session:{sessionId}

Type: List

TTL: 60 seconds

Purpose: Queue broadcast commands for SSH sessions (for POST /api/v1/sessions/{sessionId}/broadcast).

```
LPUSH broadcast:session:sess-550e8400:queue
```

```
'{"data":"c3VkbyBzeXN0ZW1jdGwgcmVzdGFydCBuZ2lueAo=", "from":"user-  
alice"}{'
```

```
EXPIRE broadcast:session:sess-550e8400:queue 60
```

18.7 Distributed Locks

Idempotency Key Cache

Key: idempotency:{userId}:{idempotencyKey}

Type: String (JSON response body)

TTL: 24 hours

Purpose: Cache responses for idempotent requests to prevent duplicate processing.

```
SET idempotency:550e8400:550e8400-idempotency-key
```

```
'{"status":201,"body":{"id":"host-abc","name":"prod-
```

```
web-01",...}}'  
EX 86400
```

Global Distributed Lock (Generic)

Key: lock:global:{resourceType}:{resourceId}

Type: String (lock token)

TTL: Variable (5–60 seconds)

Purpose: Prevent concurrent writes to the same resource across microservices.

```
SET lock:global:vault:vault-550e8400 "lock-token-nonce-xyz" NX EX  
30
```

18.8 Caching

User Context Cache

Key: cache:user:{userId}:context

Type: Hash

TTL: 60 seconds

Purpose: Cache user org/permissions context for incoming requests (reduces auth_db round-trips).

```
HSET cache:user:550e8400:context  
  org_id      "org-550e8400"  
  org_role    "admin"  
  team_ids    '["team-abc","team-def"]'  
  enforce_mfa "true"  
  status      "active"  
  cached_at   "1751125200"
```

```
EXPIRE cache:user:550e8400:context 60
```

Vault Member Cache

Key: cache:vault:{vaultId}:member:{userId}

Type: Hash

TTL: 30 seconds

Purpose: Cache vault membership for permission checks.

```
HSET cache:vault:vault-550e8400:member:550e8400
  permission "admin"
  is_owner   "true"
  cached_at  "1751125200"
```

```
EXPIRE cache:vault:vault-550e8400:member:550e8400 30
```

Host List Cache (per vault)

Key: cache:vault:{vaultId}:hosts:page:{cursor}

Type: String (JSON)

TTL: 10 seconds

Purpose: Cache paginated host list results to reduce DB load on frequently-accessed vaults.

```
SET cache:vault:vault-550e8400:hosts:page:cursor_abc
  '{"data":[...],"pagination":{"...}}'
EX 10
```

SSH Host Fingerprint Cache

Key: cache:host:{hostId}:fingerprint

Type: Hash

TTL: 1 hour

Purpose: Cache known SSH fingerprints to avoid DB lookups on every connection.

```
HSET cache:host:host-550e8400:fingerprint
  sha256      "SHA256:abc123def456..."
  ed25519     "ssh-ed25519 AAAA..."
  verified    "true"
```

```
EXPIRE cache:host:host-550e8400:fingerprint 3600
```

18.9 SFTP Session State

Key: sftp:{sftpSessionId}:state

Type: Hash

TTL: 6 hours

Purpose: Track SFTP session state and CWD.

```
HSET sftp:sftp-550e8400:state
  user_id      "550e8400-user"
  host_id      "host-550e8400"
  cwd          "/var/www/html"
  status       "connected"
  proxy_node   "proxy-node-2"
  started_at   "1751125200"

EXPIRE sftp:sftp-550e8400:state 21600
```

18.10 Collaboration State

Key: collab:{collabId}:participants
Type: Hash (field = userId, value = JSON participant state)
TTL: 24 hours
Purpose: Track active collaboration participants for WebSocket routing.

```
HSET collab:collab-550e8400:participants
  "660e8400-bob" '{"name":"Bob Jones","role":"viewer","color":"#FF6B35","connected_at":1751125200}'
  "770e8400-carol" '{"name":"Carol White","role":"contributor","color":"#4ECDC4","connected_at":1751125200}'

EXPIRE collab:collab-550e8400:participants 86400
```

18.11 Snippet Execution State

Key: snippetexec:{executionId}:state
Type: Hash
TTL: 2 hours
Purpose: Track multi-host snippet execution progress for real-time status updates.

```
HSET snippetexec:exec-550e8400:state
  snippet_id   "snip-550e8400"
  status       "running"
  total        "20"
  completed    "8"
  failed       "1"
  started_at   "1751125200"
```

```
EXPIRE snippetexec:exec-550e8400:state 7200
```

18.12 WebSocket Connection Registry

Key: `ws:user:{userId}:connections`

Type: Set (value = connection ID strings)

TTL: 24 hours

Purpose: Track all WebSocket connections for a user to enable broadcasting.

```
SADD ws:user:550e8400:connections "ws-conn-abc-proxy-1" "ws-conn-def-proxy-2"
```

```
EXPIRE ws:user:550e8400:connections 86400
```

19. Database Migrations

HelixTerminator uses [golang-migrate/migrate](#) for all database schema migrations. Each microservice manages its own migrations in `internal/db/{service}/migrations/`.

19.1 Migration Naming Convention

`{NNNNNN}_{description}.{up|down}.sql`

- `NNNNNN`: Zero-padded 6-digit sequence number
- `description`: Snake_case description of the migration
- `.up.sql`: Forward migration (apply)
- `.down.sql`: Reverse migration (rollback)

Examples:

```
000001_create_users.up.sql
000001_create_users.down.sql
000002_create_sessions.up.sql
000002_create_sessions.down.sql
000003_add_mfa_tables.up.sql
000003_add_mfa_tables.down.sql
```

19.2 Migration Tool Configuration

cmd/migrate/main.go:

```
package main

import (
    "flag"
    "fmt"
    "log"
    "os"

    "github.com/golang-migrate/migrate/v4"
    _ "github.com/golang-migrate/migrate/v4/database/postgres"
    _ "github.com/golang-migrate/migrate/v4/source/file"
)

func main() {
    service := flag.String("service", "", "Service name (auth, vault, host, etc.)")
    action := flag.String("action", "up", "Action: up, down, version, force")
    steps := flag.Int("steps", 0, "Number of steps (0 = all)")
    flag.Parse()

    dsn := os.Getenv(fmt.Sprintf("DB_%s_URL", strings.ToUpper(*service)))
    migrationsPath := fmt.Sprintf("file://internal/db/%s/migrations", *service)

    m, err := migrate.New(migrationsPath, dsn)
    if err != nil {
        log.Fatalf("Failed to init migrations: %v", err)
    }
    defer m.Close()

    switch *action {
    case "up":
        if *steps > 0 {
            err = m.Steps(*steps)
        } else {
            err = m.Up()
        }
    case "down":
        if *steps > 0 {
            err = m.Steps(-(*steps))
        }
    }
```



```

    } else {
        err = m.Down()
    }
    case "version":
        v, dirty, _ := m.Version()
        fmt.Printf("Version: %d, Dirty: %v\n", v, dirty)
    }

    if err != nil && err != migrate.ErrNoChange {
        log.Fatalf("Migration failed: %v", err)
    }
    log.Println("Migration completed successfully")
}

```

19.3 Zero-Downtime Migration Strategy

Phase 1 — Expand (backward-compatible schema change): Apply the migration while old code is still running. The new column/table must be nullable or have a default value.

Phase 2 — Deploy new application code: Deploy the new application version that reads/writes to both old and new schema.

Phase 3 — Contract (remove old columns/tables): After all instances of the old code are gone, apply the cleanup migration to remove deprecated columns.

This pattern (expand-contract / parallel change) ensures: - No downtime during migrations - Rollback is always possible - Zero-error deployments

Non-destructive migrations (always safe): - ADD COLUMN ...
 DEFAULT ... - CREATE INDEX CONCURRENTLY - CREATE TABLE IF NOT EXISTS -
 ADD CONSTRAINT ... NOT VALID → then VALIDATE CONSTRAINT separately

Risky migrations (require extra caution): - DROP COLUMN (expand-contract required) - ALTER COLUMN ... TYPE (use new column + backfill + rename) - ADD NOT NULL constraint without default (add nullable first, backfill, then add constraint) - CREATE INDEX without CONCURRENTLY (blocks writes)

19.4 Migration Files

auth_db Migrations

000001_create_extensions.up.sql

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "pg_trgm";
CREATE EXTENSION IF NOT EXISTS "citext";
```

000001_create_extensions.down.sql

```
DROP EXTENSION IF EXISTS "citext";
DROP EXTENSION IF EXISTS "pg_trgm";
DROP EXTENSION IF EXISTS "pgcrypto";
```

000002_create_users.up.sql

```
CREATE OR REPLACE FUNCTION trigger_set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = NOW();
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TABLE users (
    id                UUID PRIMARY KEY DEFAULT
                        gen_random_uuid(),
    email             CITEXT NOT NULL,
    email_verified_at TIMESTAMP WITH TIME ZONE,
    email_pending     CITEXT,
    email_pending_token VARCHAR(255),
    email_pending_expires_at TIMESTAMP WITH TIME ZONE,
    password_hash     VARCHAR(255),
    display_name      VARCHAR(100) NOT NULL,
    avatar_url        TEXT,
    bio               TEXT,
    locale            VARCHAR(20) NOT NULL DEFAULT 'en-US',
    timezone          VARCHAR(100) NOT NULL DEFAULT 'UTC',
    status            VARCHAR(20) NOT NULL DEFAULT 'active'
                        CHECK (status IN ('active',
                        'suspended', 'pending_deletion', 'deleted')),
    failed_login_attempts INTEGER NOT NULL DEFAULT 0,
    locked_until       TIMESTAMP WITH TIME ZONE,
```

```

last_login_at          TIMESTAMP WITH TIME ZONE,
last_login_ip          INET,
password_changed_at    TIMESTAMP WITH TIME ZONE,
terms_accepted_at      TIMESTAMP WITH TIME ZONE,
terms_version          VARCHAR(20),
deletion_requested_at  TIMESTAMP WITH TIME ZONE,
deletion_scheduled_at  TIMESTAMP WITH TIME ZONE,
created_at             TIMESTAMP WITH TIME ZONE NOT NULL
                        DEFAULT NOW(),
updated_at             TIMESTAMP WITH TIME ZONE NOT NULL
                        DEFAULT NOW(),
deleted_at             TIMESTAMP WITH TIME ZONE
);

CREATE UNIQUE INDEX idx_users_email ON users(email) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_users_status ON users(status) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_users_created_at ON users USING BRIN
    (created_at);

CREATE TRIGGER trg_users_updated_at
    BEFORE UPDATE ON users
    FOR EACH ROW EXECUTE FUNCTION trigger_set_updated_at();

```

000002_create_users.down.sql

```

DROP TRIGGER IF EXISTS trg_users_updated_at ON users;
DROP TABLE IF EXISTS users;
DROP FUNCTION IF EXISTS trigger_set_updated_at();

```

000003_create_user_sessions.up.sql

```

CREATE TABLE user_sessions (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
                CASCADE,
    device_id   UUID,
    ip_address  INET NOT NULL,
    user_agent  TEXT,
    location_city VARCHAR(100),
    location_country VARCHAR(10),
    mfa_verified BOOLEAN NOT NULL DEFAULT FALSE,
    mfa_method  VARCHAR(20),
    status      VARCHAR(20) NOT NULL DEFAULT 'active'
                CHECK (status IN ('active', 'expired',
                                'revoked')),

```

```

    revoked_reason    VARCHAR(100),
    last_active_at    TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    expires_at        TIMESTAMP WITH TIME ZONE NOT NULL
);

```

```

CREATE INDEX idx_user_sessions_user_id ON user_sessions(user_id);
CREATE INDEX idx_user_sessions_status ON user_sessions(status,
    expires_at)
    WHERE status = 'active';

```

000003_create_user_sessions.down.sql

```

DROP TABLE IF EXISTS user_sessions;

```

000004_create_refresh_tokens.up.sql

```

CREATE TABLE refresh_tokens (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_id        UUID NOT NULL REFERENCES user_sessions(id) ON
        DELETE CASCADE,
    user_id            UUID NOT NULL REFERENCES users(id) ON DELETE
        CASCADE,
    token_hash         VARCHAR(255) NOT NULL UNIQUE,
    family             UUID NOT NULL,
    generation         INTEGER NOT NULL DEFAULT 1,
    ip_address         INET,
    user_agent         TEXT,
    used_at            TIMESTAMP WITH TIME ZONE,
    revoked            BOOLEAN NOT NULL DEFAULT FALSE,
    revoked_at         TIMESTAMP WITH TIME ZONE,
    revoked_reason     VARCHAR(100),
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    expires_at         TIMESTAMP WITH TIME ZONE NOT NULL
);

```

```

CREATE INDEX idx_refresh_tokens_token_hash ON
    refresh_tokens(token_hash);
CREATE INDEX idx_refresh_tokens_session_id ON
    refresh_tokens(session_id);
CREATE INDEX idx_refresh_tokens_family ON refresh_tokens(family);

```

000004_create_refresh_tokens.down.sql

```

DROP TABLE IF EXISTS refresh_tokens;

```

000005_create_device_tokens.up.sql

```
CREATE TABLE device_tokens (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE  
              CASCADE,  
  name        VARCHAR(255) NOT NULL,  
  fingerprint VARCHAR(512) NOT NULL,  
  platform    VARCHAR(255),  
  user_agent   TEXT,  
  trusted      BOOLEAN NOT NULL DEFAULT TRUE,  
  last_seen_at TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT  
              NOW(),  
  last_seen_ip INET,  
  revoked      BOOLEAN NOT NULL DEFAULT FALSE,  
  revoked_at   TIMESTAMP WITH TIME ZONE,  
  created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()  
);
```

```
CREATE UNIQUE INDEX idx_device_tokens_user_fingerprint  
  ON device_tokens(user_id, fingerprint) WHERE revoked = FALSE;  
CREATE INDEX idx_device_tokens_user_id ON device_tokens(user_id);
```

000005_create_device_tokens.down.sql

```
DROP TABLE IF EXISTS device_tokens;
```

000006_create_mfa_tables.up.sql

```
CREATE TABLE mfa_totp_credentials (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE  
              CASCADE,  
  encrypted_secret TEXT NOT NULL,  
  issuer       VARCHAR(100) NOT NULL DEFAULT  
              'HelixTerminator',  
  algorithm    VARCHAR(20) NOT NULL DEFAULT 'SHA1',  
  digits       INTEGER NOT NULL DEFAULT 6,  
  period       INTEGER NOT NULL DEFAULT 30,  
  enabled      BOOLEAN NOT NULL DEFAULT TRUE,  
  last_used_at TIMESTAMP WITH TIME ZONE,  
  last_used_code VARCHAR(10),  
  created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT  
              NOW(),  
  updated_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()  
);
```

```
CREATE UNIQUE INDEX idx_mfa_totp_user_enabled ON
    mfa_totp_credentials(user_id)
    WHERE enabled = TRUE;
```

```
CREATE TABLE mfa_totp_backup_codes (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
                CASCADE,
    code_hash   VARCHAR(255) NOT NULL,
    used        BOOLEAN NOT NULL DEFAULT FALSE,
    used_at     TIMESTAMP WITH TIME ZONE,
    used_ip     INET,
    created_at  TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);
```

```
CREATE INDEX idx_backup_codes_user_id ON
    mfa_totp_backup_codes(user_id) WHERE used = FALSE;
```

```
CREATE TABLE mfa_fido2_credentials (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON
                DELETE CASCADE,
    credential_id    BYTEA NOT NULL UNIQUE,
    credential_id_b64 TEXT NOT NULL,
    name            VARCHAR(255) NOT NULL,
    public_key_cbor BYTEA NOT NULL,
    aaguid          UUID,
    sign_count      BIGINT NOT NULL DEFAULT 0,
    transports      TEXT[] DEFAULT '{}',
    backup_eligible  BOOLEAN NOT NULL DEFAULT FALSE,
    backup_state     BOOLEAN NOT NULL DEFAULT FALSE,
    attestation_type VARCHAR(50),
    attestation_data JSONB,
    last_used_at    TIMESTAMP WITH TIME ZONE,
    enabled         BOOLEAN NOT NULL DEFAULT TRUE,
    created_at      TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW(),
    updated_at      TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW()
);
```

```
CREATE INDEX idx_fido2_user_id ON mfa_fido2_credentials(user_id)
    WHERE enabled = TRUE;
```

000006_create_mfa_tables.down.sql

```
DROP TABLE IF EXISTS mfa_fido2_credentials;
DROP TABLE IF EXISTS mfa_totp_backup_codes;
DROP TABLE IF EXISTS mfa_totp_credentials;
```

000007_create_api_keys.up.sql

```
CREATE TABLE api_keys (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
              CASCADE,
  org_id      UUID,
  name        VARCHAR(100) NOT NULL,
  description  TEXT,
  key_hash    VARCHAR(255) NOT NULL UNIQUE,
  key_prefix  VARCHAR(20) NOT NULL,
  scopes      TEXT[] NOT NULL DEFAULT '{}',
  allowed_ips CIDR[] DEFAULT '{}',
  last_used_at TIMESTAMP WITH TIME ZONE,
  last_used_ip INET,
  revoked      BOOLEAN NOT NULL DEFAULT FALSE,
  revoked_at   TIMESTAMP WITH TIME ZONE,
  revoked_reason VARCHAR(255),
  expires_at   TIMESTAMP WITH TIME ZONE,
  created_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
              NOW(),
  updated_at   TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);
```

```
CREATE INDEX idx_api_keys_user_id ON api_keys(user_id) WHERE
  revoked = FALSE;
CREATE INDEX idx_api_keys_key_hash ON api_keys(key_hash);
```

000007_create_api_keys.down.sql

```
DROP TABLE IF EXISTS api_keys;
```

000008_create_login_history.up.sql

```
CREATE TABLE login_history (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE
              CASCADE,
  event_type   VARCHAR(50) NOT NULL,
  ip_address   INET,
  user_agent   TEXT,
  device_id    UUID,
```

```

        session_id          UUID,
        failure_reason       VARCHAR(255),
        metadata             JSONB DEFAULT '{}',
        occurred_at          TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                               NOW()
    );

CREATE INDEX idx_login_history_user_id ON login_history(user_id);
CREATE INDEX idx_login_history_occurred_at ON login_history USING
    BRIN (occurred_at);

```

000008_create_login_history.down.sql

```
DROP TABLE IF EXISTS login_history;
```

000009_create_sso_tables.up.sql

```

CREATE TABLE sso_providers (
    id                UUID PRIMARY KEY DEFAULT
                        gen_random_uuid(),
    org_id            UUID NOT NULL,
    provider          VARCHAR(50) NOT NULL,
    slug             VARCHAR(100) NOT NULL,
    display_name      VARCHAR(255),
    client_id         VARCHAR(512),
    encrypted_client_secret TEXT,
    discovery_url     TEXT,
    authorization_url TEXT,
    token_url        TEXT,
    userinfo_url     TEXT,
    jwks_uri         TEXT,
    scopes           TEXT[] DEFAULT ARRAY['openid', 'email',
                                           'profile'],
    attribute_mapping JSONB DEFAULT '{}',
    enabled          BOOLEAN NOT NULL DEFAULT TRUE,
    enforce_for_domain VARCHAR(255),
    created_at       TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                        NOW(),
    updated_at       TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                        NOW()
);

CREATE UNIQUE INDEX idx_sso_providers_org_provider ON
    sso_providers(org_id, provider)
    WHERE enabled = TRUE;

CREATE TABLE sso_identities (

```



```

    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL REFERENCES users(id) ON DELETE
                      CASCADE,
    provider_id       UUID NOT NULL REFERENCES sso_providers(id) ON
                      DELETE CASCADE,
    subject            VARCHAR(512) NOT NULL,
    access_token       TEXT,
    refresh_token      TEXT,
    token_expires_at  TIMESTAMP WITH TIME ZONE,
    profile_data       JSONB DEFAULT '{}',
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                      NOW(),
    updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE UNIQUE INDEX idx_sso_identities_provider_subject ON
    sso_identities(provider_id, subject);

```

000009_create_sso_tables.down.sql

```

DROP TABLE IF EXISTS sso_identities;
DROP TABLE IF EXISTS sso_providers;

```

000010_create_password_history.up.sql

```

CREATE TABLE password_history (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL REFERENCES users(id) ON DELETE
                      CASCADE,
    password_hash     VARCHAR(255) NOT NULL,
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW()
);

```

```

CREATE INDEX idx_password_history_user_id ON
    password_history(user_id);

```

000010_create_password_history.down.sql

```

DROP TABLE IF EXISTS password_history;

```

000011_create_jwt_blocklist.up.sql

```

CREATE TABLE jwt_blocklist (
    jti              VARCHAR(255) PRIMARY KEY,
    user_id          UUID NOT NULL,
    revoked_at       TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    expires_at       TIMESTAMP WITH TIME ZONE NOT NULL,

```

```
    reason          VARCHAR(100)
);

CREATE INDEX idx_jwt_blocklist_expires_at ON jwt_blocklist USING
    BRIN (expires_at);
```

000011_create_jwt_blocklist.down.sql

```
DROP TABLE IF EXISTS jwt_blocklist;
```

000012_add_users_org_id.up.sql

```
-- Expand: add nullable column first (zero-downtime)
ALTER TABLE users ADD COLUMN IF NOT EXISTS org_id UUID;
CREATE INDEX idx_users_org_id ON users(org_id) WHERE org_id IS
    NOT NULL;
```

000012_add_users_org_id.down.sql

```
DROP INDEX IF EXISTS idx_users_org_id;
ALTER TABLE users DROP COLUMN IF EXISTS org_id;
```

000013_add_users_org_id_not_null.up.sql

```
-- Contract: after backfill is complete, add NOT NULL constraint
-- Run: UPDATE users SET org_id = (SELECT id FROM organizations
    WHERE owner_id = users.id LIMIT 1) WHERE org_id IS NULL;
ALTER TABLE users ALTER COLUMN org_id SET NOT NULL;
```

000013_add_users_org_id_not_null.down.sql

```
ALTER TABLE users ALTER COLUMN org_id DROP NOT NULL;
```

000014_create_ssh_sessions.up.sql

```
-- (In session_db)
CREATE TABLE ssh_sessions (
    id                UUID NOT NULL DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL,
    host_id           UUID NOT NULL,
    vault_id          UUID NOT NULL,
    org_id            UUID NOT NULL,
    client_ip         INET NOT NULL,
    user_agent        TEXT,
    terminal_cols     SMALLINT NOT NULL DEFAULT 80,
    terminal_rows     SMALLINT NOT NULL DEFAULT 24,
```

```

terminal_type      VARCHAR(50) NOT NULL DEFAULT
                    'xterm-256color',
auth_method        VARCHAR(20),
key_id             UUID,
recording_enabled  BOOLEAN NOT NULL DEFAULT FALSE,
recording_path     TEXT,
recording_size_bytes BIGINT DEFAULT 0,
collab_enabled     BOOLEAN NOT NULL DEFAULT FALSE,
read_only          BOOLEAN NOT NULL DEFAULT FALSE,
status            VARCHAR(20) NOT NULL DEFAULT 'connecting',
reason            TEXT,
ticket_ref         VARCHAR(255),
startup_snippet_id UUID,
jump_chain         JSONB DEFAULT '[]',
exit_code          INTEGER,
disconnect_reason  TEXT,
bytes_sent         BIGINT NOT NULL DEFAULT 0,
bytes_received     BIGINT NOT NULL DEFAULT 0,
commands_count     INTEGER NOT NULL DEFAULT 0,
resize_count       INTEGER NOT NULL DEFAULT 0,
started_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
                    NOW(),
connected_at       TIMESTAMP WITH TIME ZONE,
ended_at           TIMESTAMP WITH TIME ZONE,
duration_seconds   INTEGER,
PRIMARY KEY (id, started_at)
) PARTITION BY RANGE (started_at);

```

```

CREATE TABLE ssh_sessions_2026_q2 PARTITION OF ssh_sessions
FOR VALUES FROM ('2026-04-01') TO ('2026-07-01');

```

```

CREATE TABLE ssh_sessions_2026_q3 PARTITION OF ssh_sessions
FOR VALUES FROM ('2026-07-01') TO ('2026-10-01');

```

```

CREATE TABLE ssh_sessions_2026_q4 PARTITION OF ssh_sessions
FOR VALUES FROM ('2026-10-01') TO ('2027-01-01');

```

```

CREATE INDEX idx_ssh_sessions_user_id ON ssh_sessions(user_id,
started_at DESC);

```

```

CREATE INDEX idx_ssh_sessions_host_id ON ssh_sessions(host_id,
started_at DESC);

```

```

CREATE INDEX idx_ssh_sessions_status ON ssh_sessions(status,
started_at DESC)
WHERE status IN ('connecting', 'connected');

```

000014_create_ssh_sessions.down.sql

```

DROP TABLE IF EXISTS ssh_sessions CASCADE;

```

000015_create_port_forward_rules.up.sql

```
CREATE TABLE port_forward_rules (  
  id                UUID PRIMARY KEY DEFAULT  
    gen_random_uuid(),  
  user_id           UUID NOT NULL,  
  host_id            UUID NOT NULL,  
  vault_id           UUID NOT NULL,  
  org_id             UUID NOT NULL,  
  name               VARCHAR(255) NOT NULL,  
  description        TEXT,  
  type               VARCHAR(20) NOT NULL CHECK (type IN  
    ('local', 'remote', 'dynamic')),  
  local_address      VARCHAR(255) NOT NULL DEFAULT '127.0.0.1',  
  local_port         INTEGER NOT NULL CHECK (local_port  
    BETWEEN 1 AND 65535),  
  remote_address     VARCHAR(255),  
  remote_port        INTEGER CHECK (remote_port BETWEEN 1 AND  
    65535),  
  bind_address       VARCHAR(255),  
  auto_start         BOOLEAN NOT NULL DEFAULT FALSE,  
  status             VARCHAR(20) NOT NULL DEFAULT 'inactive',  
  sort_order         INTEGER NOT NULL DEFAULT 0,  
  created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT  
    NOW(),  
  updated_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT  
    NOW(),  
  deleted_at         TIMESTAMP WITH TIME ZONE  
);
```

```
CREATE INDEX idx_pf_rules_user_id ON port_forward_rules(user_id)  
  WHERE deleted_at IS NULL;  
CREATE INDEX idx_pf_rules_host_id ON port_forward_rules(host_id)  
  WHERE deleted_at IS NULL;
```

000015_create_port_forward_rules.down.sql

```
DROP TABLE IF EXISTS port_forward_rules;
```

000016_create_hosts.up.sql

```
-- (In host_db)  
CREATE EXTENSION IF NOT EXISTS "pg_trgm";  
  
CREATE TABLE hosts (  
  id                UUID PRIMARY KEY DEFAULT  
    gen_random_uuid(),  
  vault_id           UUID NOT NULL,
```

```

group_id          UUID,
org_id            UUID NOT NULL,
created_by        UUID NOT NULL,
name              VARCHAR(255) NOT NULL,
hostname          VARCHAR(512) NOT NULL,
port              INTEGER NOT NULL DEFAULT 22 CHECK
                  (port BETWEEN 1 AND 65535),
username          VARCHAR(255),
auth_method       VARCHAR(20) NOT NULL DEFAULT 'key',
key_id            UUID,
description        TEXT,
color             VARCHAR(20),
icon              VARCHAR(50),
tags              TEXT[] NOT NULL DEFAULT '{}',
jump_host_id      UUID,
connection_timeout_seconds INTEGER NOT NULL DEFAULT 30,
keepalive_interval_seconds INTEGER NOT NULL DEFAULT 60,
status            VARCHAR(20) NOT NULL DEFAULT
                  'active',
last_connected_at  TIMESTAMP WITH TIME ZONE,
fingerprint_verified BOOLEAN NOT NULL DEFAULT FALSE,
environment_variables JSONB NOT NULL DEFAULT '{}',
custom_fields      JSONB NOT NULL DEFAULT '{}',
sort_order         INTEGER NOT NULL DEFAULT 0,
created_at         TIMESTAMP WITH TIME ZONE NOT NULL
                  DEFAULT NOW(),
updated_at         TIMESTAMP WITH TIME ZONE NOT NULL
                  DEFAULT NOW(),
deleted_at         TIMESTAMP WITH TIME ZONE
);

```

```

CREATE INDEX idx_hosts_vault_id ON hosts(vault_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_hosts_org_id ON hosts(org_id) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_hosts_tags ON hosts USING GIN (tags);
CREATE INDEX idx_hosts_name_trgm ON hosts USING GIN (name
    gin_trgm_ops) WHERE deleted_at IS NULL;

```

000016_create_hosts.down.sql

```
DROP TABLE IF EXISTS hosts;
```

000017_create_vaults.up.sql

```

-- (In vault_db)
CREATE TABLE vaults (

```

```

    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    org_id            UUID NOT NULL,
    owner_id          UUID NOT NULL,
    name              VARCHAR(255) NOT NULL,
    description        TEXT,
    color             VARCHAR(20),
    icon              VARCHAR(50),
    encrypted          BOOLEAN NOT NULL DEFAULT TRUE,
    sync_enabled       BOOLEAN NOT NULL DEFAULT TRUE,
    item_count         INTEGER NOT NULL DEFAULT 0,
    storage_bytes      BIGINT NOT NULL DEFAULT 0,
    version            BIGINT NOT NULL DEFAULT 1,
    created_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    updated_at         TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
    deleted_at         TIMESTAMP WITH TIME ZONE
);

```

```

CREATE INDEX idx_vaults_org_id ON vaults(org_id) WHERE deleted_at
    IS NULL;
CREATE INDEX idx_vaults_owner_id ON vaults(owner_id) WHERE
    deleted_at IS NULL;

```

000017_create_vaults.down.sql

```

DROP TABLE IF EXISTS vaults;

```

000018_create_audit_events.up.sql

```

-- (In audit_db)
CREATE TABLE audit_events (
    id                UUID NOT NULL DEFAULT gen_random_uuid(),
    seq               BIGSERIAL,
    org_id            UUID NOT NULL,
    event_type        VARCHAR(100) NOT NULL,
    user_id           UUID,
    resource_type     VARCHAR(50),
    resource_id       UUID,
    resource_name     VARCHAR(512),
    outcome            VARCHAR(20) NOT NULL DEFAULT 'success',
    ip_address        INET,
    user_agent        TEXT,
    session_id        UUID,
    source_service    VARCHAR(50) NOT NULL,
    metadata          JSONB NOT NULL DEFAULT '{}',
    hash              VARCHAR(128) NOT NULL,

```

```

prev_hash          VARCHAR(128),
occurred_at        TIMESTAMP WITH TIME ZONE NOT NULL,
recorded_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT NOW(),
PRIMARY KEY (id, occurred_at)
) PARTITION BY RANGE (occurred_at);

```

```

CREATE TABLE audit_events_2026_06 PARTITION OF audit_events
FOR VALUES FROM ('2026-06-01') TO ('2026-07-01');
CREATE TABLE audit_events_2026_07 PARTITION OF audit_events
FOR VALUES FROM ('2026-07-01') TO ('2026-08-01');

```

```

CREATE INDEX idx_audit_events_org_id ON audit_events(org_id,
occurred_at DESC);
CREATE INDEX idx_audit_events_occurred_at ON audit_events USING
BRIN (occurred_at);

```

000018_create_audit_events.down.sql

```

DROP TABLE IF EXISTS audit_events CASCADE;

```

000019_create_organizations.up.sql

```

-- (In org_db)
CREATE EXTENSION IF NOT EXISTS "citext";

CREATE TABLE organizations (
    id                                UUID PRIMARY KEY DEFAULT
        gen_random_uuid(),
    name                             VARCHAR(255) NOT NULL,
    slug                             CITEXT NOT NULL UNIQUE,
    domain                           CITEXT,
    domain_verified                   BOOLEAN NOT NULL DEFAULT FALSE,
    plan                              VARCHAR(50) NOT NULL DEFAULT
        'free',
    max_members                       INTEGER NOT NULL DEFAULT 5,
    enforce_mfa                       BOOLEAN NOT NULL DEFAULT FALSE,
    session_recording_required        BOOLEAN NOT NULL DEFAULT FALSE,
    owner_id                         UUID NOT NULL,
    status                            VARCHAR(20) NOT NULL DEFAULT
        'active',
    settings                          JSONB NOT NULL DEFAULT '{}',
    created_at                        TIMESTAMP WITH TIME ZONE NOT
        NULL DEFAULT NOW(),
    updated_at                        TIMESTAMP WITH TIME ZONE NOT
        NULL DEFAULT NOW(),
    deleted_at                        TIMESTAMP WITH TIME ZONE
);

```

```
CREATE INDEX idx_organizations_slug ON organizations(slug) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_organizations_owner_id ON
    organizations(owner_id);
```

000019_create_organizations.down.sql

```
DROP TABLE IF EXISTS organizations;
```

000020_create_snippets.up.sql

```
-- (In snippet_db)
CREATE EXTENSION IF NOT EXISTS "pg_trgm";

CREATE TABLE snippets (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    vault_id          UUID NOT NULL,
    org_id            UUID NOT NULL,
    created_by        UUID NOT NULL,
    category_id       UUID,
    name              VARCHAR(255) NOT NULL,
    description        TEXT,
    content            TEXT NOT NULL,
    language          VARCHAR(50) NOT NULL DEFAULT 'bash',
    tags              TEXT[] NOT NULL DEFAULT '{}',
    parameters        JSONB NOT NULL DEFAULT '[]',
    shared            BOOLEAN NOT NULL DEFAULT FALSE,
    pinned            BOOLEAN NOT NULL DEFAULT FALSE,
    executions_count  BIGINT NOT NULL DEFAULT 0,
    last_executed_at  TIMESTAMP WITH TIME ZONE,
    fts_vector         TSVECTOR,
    created_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    updated_at        TIMESTAMP WITH TIME ZONE NOT NULL DEFAULT
        NOW(),
    deleted_at        TIMESTAMP WITH TIME ZONE
);

CREATE INDEX idx_snippets_vault_id ON snippets(vault_id) WHERE
    deleted_at IS NULL;
CREATE INDEX idx_snippets_fts ON snippets USING GIN (fts_vector)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_snippets_name_trgm ON snippets USING GIN (name
    gin_trgm_ops) WHERE deleted_at IS NULL;

CREATE OR REPLACE FUNCTION snippets_fts_update()
```



```

RETURNS TRIGGER AS $$
BEGIN
    NEW.fts_vector :=
        setweight(to_tsvector('english', coalesce(NEW.name, '')),
            'A') ||
        setweight(to_tsvector('english', coalesce(NEW.description,
            '')), 'B') ||
        setweight(to_tsvector('english', coalesce(NEW.content, '')),
            'C');
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_snippets_fts
    BEFORE INSERT OR UPDATE ON snippets
    FOR EACH ROW EXECUTE FUNCTION snippets_fts_update();

```

000020_create_snippets.down.sql

```

DROP TRIGGER IF EXISTS trg_snippets_fts ON snippets;
DROP FUNCTION IF EXISTS snippets_fts_update();
DROP TABLE IF EXISTS snippets;

```

20. Performance Indexes

This section documents the comprehensive indexing strategy for all HelixTerminator databases. Indexes are chosen based on actual query patterns observed in production workloads.

20.1 Index Strategy Overview

Index selection criteria: 1. **Selectivity:** Index columns with high cardinality first. 2. **Query frequency:** Index columns that appear in WHERE clauses of frequent queries. 3. **Write overhead:** Every index adds ~10-30% write overhead; only create indexes that pay their cost. 4. **Composite indexes:** Leading column matters — place the most selective or equality-matched column first. 5. **Partial indexes:** Use WHERE conditions to reduce index size for sparse columns (e.g., WHERE deleted_at IS NULL).

20.2 Composite Indexes for Common Query Patterns

auth_db

```
-- Find all active sessions for a user, ordered by creation time  
    (session list page)
```

```
CREATE INDEX idx_user_sessions_user_status_created  
    ON user_sessions(user_id, status, created_at DESC)  
    WHERE status = 'active';
```

```
-- Find all non-revoked API keys for a user+org combination
```

```
CREATE INDEX idx_api_keys_user_org_active  
    ON api_keys(user_id, org_id, created_at DESC)  
    WHERE revoked = FALSE;
```

```
-- Login history for a user with time range filter (audit page)
```

```
CREATE INDEX idx_login_history_user_type_time  
    ON login_history(user_id, event_type, occurred_at DESC);
```

```
-- SSO identity lookup by provider and subject (login flow)
```

```
CREATE INDEX idx_sso_identities_provider_subject_user  
    ON sso_identities(provider_id, subject, user_id);
```

host_db

```
-- Host listing with vault + status + sort (main host list)
```

```
CREATE INDEX idx_hosts_vault_status_sort  
    ON hosts(vault_id, status, sort_order ASC, name ASC)  
    WHERE deleted_at IS NULL;
```

```
-- Host listing with group filter (group view)
```

```
CREATE INDEX idx_hosts_group_sort  
    ON hosts(group_id, sort_order ASC)  
    WHERE deleted_at IS NULL AND group_id IS NOT NULL;
```

```
-- Recent connections across all hosts for a user (dashboard)
```

```
CREATE INDEX idx_host_conn_history_user_recent  
    ON host_connection_history(user_id, started_at DESC, host_id);
```

```
-- Recent connections to a specific host (host detail page)
```

```
CREATE INDEX idx_host_conn_history_host_recent  
    ON host_connection_history(host_id, started_at DESC, user_id);
```

```
-- Hosts by tag (tag filtering)
```

```

CREATE INDEX idx_hosts_vault_tags
  ON hosts USING GIN (tags)
  WHERE deleted_at IS NULL;

-- Jump host chain lookup
CREATE INDEX idx_hosts_jump_host
  ON hosts(jump_host_id, vault_id)
  WHERE jump_host_id IS NOT NULL AND deleted_at IS NULL;

-- Host group hierarchy traversal
CREATE INDEX idx_host_groups_parent_vault_sort
  ON host_groups(parent_id, vault_id, sort_order ASC)
  WHERE deleted_at IS NULL;

```

session_db

```

-- Active sessions for a host (live monitoring view)
CREATE INDEX idx_ssh_sessions_host_active
  ON ssh_sessions(host_id, status, started_at DESC)
  WHERE status IN ('connecting', 'connected');

-- Session list for an org with time filter (admin view)
CREATE INDEX idx_ssh_sessions_org_time
  ON ssh_sessions(org_id, started_at DESC, user_id);

-- Recording availability check
CREATE INDEX idx_session_recordings_session
  ON session_recordings(session_id, processed)
  WHERE processed = TRUE;

-- SFTP transfers for a session (transfer log)
CREATE INDEX idx_sftp_transfers_session_time
  ON sftp_transfers(sftp_session_id, transferred_at DESC);

-- Port forward rules by host and status (proxy startup)
CREATE INDEX idx_pf_rules_host_status
  ON port_forward_rules(host_id, status, auto_start)
  WHERE deleted_at IS NULL;

```

audit_db

```

-- Audit event query: org + time range (most common query pattern)
CREATE INDEX idx_audit_events_org_time
  ON audit_events(org_id, occurred_at DESC);

```

```

-- Audit event query: org + event_type + time range
CREATE INDEX idx_audit_events_org_type_time
  ON audit_events(org_id, event_type, occurred_at DESC);

-- Audit event query: org + user + time range
CREATE INDEX idx_audit_events_org_user_time
  ON audit_events(org_id, user_id, occurred_at DESC)
  WHERE user_id IS NOT NULL;

-- Audit event query: resource lookup (investigate specific
  resource)
CREATE INDEX idx_audit_events_resource_time
  ON audit_events(resource_type, resource_id, occurred_at DESC)
  WHERE resource_type IS NOT NULL;

-- Audit event query: outcome filter (failure analysis)
CREATE INDEX idx_audit_events_org_outcome_time
  ON audit_events(org_id, outcome, occurred_at DESC)
  WHERE outcome = 'failure';

```

snippet_db

```

-- Snippet list: vault + category (main list)
CREATE INDEX idx_snippets_vault_category_sort
  ON snippets(vault_id, category_id, name ASC)
  WHERE deleted_at IS NULL;

-- Snippet list: most recently used
CREATE INDEX idx_snippets_vault_last_executed
  ON snippets(vault_id, last_executed_at DESC NULLS LAST)
  WHERE deleted_at IS NULL;

-- Execution results: lookup by execution ID + status
CREATE INDEX idx_exec_results_exec_status
  ON snippet_execution_results(execution_id, status);

```

org_db

```

-- Member list: org + role + status
CREATE INDEX idx_org_members_org_role_status
  ON org_members(org_id, role, status, joined_at DESC);

-- Member lookup: find all orgs for a user

```

```

CREATE INDEX idx_org_members_user_status
  ON org_members(user_id, status);

-- Invitation lookup: pending invitations for an org
CREATE INDEX idx_invitations_org_status_expires
  ON invitations(org_id, status, expires_at DESC)
  WHERE status = 'pending';

-- Team membership: all teams for a user
CREATE INDEX idx_team_members_user_team
  ON team_members(user_id, team_id);

```

20.3 Partial Indexes for Soft-Delete Patterns

Partial indexes with `WHERE deleted_at IS NULL` are applied to every table with soft deletion. This dramatically reduces index size because deleted records (typically 5–15% of total) are excluded.

```

-- Soft-delete partial indexes (applied to all soft-delete tables)
-- These are already included in the schema definitions above.
-- The pattern is consistently: CREATE INDEX ... WHERE deleted_at
  IS NULL

-- Additional covering partial indexes for read performance:

-- Hosts: active hosts with key columns for list rendering
CREATE INDEX idx_hosts_active_list_covering
  ON hosts(vault_id, sort_order, name, hostname, status,
    last_connected_at)
  WHERE deleted_at IS NULL AND status != 'archived';

-- API keys: non-expired, non-revoked keys for lookup
CREATE INDEX idx_api_keys_active
  ON api_keys(key_hash, user_id, scopes)
  WHERE revoked = FALSE AND (expires_at IS NULL OR expires_at >
    NOW());

```

20.4 GIN Indexes for JSONB Columns

```

-- Hosts: environment_variables (for searching hosts by env var
  key)
CREATE INDEX idx_hosts_env_vars_gin
  ON hosts USING GIN (environment_variables)
  WHERE deleted_at IS NULL;

```

```

-- Hosts: custom_fields (for custom attribute search)
CREATE INDEX idx_hosts_custom_fields_gin
  ON hosts USING GIN (custom_fields)
  WHERE deleted_at IS NULL;

-- Audit events: metadata search (IP address, resource metadata)
CREATE INDEX idx_audit_events_metadata_gin
  ON audit_events USING GIN (metadata);

-- Snippets: parameters (search snippets with specific parameter
  names)
CREATE INDEX idx_snippets_parameters_gin
  ON snippets USING GIN (parameters)
  WHERE deleted_at IS NULL;

-- Organizations: settings (feature flags, custom configuration)
CREATE INDEX idx_organizations_settings_gin
  ON organizations USING GIN (settings);

-- Workspace layout (find workspaces by host ID embedded in layout
  JSON)
CREATE INDEX idx_workspaces_layout_gin
  ON workspaces USING GIN (layout)
  WHERE deleted_at IS NULL;

-- SSH sessions: jump_chain (find sessions through a specific jump
  host)
CREATE INDEX idx_ssh_sessions_jump_chain_gin
  ON ssh_sessions USING GIN (jump_chain);

-- PKI certificates: extensions
CREATE INDEX idx_certificates_extensions_gin
  ON certificates USING GIN (extensions);

-- Roles: permissions array
CREATE INDEX idx_roles_permissions_gin
  ON roles USING GIN (permissions);

```

20.5 BRIN Indexes for Time-Series Data

BRIN (Block Range INdex) indexes are used for append-only, time-ordered tables where rows are inserted in roughly chronological order. They are tiny (a few KB for millions of rows) but very effective for range queries.

```
-- BRIN indexes are appropriate for naturally time-ordered tables:
```

```
-- auth_db
```

```
CREATE INDEX idx_users_created_at_brin ON users USING BRIN  
    (created_at);
```

```
CREATE INDEX idx_login_history_occurred_at_brin ON login_history  
    USING BRIN (occurred_at);
```

```
CREATE INDEX idx_refresh_tokens_expires_at_brin ON refresh_tokens  
    USING BRIN (expires_at);
```

```
-- host_db
```

```
CREATE INDEX idx_host_conn_history_started_at_brin  
    ON host_connection_history USING BRIN (started_at);
```

```
-- session_db
```

```
CREATE INDEX idx_ssh_sessions_started_at_brin ON ssh_sessions  
    USING BRIN (started_at);
```

```
CREATE INDEX idx_session_events_occurred_at_brin ON  
    session_events USING BRIN (occurred_at);
```

```
CREATE INDEX idx_sftp_transfers_transferred_at_brin ON  
    sftp_transfers USING BRIN (transferred_at);
```

```
-- audit_db
```

```
CREATE INDEX idx_audit_events_occurred_at_brin ON audit_events  
    USING BRIN (occurred_at);
```

```
CREATE INDEX idx_audit_events_recorded_at_brin ON audit_events  
    USING BRIN (recorded_at);
```

```
-- keychain_db
```

```
CREATE INDEX idx_key_usage_occurred_at_brin ON key_usage_log  
    USING BRIN (occurred_at);
```

```
-- vault_db
```

```
CREATE INDEX idx_vault_audit_occurred_at_brin ON  
    vault_audit_events USING BRIN (occurred_at);
```

BRIN index parameters: - Default pages_per_range = 128 (1 MB ranges)
- For high-insertion-rate tables: WITH (pages_per_range = 64) - For long-term archives: WITH (pages_per_range = 256)

20.6 pg_trgm Indexes for Full-Text Search

Trigram indexes enable efficient LIKE '%pattern%' and similarity search. They back the ?q= search parameter on host, snippet, and key endpoints.

```
-- host_db: trigram search on name and hostname
```

```
CREATE INDEX idx_hosts_name_trgm ON hosts USING GIN (name  
    gin_trgm_ops)
```

```

    WHERE deleted_at IS NULL;
CREATE INDEX idx_hosts_hostname_trgm ON hosts USING GIN (hostname
    gin_trgm_ops)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_hosts_description_trgm ON hosts USING GIN
    (description gin_trgm_ops)
    WHERE deleted_at IS NULL AND description IS NOT NULL;
CREATE INDEX idx_host_groups_name_trgm ON host_groups USING GIN
    (name gin_trgm_ops)
    WHERE deleted_at IS NULL;

-- keychain_db: trigram search on key names
CREATE INDEX idx_ssh_keys_name_trgm ON ssh_keys USING GIN (name
    gin_trgm_ops)
    WHERE deleted_at IS NULL;

-- snippet_db: trigram for content search
CREATE INDEX idx_snippets_name_trgm ON snippets USING GIN (name
    gin_trgm_ops)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_snippets_content_trgm ON snippets USING GIN
    (content gin_trgm_ops)
    WHERE deleted_at IS NULL;

-- vault_db: vault name search
CREATE INDEX idx_vaults_name_trgm ON vaults USING GIN (name
    gin_trgm_ops)
    WHERE deleted_at IS NULL;

-- org_db: organization and team name search
CREATE INDEX idx_organizations_name_trgm ON organizations USING
    GIN (name gin_trgm_ops)
    WHERE deleted_at IS NULL;
CREATE INDEX idx_teams_name_trgm ON teams USING GIN (name
    gin_trgm_ops)
    WHERE deleted_at IS NULL;

-- Similarity threshold configuration:
-- SET pg_trgm.similarity_threshold = 0.3;
-- Used in queries like: WHERE name % 'search_term' (similarity
    operator)
-- Or: WHERE name ILIKE '%search_term%' (uses GIN trigram
    automatically)

```

20.7 Expression and Covering Indexes

```

-- Lowercase email lookup (case-insensitive search without CITEXT)
-- (Not needed if using CITEXT, but useful for non-CITEXT columns)

```



```

CREATE INDEX idx_users_email_lower ON users(lower(email::TEXT))
  WHERE deleted_at IS NULL;

-- Expired sessions cleanup (maintenance queries)
CREATE INDEX idx_user_sessions_expired
  ON user_sessions(expires_at)
  WHERE status = 'active' AND expires_at < NOW();

-- API key by prefix (for user display – no full hash needed)
CREATE INDEX idx_api_keys_prefix ON api_keys(key_prefix, user_id)
  WHERE revoked = FALSE;

-- Vault items by type (counting items per type for vault stats)
CREATE INDEX idx_vault_items_vault_type
  ON vault_items(vault_id, item_type)
  WHERE is_deleted = FALSE;

-- Covering index for session list rendering (avoids heap fetch
  for common columns)
CREATE INDEX idx_ssh_sessions_list_covering
  ON ssh_sessions(user_id, started_at DESC)
  INCLUDE (host_id, status, recording_enabled, ticket_ref,
    ended_at)
  WHERE started_at > NOW() - INTERVAL '90 days';

-- PKI certificate validity check
CREATE INDEX idx_certificates_validity
  ON certificates(valid_before, valid_after, revoked)
  WHERE revoked = FALSE AND valid_before > NOW();

```

20.8 Index Maintenance

pg_stat_user_indexes is monitored weekly. Indexes with zero `idx_scan` for 30 days are candidates for removal after review.

REINDEX CONCURRENTLY is run monthly for: - Tables with high UPDATE rates (sessions, hosts, vault_items) - After large bulk operations

VACUUM ANALYZE is configured via autovacuum with aggressive settings for: - audit_events partitions: `autovacuum_vacuum_scale_factor = 0.01` - session_events partitions: `autovacuum_vacuum_scale_factor = 0.005` - host_connection_history partitions: `autovacuum_analyze_scale_factor = 0.01`

Partition maintenance script (run monthly via cron):

```
-- Create next quarter's partition in advance
CREATE TABLE IF NOT EXISTS ssh_sessions_2027_q1 PARTITION OF
    ssh_sessions
    FOR VALUES FROM ('2027-01-01') TO ('2027-04-01');

-- Archive old partitions (detach after retention period)
ALTER TABLE host_connection_history DETACH PARTITION
    host_connection_history_2025_q1;
-- Move to cold storage or drop

-- Update statistics on large tables after bulk operations
ANALYZE VERBOSE ssh_sessions;
ANALYZE VERBOSE audit_events;
ANALYZE VERBOSE host_connection_history;
```

End of HelixTerminator API & Database Specification

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